

ZTE ZXA10 C300 CONFIGURATION MANUAL



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ZXA10 C300

Optical Access Convergence Equipment

Configuration Manual (CLI)

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About This Manual

Purpose

The ZXA10 C300 Optical Access Convergence Equipment (ZXA10 C300 for short) is a full-service optical access platform. It supports the video, data, voice, TDM, and CATV services. The ZXA10 C300 can be connected with MDUs/ONUs/ONTs through various networking technologies.

This manual provides detailed information about configurations (CLI) on the ZXA10 C300 Optical Access Convergence Equipment.

Intended Audience

This document is intended for:

- Debugging engineer
- Maintenance engineer

What Is in This Manual

This manual contains the following chapters:

Chapter	Summary
1, Basic Configuration	Describes basic configuration.
2, GPON/XGPON Service Configuration	Describes GPON and XG-PON service configuration.
3, CES Service Configuration	Describes CES service configuration.
4, P2P Service Configuration	Describes P2P service configuration.
5, VLAN Configuration	Describes VLAN configuration.
6, IPTV Configuration	Describes IPTV configuration.
7, QoS Configuration	Describes QoS configuration.
8, ACL Configuration	Describes ACL configuration.
9, NTP Configuration	Describes NTP configuration.
10, STP Configuration	Describes STP configuration.
11, DHCP Configuration	Describes DHCP configuration.
12, Uplink Protection Configuration	Describes uplink protection configuration.
13, PON Protection Configuration	Describes PON protection configuration.

Chapter	Summary
14, Access Security Configuration	Describes access security configuration.
15, System Security Configuration	Describes system security configuration.
16, MPLS Configuration	Describes MPLS configuration.
17, OAM Configuration	Describes OAM configuration.
18, Route Protocol Configuration	Describes route protocol configuration
19, Clock Configuration	Describes clock configuration.

Conventions

This manual uses the following typographical conventions:

Typeface	Meaning
	Note: provides additional information about a certain topic.

Chapter 1

Basic Configuration

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1.1 Management Methods

The ZXA10 C300 supports the following management methods:

- Login Through HyperTerminal
Before configuring the in-band or out-of-band [NM](#), you can only manage the ZXA10 C300 through HyperTerminal.
- Login Through Telnet
After configuring the in-band or out-of-band NM (refer to [1.2 NM Configuration](#)), you can manage the ZXA10 C300 through Telnet.
- Login Through [NMS](#)
After configuring the in-band or out-of-band NM, you can manage the ZXA10 C300 through NMS.

This manual describes the [CLI](#) configuration after login through HyperTerminal or Telnet.

1.1.1 Login Through HyperTerminal

Perform this procedure to log in to the ZXA10 C300 through HyperTerminal.

Context

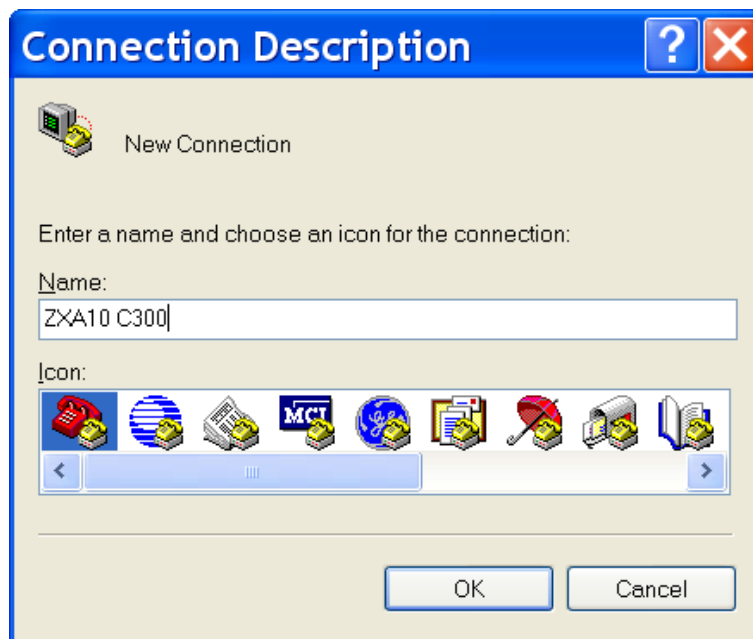
When you log in to the ZXA10 C300 through HyperTerminal, the user name and password are case-sensitive.

This topic takes the Windows XP operating system as the example.

Steps

1. In Windows XP, click **Start > All Programs > Accessories > Communications > HyperTerminal**. The **Connection Description** dialog box is displayed, as shown in [Figure 1-1](#).

Figure 1-1 Connection Description



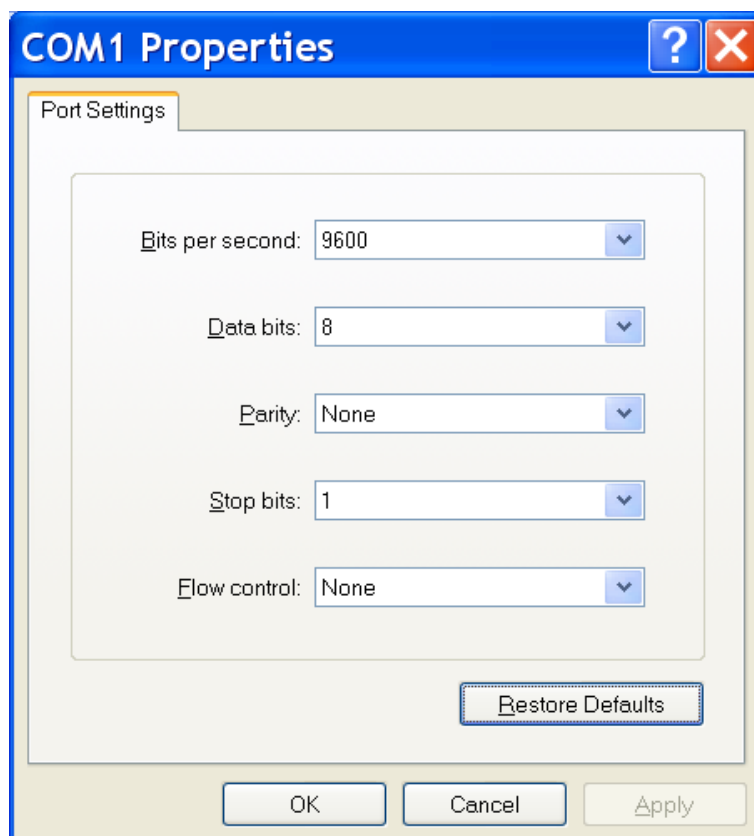
2. Fill in **Name** and click **OK**. The **Connect To** dialog box is displayed, as shown in Figure 1-2.

Figure 1-2 Connect To



3. Select **COM1** or **COM2**, and then click **OK**. The **COM1 Properties** (or **COM2 Properties**) dialog box is displayed.
4. Click **Restore Defaults**, as shown in Figure 1-3, and then click **OK**.

Figure 1-3 COM1 Properties



5. If the system runs properly, the **HyperTerminal** window is displayed. The system enters operator mode (ZXAN>). Enter the **enable** command and the password to enter administrator mode (ZXAN#), as shown below.

```
*****
Welcome to ZXAN product C300 of ZTE Corporation
*****
ZXAN>enable
Password:
ZXAN#
```

– End of Steps –

1.1.2 Login Through Telnet

Perform this procedure to log in to the ZXA10 C300 through Telnet.

Prerequisite

Before this operation, make sure that:

- The in-band or out-of-band **NM IP** address is configured.
- The Telnet computer can ping the in-band or out-of-band NM IP address.

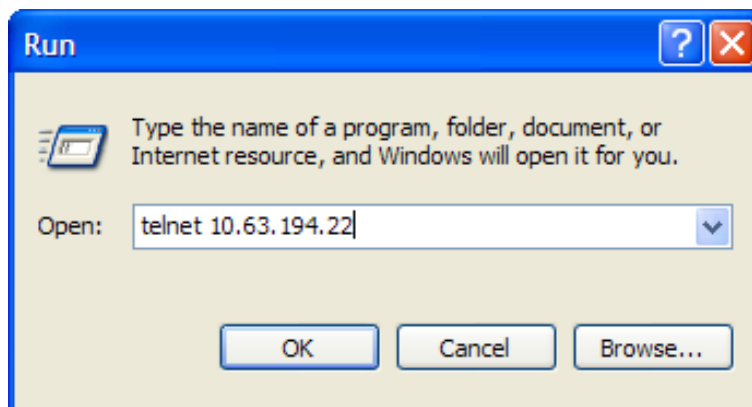
Context

When you log in to the ZXA10 C300 through Telnet, the user name and password are case-sensitive.

Steps

1. In Windows, click **Start > Run** to display the **Run** dialog box, as shown in [Figure 1-4](#).

Figure 1-4 Run Dialog Box



2. In the dialog box, enter `Telnet x.x.x.x`, where `x.x.x.x` is the [NE](#) IP address. Click **OK** to start the Telnet client.
3. If the connection is proper, the login dialog box is displayed. Enter the user name (`zte`) and password (`ZTEzte123`) to enter operator mode (`ZXAN>`), and then enter the **enable** command and the password to enter administrator mode (`ZXAN#`), as shown below.

```
*****
Welcome to ZXAN product C300 of ZTE Corporation
*****

Username:zte
Password:
ZXAN>enable
Password:
ZXAN#
```

– End of Steps –

1.1.3 Login Through NMS

Before logging in to the device through the [NMS](#), install the SQL Server database and the NetNumen U31 NMS software.

To log in to the NMS, start the SQL Server database, NMS server, and NMS client.

After creating the ZXA10 C300 [NE](#), you can manage the ZXA10 C300 through the NMS.

1.2 NM Configuration

The ZXA10 C300 supports in-band NM and out-of-band NM.

- In in-band NM mode, the ZXA10 C300 accesses the IP network via the service channel (uplink port) to transmit NM information. The in-band NM mode is usually used in practical engineering.
- In out-of-band NM mode, the ZXA10 C300 accesses the NMS via the **10/100M** port on the switching and control card. The non-service channel is used to transmit the management information so that the management channel and service channel are separated. The out-of-band NM mode is usually used in local management and maintenance.

1.2.1 Configuring the In-Band NM

In in-band [NM](#) mode, the NM information is transmitted via the service channel of the equipment. The in-band NM mode supports flexible networking and requires no additional equipment.

Prerequisite

Before this operation, make sure that:

- You have logged in to the ZXA10 C300 through HyperTerminal and entered administrator mode.
- The uplink card has been added. (Refer to [1.3.3 Adding a Card.](#))

Configuration Data

[Table 1-1](#) lists the configuration data of the in-band NM.

Table 1-1 Configuration Data of the In-Band NM

Item	Data
Uplink port	gei_1/21/1
In-band NM VLAN	VLAN ID: 1000
In-band NM IP address	10.1.1.1/24
Next hop IP address	10.1.1.254/24
NM server (SNMP server)	IP address: 10.2.1.1/24 Version: V2C Community name: public Alarm level: notifications UDP port: 162

Steps

1. Enter global configuration mode.

```
ZXAN#configure terminal
```

Enter configuration commands, one per line. End with CTRL/Z.

ZXAN(config)#

2. Add the uplink port to the in-band NM VLAN.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 1000 tag
ZXAN(config-if)#exit
```



Note:

When you use the **switchport vlan** command to configure the port VLAN, the system automatically creates the VLAN.

3. Configure the in-band NM IP address.

```
ZXAN(config)#interface vlan 1000
ZXAN(config-if)#ip address 10.1.1.1 255.255.255.0
ZXAN(config-if)#exit
```



Note:

The out-of-band and in-band NM IP addresses cannot be in the same network segment.

4. Configure the in-band NM route.

```
ZXAN(config)#ip route 10.2.1.0 255.255.255.0 10.1.1.254
```

5. Configure the SNMP community name.

```
ZXAN(config)#snmp-server community public view allview rw
```



Note:

The SNMP community name should be consistent with that on the [NMS](#).

6. Configure the IP address of the SNMP server.

```
ZXAN(config)#snmp-server host 10.2.1.1 version 2c public enable notifications target-addr-name zte target-param-name zte udp-port 162
```

7. Save the configuration data.

```
ZXAN(config)#exit
```

```
ZXAN#write
```

– End of Steps –

1.2.2 Configuring the Out-of-Band NM

In out-of-band **NM** mode, the non-service channel is used to transmit the management information so that the management channel and service channel are separated. Compared with the in-band NM mode, the out-of-band NM mode provides more reliable equipment management channel. When the ZX A10 C300 is faulty, the network equipment information can be located in time and monitored in real time.

Prerequisite

You have logged in to the ZX A10 C300 through HyperTerminal and entered administrator mode.

Configuration Data

Table 1-2 lists the configuration data of the out-of-band NM.

Table 1-2 Configuration Data of the Out-of-Band NM

Item	Data
Out-of-band NM IP address	11.1.1.1/24
Next hop IP address	11.1.1.254/24
NM server (SNMP server)	IP address: 10.2.1.1/24 Version: V2C Community name: public Alarm level: notifications UDP port: 162

Steps

1. Enter global configuration mode.

```
ZXAN#configure terminal
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#
```

2. Configure the out-of-band NM IP address.

```
ZXAN(config)#interface mng1
ZXAN(config-if)#ip address 11.1.1.1 255.255.255.0
ZXAN(config-if)#exit
```

**Note:**

The out-of-band and in-band NM IP addresses cannot be in the same network segment.

3. Configure the out-of-band NM route.

```
ZXAN(config)#ip route 10.2.1.0 255.255.255.0 11.1.1.254
```

4. Configure the SNMP community name.

```
ZXAN(config)#snmp-server community public view allview rw
```

**Note:**

The SNMP community name should be consistent with that on the [NMS](#).

5. Configure the IP address of the SNMP server.

```
ZXAN(config)#snmp-server host 10.2.1.1 version 2c public enable notifications target-addr-name zte target-param-name zte udp-port 162
```

6. Save the configuration data.

```
ZXAN(config)#exit  
ZXAN#write
```

– End of Steps –

1.3 Physical Configuration

1.3.1 Adding a Rack

When commissioning the ZXA10 C300, you need to add a rack.

Context

The ZXA10 C300 supports the following two types of racks:

- ETSI 21: 21-inch rack
- IEC 19: 19-inch rack

Steps

1. Enter global configuration mode.

```
ZXAN#configure terminal
```

Enter configuration commands, one per line. End with CTRL/Z.

```
ZXAN(config)#
```

2. Add the rack.

```
ZXAN(config)#add-rack rackno 1 racktype ETSI21
```



Note:

The ZXA10 C300 supports only one rack currently, and thus *rackno* can only be 1.

3. (Optional) Query the rack configuration.

```
ZXAN(config)#show rack
```

Rack	RackType	SupShelfNum	CfgShelfNum

1	ETSI21	1	1



Note:

'SupShelfNum' is the maximum shelf number supported by the rack.

– End of Steps –

1.3.2 Adding a Shelf

When commissioning the ZXA10 C300, you need to add a shelf in a rack.

Prerequisite

The rack has been added.

Context

The ZXA10 C300 supports the following two types of shelves:

- ETSI_SHELF: 21-inch shelf
- IEC_SHELF: 19-inch shelf

Steps

1. Enter global configuration mode.

```
ZXAN#configure terminal
```

```
Enter configuration commands, one per line. End with CTRL/Z.
```

```
ZXAN(config)#
```

2. Add the shelf.

```
ZXAN(config)#add-shelf shelfno 1 shelftype ETSI_SHELF
```



Note:

The ZXA10 C300 supports only one shelf currently, and thus *shelfno* can only be 1.

3. (Optional) Query the shelf configuration.

```
ZXAN(config)#show shelf
```

Rack	Shelf	ShelfType	CleiCode	Serial-Number
1	1	ETSI_SHELF	UNKNOWN	10011001C3000090 000000000000

– End of Steps –

Result

After the shelf is added, the system will automatically add two switching and control cards.

```
ZXAN(config)#show card
```

Rack	Shelf	Slot	CfgType	RealType	Port	HardVer	SoftVer	Status
1	1	10	SCXN	SCXN	4	120900	V2.0.0	INSERVICE
1	1	11	SCXN	SCXN	4	120900	V2.0.0	STANDBY

1.3.3 Adding a Card

When replacing the card type during commissioning or capacity expansion, you need to add a card.

Context

Table 1-3 describes the card configurations for a 19-inch shelf.

Table 1-3 Card Configurations for a 19-Inch Shelf

Slot Number	Card Type
0/1	Power card
2–9	PON interface card, TDM interface card, Ethernet interface card, and P2P interface card
10/11	Switching and control card
12–17	PON interface card, TDM interface card, Ethernet interface card, and P2P interface card
18	Common interface card
19/20	Uplink card

Table 1-4 describes the card configurations for a 21-inch shelf.

Table 1-4 Card Configurations for a 21-Inch Shelf

Slot Number	Card Type
0/1	Power card
2–9	PON interface card, TDM interface card, Ethernet interface card, and P2P interface card
10/11	Switching and control card
12–19	PON interface card, TDM interface card, Ethernet interface card, and P2P interface card
20	Common interface card
21/22	Uplink card

Steps

1. Enter global configuration mode.

```
ZXAN#configure terminal
```

Enter configuration commands, one per line. End with CTRL/Z.

```
ZXAN(config)#
```

2. Add cards.

```
ZXAN(config)#add-card slotno 15 GTGO
```

```
ZXAN(config)#add-card slotno 21 GUSQ
```

3. (Optional) Query the card configuration.

```
ZXAN(config)#show card
```

Rack	Shelf	Slot	CfgType	RealType	Port	HardVer	SoftVer	Status
1	1	0	PRWG	PRWG	0			INSERVICE
1	1	1	PRWG	PRWG	0			INSERVICE
1	1	10	SCXN	SCXN	4	120900	V2.0.0	INSERVICE
1	1	11	SCXN	SCXN	4	120900	V2.0.0	STANDBY
1	1	15	GTGO	GTGOG	8	120301	V2.0.0	INSERVICE
1	1	21	GUSQ	GUSQ	4	090200	V2.0.0	INSERVICE

– End of Steps –

1.3.4 Enabling the PnP Function

The ZXA10 C300 supports the plug and play (PnP) function of the card. By default, the PnP function of the ZXA10 C300 is enabled.

Steps

1. In global configuration mode, enable the PnP function.

```
ZXAN#configure terminal
```

Enter configuration commands, one per line. End with CTRL/Z.

```
ZXAN(config)#set-pnp enable
```

2. (Optional) Query the PnP status.

```
ZXAN#show pnp
```

Equipment PNP function is enable.

3. (Optional) Query the card configuration.

```
ZXAN(config)#show card
```

Rack	Shelf	Slot	CfgType	RealType	Port	HardVer	SoftVer	Status
1	1	0	PRWG	PRWG	0			INSERVICE
1	1	1	PRWG	PRWG	0			INSERVICE
1	1	10	SCXN	SCXN	4	120900	V2.0.0	INSERVICE
1	1	11	SCXN	SCXN	4	120900	V2.0.0	STANDBY
1	1	15	GTGO	GTGOG	8	120301	V2.0.0	INSERVICE
1	1	21	GUSQ	GUSQ	4	090200	V2.0.0	INSERVICE

– End of Steps –

1.3.5 Showing Cards

The card information includes slot No., card type, number of ports, hardware version, software version, and status.

Context

Table 1-5 describes the card status of the ZXA10 C300.

Table 1-5 Card Status Description

Status	Description
INSERVICE	The card is working normally.
CONFIGING	The card is being configured.
CONFIGFAILED	The service configuration for the card fails.
DISABLE	The card is added and is online, but the system fails to receive the card information.
HWONLINE	The card of incorrect version is inserted into the shelf so that it does not run normally.
OFFLINE	The card is added but is offline.
STANDBY	The card is in standby state.
TYPENISMATCH	The card type is different from the configured type.
NOPOWER	The power card is not powered on.

Steps

1. Query all the cards.

```
ZXAN#show card
```

Rack	Shelf	Slot	CfgType	RealType	Port	HardVer	SoftVer	Status
1	1	0	PRWG	PRWG	0			INSERVICE
1	1	1	PRWG	PRWG	0			INSERVICE
1	1	10	SCXN	SCXN	4	120900	V2.0.0	INSERVICE
1	1	11	SCXN	SCXN	4	120900	V2.0.0	STANDBY
1	1	15	GTGO	GTGOG	8	120301	V2.0.0	INSERVICE
1	1	21	GUSQ	GUSQ	4	090200	V2.0.0	INSERVICE

2. Query a certain card.

```
ZXAN#show card slotno 10
```

```
Config-Type      : SCXN      Status      : INSERVICE
Port-Number      : 4
Cpu-Alarm-Threshold : 100%
Mem-Alarm-Threshold : 100%
```

```
Real-Type      : SCXN      Serial-Number : 269644400040
Phy-Mem-Size   : 1024MB    Main-CPU      : PowerPC Processor
PCB-VER        : 120900
Cpld-VER       : V1.4      Fpga-VER       :
OtherfirewareVER:
```

```
BootROM-VER     : V4.0.0    2013-03-22 10:50:14
Software-VER    : V2.0.0    2013-05-17 15:06:01
Cpu-Usage       : 18%
Mem-Usage       : 36%
Uptime          : 6 Days, 7 Hours, 23 Minutes, 42 Seconds
```

– End of Steps –

1.3.6 Deleting a Card

When replacing a card with another card of different type, you need to delete the existing card before adding a new card.

Steps

1. Enter global configuration mode.

```
ZXAN#configure terminal
Enter configuration commands, one per line. End with CTRL/Z.
ZXAN(config)#
```

2. Delete the card.

```
ZXAN(config)#del-card slotno 15
Confirm to delete card? [yes/no]:y
```

3. (Optional) Query the card configuration.

```
ZXAN(config)#show card
```

Rack	Shelf	Slot	CfgType	RealType	Port	HardVer	SoftVer	Status
1	1	0	PRWG	PRWG	0			INSERVICE
1	1	1	PRWG	PRWG	0			INSERVICE
1	1	10	SCXN	SCXN	4	120900	V2.0.0	INSERVICE
1	1	11	SCXN	SCXN	4	120900	V2.0.0	STANDBY
1	1	21	GUSQ	GUSQ	4	090200	V2.0.0	INSERVICE

– End of Steps –

1.3.7 Resetting a Card

You can reset the card to rectify a fault or clear an alarm. For example, when the ZXA10 C300 reports an "abnormal card state" alarm, you can clear the alarm by resetting the card.

Steps

1. In administrator mode, reset the card.

```
ZXAN#reset-card slotno 15
Confirm to reset card? [yes/no]:y
```

– End of Steps –

1.3.8 Swapping the Active/Standby Switching and Control Cards

When the active switching and control card is faulty, you can switch the service to the standby switching and control card to ensure normal service by swapping the active and standby switching and control cards.

Steps

1. In administrator mode, swap the active and standby switching and control cards.

```
ZXAN#swap
Confirm to master swap? [yes/no]:y
```

**Note:**

You can also swap the active and standby switching and control cards through the following methods:

- Pull out the active switching and control card.
- Press the **RST** button on the active switching and control card.

– End of Steps –

1.3.9 Configuring Fans

This section describes how to configure the fan parameters, such as working mode, speed, and temperature threshold.

Context

The ZXA10 C300 supports configuration of the following fan parameters:

- Working mode
 - Temperature-control mode
 - Fixed-speed mode
- Fan speed level

The options are 0 – 4. This parameter is valid only when the fans are in fixed-speed working mode.
- Fan speed percent

The fan speed is the maximum fan speed multiplied with the fan speed percent.
- High-temperature threshold

When the ambient temperature is higher than the high-temperature threshold, the ZXA10 C300 reports a high-temperature alarm and disables the interface card. When the ambient temperature is lower than the high-temperature threshold, the ZXA10 C300 enables the interface card again.

Steps

1. In global configuration mode, configure the fan working mode.

```
ZXAN(config)#fan control temp_level 30 40 50 60
```

**Note:**

The ZXA10 C300 supports four temperature levels.

You can use the **fan control fixed-speed** command to set fixed-speed working mode, and then use the **fan speed** command to set the fan speed level.

2. Configure the speed percent of each level.

```
ZXAN(config)#fan speed-percent-set 25 36 50 75
```

3. Configure the high-temperature threshold.

```
ZXAN(config)#fan high-threshold 70
```

4. (Optional) Query the fan configuration.

```
ZXAN(config)#show fan
Shelf                : 1
epm                  : disable
FanControlType       : temperature-control
TemperatureThreshold : 30 40 50 60 (deg c)
FanSpeedLevelPercent : 25% 36% 50% 75%
HighTemperatureThreshold : 70 (deg c)
Environment Temperature : 61 (deg c)
HighTemperatureProtection : Threshold : N/A.(deg c)
                        RestartTime: N/A.(Minute)
Upper Fanboard Status : online
All fan units actual status:
```

```
-----
FanUnitId   SpeedLevel   ShiftSpeed(RPM)
-----
1           full        2260
2           full        2320
3           full        2300
4           full        2480
-----
```

– End of Steps –

1.4 System Configuration

1.4.1 Configuring the System Time

After the system time is configured, you can query [CLI](#) logs and alarms logs in specific time for troubleshooting.

Context

The ZXA10 C300 software maintains the system time. When the NE is powered on, the system acquires the hardware clock and initializes the system time of the NE.

Steps

1. In global configuration mode, configure the time zone.

```
ZXAN(config)#clock timezone utc 8
ZXAN(config)#exit
```

2. In administrator mode, configure the system time.

```
ZXAN#clock set 08:00:00 may 7 2013
```

3. (Optional) Query the system time.

```
ZXAN#show clock
08:01:55 Mon May 7 2013 utc
```

– End of Steps –

1.4.2 Configuring the Version Auto-Update Function

The ZXA10 C300 supports the periodic version auto-update function, that is, the ZXA10 C300 checks the consistency between the version files with the version files on the file server and updates the version files according the auto-update policy whenever there is an inconsistency.

Configuration Data

Table 1-6 lists the configuration data of auto-update function.

Table 1-6 Configuration Data of Auto-Update Function

Item	Data
File server	● IP address: 10.1.1.1 ● User name: zte ● Password: zte
Auto-update check period	● Starting time: 15:00:00, March 5, 2013 ● Interval: 24 hour
Auto-update policy	● Version backup: enable ● Version activate: enable

Steps

1. In global configuration mode, configure the version file server.

```
ZXAN(config)#file-server auto-update server-index 1 ftp ipaddress 10.1.1.1 user
zte password zte
```

2. Configure the starting time and interval of the version auto-update check period.

```
ZXAN(config)#auto-update check-period 15:00:00 mar 5 2013 interval 24
```

3. Enable the auto-backup function and card-version-update function in the auto-update process.

```
ZXAN(config)#auto-update backup enable
ZXAN(config)#auto-update activate enable
```

4. (Optional) Query the file server configuration.

```
ZXAN(config)#show file-server auto-update
ServerIndex : 1
ProtocolType: Ftp                      Server-IPAddr: 10.1.1.1
Username    : zte
Password    : *****
Path        :

ServerIndex : 2
Not configuration
```

5. (Optional) Query the auto-update check-period configuration.

```
ZXAN(config)#show auto-update check-period configure
Enable      Start-time                Interval(hours)
-----
enable      2013-03-05 15:00:00      24
```

6. (Optional) Query the auto-update configuration.

```
ZXAN#show auto-update configure
Backup      Active
-----
enable      enable
```

– End of Steps –

1.4.3 Configuring the Auto-Backup Function

The ZXA10 C300 supports conditional auto-backup for configuration file, log file, and version files.

Configuration Data

Table 1-7 lists the configuration data of auto-backup function.

Table 1-7 Configuration Data of Auto-Backup Function

Item	Data
File server	● File type: all ● IP address: 10.1.1.1 ● Path: bak ● User name: zte ● Password: zte

Item	Data
Auto-backup condition	● Configuration changed: enable ● Hold-off time: 1 hour ● Maximum hold-off time: 2 hour

To configure the auto-backup function, perform the following steps:

Steps

1. In global configuration mode, configure the backup file server.

```
ZXAN(config)#file-server auto-backup all server-index 1 ftp ipaddress 10.1.1.1
path bak user zte password zte
```

2. Configure the condition and interval of the version auto-backup check point.

```
ZXAN(config)#auto-backup condition cfg-changed hold-off-time 1 max-hold-off-time 2
```

3. Query the file server configuration.

```
ZZXAN(config)#show file-server auto-backup
FileType      : Cfg
ServerIndex   : 1
ProtocolType: Ftp                               Server-IPAddr: 10.1.1.1
Username      : zte
Password      : *****
Path          : bak

FileType      : Cfg
ServerIndex   : 2
Not configuration
```

```
FileType      : Log
ServerIndex   : 1
ProtocolType: Ftp                               Server-IPAddr: 10.1.1.1
Username      : zte
Password      : *****
Path          : bak
```

```
FileType      : Log
ServerIndex   : 2
Not configuration
```

```
FileType      : Img
ServerIndex   : 1
ProtocolType: Ftp                               Server-IPAddr: 10.1.1.1
Username      : zte
Password      : *****
Path          : bak
```

```

FileType      : Img
ServerIndex   : 2
Not configuration

```

4. (Optional) Query the auto-backup condition configuration.

```

ZXAN(config)#show auto-backup condition configure
Cfg-changed      Hold-off-time (hours)      Max-hold-off-time (hours)
-----
enable           1                          2

```

– End of Steps –

1.4.4 Configuring the Auto-Save Function

The ZXA10 C300 supports saving configuration automatically.

Steps

1. In global configuration mode, enable the auto-save function.

```
ZXAN(config)#auto-write enable
```

2. Configure the time for auto-save operation.

```
ZXAN(config)#auto-write 02:00:00 may 5 2013
```

3. (Optional) Query the auto-save configuration.

```
ZXAN(config)#show auto-write
```

```

auto-write global configuration:
-----
auto-write enable
auto-write 02:00:00 May 5 2013

```

– End of Steps –

1.5 User Management

Users (operators) refer to the personnel who manage and maintain the ZXA10 C300 after logging in to it through CLI terminals, including console port, telnet, or security shell (SSH).

The user management defines 16 privilege levels (0 – 15). [Table 1-8](#) describes user privileges.

Table 1-8 User Privilege Description

Privilege Level	Description
0–1	When the user logs in and enters operator mode, he can type the enable command and the password to enter privilege mode (privilege level is 15), and uses any commands.

Privilege Level	Description
2–9	When the user logs in and enters the administrator mode, he can use the commands of level 0 – 9.
10–15	When the user logs in and enters the administrator mode, he can use the commands of level 0 – 15. The user can manages user accounts.

- A user whose privilege level is 0 can only use the commands of level 0.
- A user whose privilege level is 1 can only use the commands of level 0–1.
- A user whose privilege level is 2 can only use the commands of level 0–2, and so on.
- A user whose privilege level is 15 can only use the commands of level 0–15.

1.5.1 Adding a User

When you add a user, you need to configure user properties, including the user name, password and privilege.

Context

Table 1-9 describes user properties.

Table 1-9 User Properties Description

Property	Description
Username	1 – 16 printable characters (no space), case sensitive The user name must be unique on the ZXA10 C300.
Password	8–32 characters, should contain characters from at least three categories: lower-case, capitals, digits or special characters.
Max-session	Maximum session number, 1 – 16
Privilege	0 – 15

The ZXA10 C300 supports maximum 20 users.

The default user on the ZXA10 C300 is zte, whose password is ZTEzte123, and the privilege is 1.

Steps

1. Add a user.

```
ZXAN(config)#username abc password Abc12345 privilege 10
```

2. (Optional) Query the user configuration.

```
ZXAN(config)#show username
cli user global configuration
```

```
-----
```

```
name      sessions  pri  OperStatus  login-begin  login-end  expire-date
```

```

-----
zte          16      1   Normal    00:00:00    23:59:59    2099-12-31 23:59:59
admin        16      1   Normal    00:00:00    23:59:59    2099-12-31 23:59:59
123          16      1   Normal    00:00:00    23:59:59    2099-12-31 23:59:59
abc          16     10   Normal    00:00:00    23:59:59    2099-12-31 23:59:59

```

– End of Steps –

1.5.2 Modifying a User

It is recommended to modify user password and privilege in time to ensure the security.

Context

Only the user whose privilege is 15 can modify other users.

Steps

1. Modify the user password and privilege.

```
ZXAN(config)#username abc password Abcabcl23 privilege 15
```

2. (Optional) Query the user configuration.

```
ZXAN(config)#show username
cli user global configuration
```

```

-----
name      sessions  pri  OperStatus  login-begin  login-end  expire-date
-----
zte        16       1   Normal     00:00:00     23:59:59   2099-12-31 23:59:59
admin      16       1   Normal     00:00:00     23:59:59   2099-12-31 23:59:59
123        16       1   Normal     00:00:00     23:59:59   2099-12-31 23:59:59
abc        16      15   Normal     00:00:00     23:59:59   2099-12-31 23:59:59

```

– End of Steps –

1.5.3 Deleting a User

It is recommended to delete idle users to ensure the security.

Context

Only the user whose privilege is 15 can delete other users.

Steps

1. Delete the user.

```
ZXAN(config)#no username abc
```

2. (Optional) Query the user configuration.

```

ZXAN(config)#show username
cli user global configuration
-----

name      sessions  pri  OperStatus  login-begin  login-end  expire-date
-----
zte       16         1   Normal     00:00:00     23:59:59   2099-12-31 23:59:59
admin     16         1   Normal     00:00:00     23:59:59   2099-12-31 23:59:59
123       16         1   Normal     00:00:00     23:59:59   2099-12-31 23:59:59

```

– End of Steps –

1.5.4 Disconnecting an Online User

When the number of online users reaches the limit, you can disconnect the specific online user.

Context

Only the user whose privilege is 5 – 15 can disconnect online users.

Steps

1. Query the online users.

```

ZXAN#show users

Line      User      Host(s)      Idle      Location
---
66 vty 0  zte      idle        00:50:48   10.63.192.213
67 vty 1  zte      idle        00:13:27   10.63.78.129
* 69 vty 3  abc      idle        00:00:00   10.60.113.35

```

2. Query the TCP connections.

```

ZXAN#show tcp brief

TCB          Local Address      Foreign Address     State
---
410088992    10.63.192.225.23   10.60.113.35.2053   ESTAB
417166736    10.63.192.225.23   10.63.78.129.1617   ESTAB
410187888    10.63.192.225.23   10.63.192.213.3641   ESTAB

```

3. Disconnect the online user.

- Disconnect the online user by teletypewriter (TTY) line.

```
ZXAN#clear tcp line 69
```

- Disconnect the online user by IP address.

```
ZXAN#clear tcp connect 10.63.192.225 23 10.63.192.213 3641
```

– End of Steps –

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Chapter 2

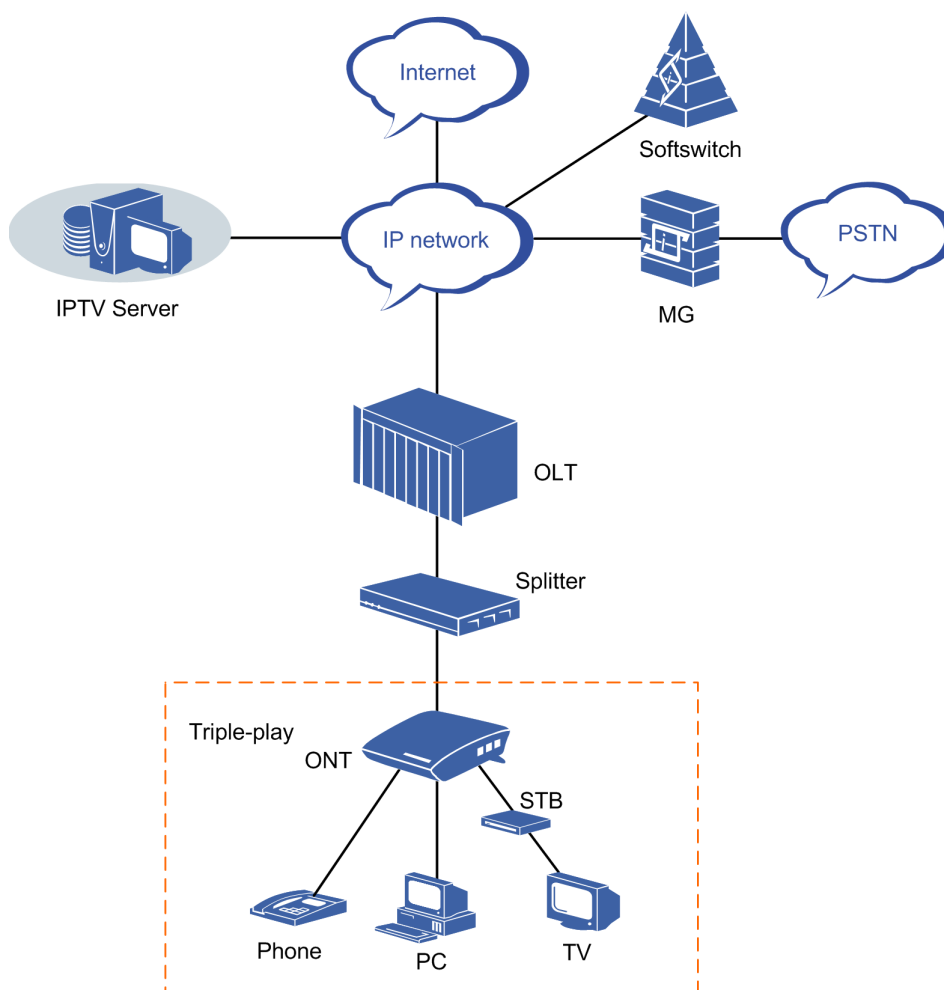
GPON/XG-PON Service Configuration

The Gigabit Passive Optical Network ([GPON](#)) and XG-PON access are a flexible access technologies that provide super bandwidth access in both broadband and narrowband service environments. It supports multiple rate modes and uses a single optical fiber to provide the subscriber with the voice, data, and video services.

The principles of the GPON service and XG-PON service are the same except the downstream bandwidth.

- GPON service: 2.488 Gbps
- XG-PON service: 9.553 Gbps

[Figure 2-1](#) shows the GPON/XG-PON service networking diagram.

Figure 2-1 GPON/XG-PON Service Networking Diagram**Table of Contents**

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2.1 Configuring the GPON ONU Type Profile

Before authenticating the GPON optical network unit (ONU), you need to create an ONU type profile if the ONU type does not exist.

Context

The ZXA10 C300 supports the following default GPON ONU types.

- ZTE-F601
- ZTE-F621
- ZTE-F622
- ZTE-F625
- ZTE-F628
- ZTE-F640
- ZTE-F641

The ZXA10 C300 supports the only one type of XG-PON ONU: ZTE-F220.

On the ZXA10 C300, ZTE-9806, ZTE-F822, and ZTE-F820 are EPON ONU types. If you need to configure the corresponding GPON ONU type, you can use ZTEG-9806H, ZTEG-F822, and ZTEG-F820 respectively.

You can use the **show onu-type gpon** command to query the default GPON ONU types.

Configuration Data

Table 2-1 lists the configuration data of the GPON ONU type.

Table 2-1 Configuration Data of the GPON ONU Type

Item	Data
ONU type	ZTEG-F620
ONU description	4ETH,2POTS
Maximum T-CONT	7
Maximum number of GEM ports	32
Maximum number of switch units per slot	1
Maximum number of flows per switch unit	8
Number of user ports	ETH: 4 POTS: 2

Steps

1. Enter global configuration mode.

```
ZXAN#configure terminal
```

Enter configuration commands, one per line. End with CTRL/Z.

```
ZXAN(config)#
```

2. In PON configuration mode, create an ONU type profile.

```
ZXAN(config)#pon
ZXAN(config-pon)#onu-type ZTEG-F620 gpon description 4ETH,2POTS max-tcont 7
max-gemport 32 max-switch-perslot 1 max-flow-perswitch 8
```

**Note:**

On the ZXA10 C300, the **GPON** and **EPON** ONU types must be different.

3. Configure the user port of the ONU type.

```
ZXAN(config-pon)#onu-type-if ZTEG-F620 eth_0/1-4
ZXAN(config-pon)#onu-type-if ZTEG-F620 pots_0/1-2
```

4. (Optional) Query the configured ONU type profile.

```
ZXAN(config-pon)#show onu-type gpon ZTEG-F620
```

```
ONU type name:          ZTE-F620
PON type:               gpon
Description:            4ETH, 4POTS
Max T-CONT:             7
Max GEM port:           32
Max switch per slot:    1
Max flow per switch:    8
Max iphost:             2
Service ability N:l:    support
Service ability l:M:    support
Service ability l:P:    support
WIFI mgmt via non OMCI: disable
OMCI send mode:         async
Default multicast range: none
```

– End of Steps –

2.2 Authenticating the GPON ONU

Before configuring its services, you need to authenticate the GPON ONU that is online initially.

Prerequisite

The GPON ONU type profile has been configured by default or manually.

Context

The ZXA10 C300 supports the following ONU authentication modes:

- SN authentication

Using the ONU SN for authentication

- Password authentication

Using the ONU password for authentication

- SN + password authentication

Using the ONU SN and password for authentication

Configuration Data

Table 2-2 lists the configuration data for GPON ONU authentication.

Table 2-2 Configuration Data for GPON ONU Authentication

Item	Data
ONU ID	1
ONU type	ZTEG-F620
SN	ZTEG00000002

Steps

1. Query the unauthenticated ONU.

```
ZXAN(config)#show gpon onu uncfg gpon-olt_1/5/1
```

```
OnuIndex          Sn                      State
```

```
-----
gpon-onu_1/5/1:1      ZTEG00000002          unknown
```

2. In Optical Line Terminal (OLT) interface mode, authenticate the ONU.

```
ZXAN(config)#interface gpon-olt_1/5/1
```

```
ZXAN(config-if)#onu 1 type ZTEG-F620 sn ZTEG00000002
```

```
[Successful]
```

```
ZXAN(config-if)#exit
```

3. (Optional) Query the authenticated ONU.

```
ZXAN(config)#show gpon onu state gpon-olt_1/5/1
```

```
OnuIndex          Admin State  Omcc State   O7 State     Phase State
```

```
-----
gpon-onu_1/5/1:1      enable      enable       operation     working
```

Table 2-3 describes the ONU phase states.

Table 2-3 Descriptions of ONU Phase States

State	Description
offline	The OLT does not find the ONU because the ONU is offline.
logging	The OLT has found the ONU and is measuring the distance.
syncMib	The OLT has measured the distance to the ONU and is synchronizing data.

State	Description
working	The data synchronization completes, and you can configure services.
LOS	The fiber link between the OLT and ONU is faulty so that the ONU is offline.
DyingGasp	The ONU is powered off.

– End of Steps –

2.3 Configuring the T-CONT Bandwidth Profile

The T-CONT bandwidth profile describes the T-CONT flow parameters. By specifying the T-CONT bandwidth profile, you can implement the T-CONT flow control.

Context

The ZXA10 C300 supports 512 transmission container (T-CONT) profiles.

There are the following five types of upstream bandwidths:

- Fixed bandwidth (FBW)
- Assured bandwidth (ABW)
- Non-assured bandwidth
- Best-effort bandwidth
- Maximum bandwidth (MBW)

The priorities of fixed bandwidth, assured bandwidth, non-assured bandwidth, best-effort bandwidth, and maximum bandwidth are in descending order.

A T-CONT bandwidth profile may contains one or multiple types of bandwidths. Five types of T-CONT bandwidth profiles are as follows:

- Fixed bandwidth (type 1)

Type 1 includes only fixed bandwidth. Type 1 has fixed bandwidth and timeslot. It is applicable to the service that is sensitive to delay and jitter and has fixed or stable flow rate, such as the voice service.

- Assured bandwidth (type 2)

Type 2 includes only assured bandwidth. Type 2 has fixed bandwidth but not timeslot. It is applicable to the service that is insensitive to delay and jitter and has limited flow rate, such as the video on demand (VOD) service.

- Assured and non-assured bandwidths (type 3)

Type 3 includes assured and non-assured bandwidths. It has the assured minimum bandwidth and shares the excess bandwidth dynamically. Meanwhile, it is constrained by the maximum bandwidth. It is applicable to the service that requires service assurance but has a large volume of flow burst, such as the subscription download service.

- Best-effort bandwidth (type 4)

Type 4 includes only best-effort bandwidth. After the fixed bandwidth, assured bandwidth, and non-assured bandwidth are allocated, type 4 competes for the excess bandwidth. It is applicable to the service that is insensitive to delay and jitter, such as the Web browse service.

- Support all (type 5)

Type 5 combines the four types and has the characteristics of the four types. It is applicable to most service streams.

The summary of fixed bandwidth and assured bandwidth on a PON port must be no more than 1 Gbps. [Table 2-4](#) lists the parameters of the default T-CONT bandwidth profile.

Table 2-4 Parameters of the Default T-CONT Profile

Parameter	Value
Bandwidth type	1
FBW	10000 kbps
ABW	0
MBW	0

Configuration Data

[Table 2-5](#) lists the configuration data for the T-CONT bandwidth profile.

Table 2-5 Configuration Data for the T-CONT Profile

Item	Data
T-CONT bandwidth profile 1	Profile name: 20M Bandwidth type: type 5 Fixed bandwidth: 2000 kbps Assured bandwidth: 5000 kbps Maximum bandwidth: 20000 kbps
T-CONT bandwidth profile 2	Profile name: 15M Bandwidth type: type 4 Maximum bandwidth: 15000 kbps
T-CONT bandwidth profile 3	Profile name: 10M Bandwidth type: type 3 Assured bandwidth: 5000 kbps Maximum bandwidth: 10000 kbps
T-CONT bandwidth profile 4	Profile name: 5M Bandwidth type: type 2 Assured bandwidth: 5000 kbps

Item	Data
T-CONT bandwidth profile 5	Profile name: 2M Bandwidth type: type 1 Fixed bandwidth: 2000 kbps

Steps

1. In GPON configuration mode, create a T-CONT bandwidth profile.

```

ZXAN(config)#gpon
ZXAN(config-gpon)#profile tcont 20M type 5 fixed 2000 assured 5000 maximum 20000
[Successful]
ZXAN(config-gpon)#profile tcont 15M type 4 maximum 15000
[Successful]
ZXAN(config-gpon)#profile tcont 10M type 3 assured 5000 maximum 10000
[Successful]
ZXAN(config-gpon)#profile tcont 5M type 2 assured 5000
[Successful]
ZXAN(config-gpon)#profile tcont 2M type 1 fixed 2000
[Successful]

```

2. (Optional) Query the T-CONT bandwidth profile configuration.

```

ZXAN(config-gpon)#show gpon profile tcont
Name :default
Type          FBW(kbps)  ABW(kbps)  MBW(kbps)
1             10000     0           0
Name :20M
Type          FBW(kbps)  ABW(kbps)  MBW(kbps)
5             2000      5000       20000
Name :15M
Type          FBW(kbps)  ABW(kbps)  MBW(kbps)
4             0         0          15000
Name :10M
Type          FBW(kbps)  ABW(kbps)  MBW(kbps)
3             0         5000       10000
Name :5M
Type          FBW(kbps)  ABW(kbps)  MBW(kbps)
2             0         5000       0
Name :2M
Type          FBW(kbps)  ABW(kbps)  MBW(kbps)
1             2000      0           0

```

– End of Steps –

2.4 Configuring the GPON ONU IP Profile

Using the GPON ONU IP profile, you can configure IP addresses for GPON ONUs in batches.

Context

The ZXA10 C300 supports the following three IP address allocation modes:

- Static allocation mode
- Dynamic Host Configuration Protocol (DHCP) mode
- Point to Point Protocol over Ethernet (PPPoE) mode

One ONU can use only one IP address allocation mode.

The ONU IP profile is applicable to only the static allocation mode.

Configuration Data

Table 2-6 lists the configuration data of the GPON ONU IP profile.

Table 2-6 Configuration Data of the GPON ONU IP Profile

Item	Data
Profile name	ip-test
IP address allocation mode	static
Gateway IP address	1.2.3.1

Steps

1. In GPON configuration mode, configure the ONU IP profile.

```
ZXAN(config)#gpon
ZXAN(config-gpon)#onu profile ip ip-test gateway 1.2.3.1
```

2. (Optional) Query the ONU IP profile.

```
ZXAN(config-gpon)#show gpon onu profile ip
Profile name:  ip-test
Gateway:       1.2.3.1
Primary DNS:   0.0.0.0
Secondary DNS: 0.0.0.0
```

– End of Steps –

2.5 Configuring the GPON ONU VLAN Profile

Using the GPON ONU VLAN profile, you can configure VLANs for GPON ONUs in batches.

Configuration Data

Table 2-7 lists the configuration data of the GPON VLAN profile.

Table 2-7 Configuration Data of the GPON VLAN Profile

Item	Data
Profile name	vlan-test
Tag mode	Tag
VLAN ID	300
Priority	7

Steps

1. In GPON configuration mode, configure the ONU VLAN profile.

```
ZXAN(config)#gpon
ZXAN(config-gpon)#onu profile vlan vlan-test tag-mode tag cvlan 300 pri 7
```

2. (Optional) Query the ONU VLAN profile.

```
ZXAN(config-gpon)#show gpon onu profile vlan
Profile name:  vlan-test
Tag mode:      tag
CVLAN:        300
CVLAN priority:7
```

– End of Steps –

2.6 Configuring the VoIP Access Code Profile

The [VoIP](#) access code profile can be used to configure access codes for VoIP advanced services, which are based on [SIP](#), for [GPON ONUs](#) in batches.

Context

You can set up relation between access codes and corresponding services on ONUs by configuring a VoIP access code profile. When a subscriber dials an access code, the corresponding service is activated on the ONU (SIP agent), and then processed according to the service procedure.

Configuration Data

[Table 2-8](#) lists the configuration data of the [VoIP](#) service application profile.

Table 2-8 Configuration Data of the VoIP Access Code Profile

Item	Data
Profile name	abc
Access code for call hold	***

Steps

1. In GPON configuration mode, configure the VoIP access code profile.

```
ZXAN(config)#gpon
ZXAN(config-gpon)#onu profile voip-accesscode abc call-hold ***
```

2. (Optional) Query the VoIP access code profile.

```
ZXAN(config-gpon)#show gpon onu profile voip-accesscode
Profile-name:                abc
cancel-call-waiting:
call-hold:                   ***
call-park:
cid-activate:
cid-deactivate:
no-disturb-activation:
no-disturb-deactivation:
no-disturb-pin-change:
emergency-srv-num:
intercom-service:
unattend-blind-call-transfer:
attend-call-transfer:
```

– End of Steps –

2.7 Configuring the VoIP Service Application Profile

The [VoIP](#) service application profile can be used to configure VoIP advanced services, which are based on [SIP](#), for [GPON ONU](#)s in batches.

Configuration Data

[Table 2-9](#) lists the configuration data of the [VoIP](#) service application profile.

Table 2-9 Configuration Data of the VoIP Service Application Profile

Item	Data
Profile name	voip-service
Call waiting	Enable
Call transfer	Enable
Call hold	Enable
3-way call	Enable

Steps

1. In GPON configuration mode, configure the VoIP service application profile.

```
ZXAN(config)#gpon
ZXAN(config-gpon)#onu profile voip-appsrv voip-service call-waiting enable
call-transfer enable call-hold enable 3way enable
```

2. (Optional) Query the VoIP service application profile.

```

ZXAN(config-gpon)#show gpon onu profile voip-appsrv
Profile-name:                voip-service
calling-num:                 disable
calling-name:                disable
cid-blocking:                disable
cid-num-permanent-status:    disable
cid-name-permanent-status:   disable
anonymous-cid-blocking:     disable
call-wating:                 enable
cid-announcement:            disable
3way:                        enable
call-transfer:               enable
call-hold:                   enable
call-park:                   disable
no-disturb:                  disable
emergency-call-flash:        disable
emergency-originate-hold:    disable
6way:                         disable
message-waiting-splash-ring: disable
message-wating-special-dialtone: disable
message-waiting-visual-ind:  disable
call-forwarding-ind:         disable
direct-connect-feature:      disable
dialtone-delay:              disable
direct-connect-uri:
    Validation scheme:        disable
    Username:
    Password:
    Realm:
bridge-line-agent-uri:
    Validation scheme:        disable
    Username:
    Password:
    Realm:
conference-factory-uri:
    Validation scheme:        disable
    Username:
    Password:
    Realm:

```

– End of Steps –

2.8 Configuring the Dial Plan

A dial plan establishes the expected number and pattern of digits for a telephone number, which includes country codes, access codes, area codes and all combinations of digits dialed.

Steps

1. In GPON configuration mode, create the dial plan.

```
ZXAN(config)#gpon
ZXAN(config-gpon)#onu profile dial-plan test
```

2. Configure the digit-map of the dial plan.

```
ZXAN(config-gpon)#onu profile dial-plan test digit-map X*.X.#|#X*.X.##
```

3. (Optional) Query the dial plan configuration.

```
ZXAN(config-gpon)#show gpon onu profile dial-plan test
Profile name:      test
Critical timeout:4000
Partial timeout: 16000
Format:            H.248
Digit map:         X*.X.#|#X*.X.##
```

– End of Steps –

2.9 Configuring the GPON SIP Profile

Using the GPON SIP profile, you can configure the GPON ONU SIP parameters for GPON ONUs in batches.

Prerequisite

Before this operation, make sure that:

- The access code profile has been configured.
- The service application profile has been configured.
- The dial plan table has been configured.

Configuration Data

Table 2-10 lists the configuration data of the Session Initiation Protocol (SIP) profile.

Table 2-10 Configuration Data of the GPON SIP Profile

Item	Data
Profile name	sip-test
Access code profile	abc
Service application profile	voip-service

Item	Data
Dial plan table	test
IP address of the proxy server	1.2.3.1

Steps

1. In GPON configuration mode, configure the SIP profile.

```
ZXAN(config)#gpon
ZXAN(config-gpon)#onu profile sip sip-test proxy 1.2.3.1
ZXAN(config-gpon)#onu profile sip sip-test accesscode abc
ZXAN(config-gpon)#onu profile sip sip-test appsrv voip-service
ZXAN(config-gpon)#onu profile sip sip-test dial-plan test
```

2. (Optional) Query the SIP profile.

```
ZXAN(config-gpon)#show gpon onu profile sip sip-test
Profile name :                sip-test
Proxy server:                1.2.3.1
Outbound proxy:              1.2.3.1
Registrar:                  1.2.3.1
Validation scheme:           disable
UDP port:                    5060
DSCP/TOS:                    0
Media UDP port:              5060
Media DSCP/TOS:              0
DNS1:                        0.0.0.0
DNS2:                        0.0.0.0
Registration expiration time: 3600(s)
Re-registration time:        360(s)
Softswitch vendor:
Dial plan table name:        test
Release timer:               10(s)
ROH timer:                   15(s)
Link test:                   disable
Link test interval:          N/A
appsrv:                      voip-service
accesscode:                  abc
```

– End of Steps –

2.10 Configuring the GPON MGC Profile

Using the GPON MGC profile, you can configure MGC parameters for GPON ONUs in batches.

Context

The ZXA10 C300 supports the following two Media Gateway Controller (MGC) protocols.

- MGCP
- H.248

Configuration Data

Table 2-11 lists the configuration data of the GPON MGC profile.

Table 2-11 Configuration Data of the GPON MGC Profile

Item	Data
Profile name	mgc-test
Active server IP address	1.2.3.1
Standby server IP address	1.2.3.2
User TID	Prefix: user Postfix length: 5 Postfix start number: 1
RTP TID	Prefix: rtp Postfix length: 5 Postfix start number: 1

Steps

1. In GPON configuration mode, configure the active and standby MGC servers.

```
ZXAN(config)#gpon
ZXAN(config-gpon)#onu profile mgc mgc-test server1 1.2.3.1
ZXAN(config-gpon)#onu profile mgc mgc-test server2 1.2.3.2
```

2. In GPON configuration mode, configure the user Terminal Identification (TID) and Real-time Transport Protocol (RTP) TID.

```
ZXAN(config-gpon)#onu profile mgc mgc-test user-tid prefix user postfix-len 5
postfix-start 1
ZXAN(config-gpon)#onu profile mgc mgc-test rtp-tid prefix rtp postfix-len 5
postfix-start 1
```

3. (Optional) Query the MGC profile.

```
ZXAN(config-gpon)#show gpon onu profile mgc
Profile name:          mgc-test
Server1:              1.2.3.1
  Validation scheme:   disable
Username:             N/A
Password:             N/A
Realm:               N/A
Server2:              1.2.3.2
  Validation scheme:   disable
```

```
Username: N/A
Password: N/A
Realm: N/A
UDP port: 2944
DSCP/TOS: 0
Media UDP port: 2944
Media DSCP/TOS: 0
Message format: text long
Version: 1
Maximum retry time: 0(s)
Maximum retry attempts: 0(s)
Service change delay: 0(s)
Softswitch vendor:
User TID:
  Prefix: user
  Postfix length: 5
  Postfix start number: 1
RTP TID:
  Prefix: rtp
  Postfix length: 5
  Postfix start number: 1
Heart beat: service change
RTP link detect: disable
Number shortest match: disable
Digit map long timer: 20000(ms)
Digit map short timer: 5000(ms)
Digit map start timer: 10000(ms)
Heart beat interval: 60000(ms)
Rereg timer min: 60000(ms)
Rereg timer max: 120000(ms)
Regmsg retrans timer: 2000(ms)
Total retrans timer: 10000(ms)
```

– End of Steps –

2.11 Configuring the GPON Broadband Service

After you configure the GPON broadband service, the subscriber can access the Internet.

Prerequisite

- The GPON ONU has been authenticated.
- The T-CONT bandwidth profile has been configured.

Configuration Data

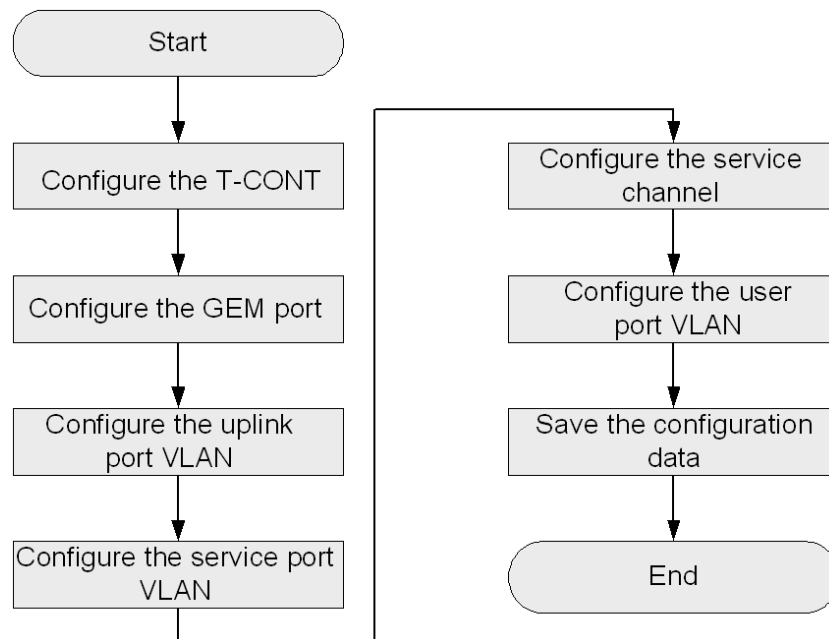
Table 2-12 lists the configuration data of the GPON broadband service.

Table 2-12 Configuration Data of the GPON Broadband Service

Item	Data
Service VLAN ID	100
Service priority	0
Uplink port	gei_1/22/1
Service port	ONU interface: gpon-onu_1/5/1:1 Service-port ID: 1 Virtual port ID: 1
T-CONT	Index: 1 Name: T1 T-CONT bandwidth profile: 10M
GEM Port	Index: 1 Name: gemport1 T-CONT index: 1
Service channel	Name: HSI GEM port index: 1 Priority: 0 VLAN ID: 100
User port VLAN	Port: eth_0/1 VLAN mode: tag (The untagged upstream packet is tagged with PVID.) VLAN ID: 100 Priority: 0

Configuration Flowchart

Figure 2-2 shows the configuration flowchart of the GPON broadband service.

Figure 2-2 Configuration Flowchart of the GPON Broadband Service**Steps**

1. In ONU interface mode, configure the T-CONT.

```

ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#tcont 1 name T1 profile 10M

```

2. Configure the GEM port.

```

ZXAN(config-if)#gemport 1 name gemport1 tcont 1
ZXAN(config-if)#exit

```

3. In uplink interface configuration mode, configure the uplink port VLAN.

```

ZXAN(config)#interface gei_1/22/1
ZXAN(config-if)#switchport vlan 100 tag
ZXAN(config-if)#exit

```

4. In ONU interface mode, configure the service port VLAN.

```

ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#service-port 1 vport 1 user-vlan 100 vlan 100
ZXAN(config-if)#exit

```

**Note:**

By default, the mapping mode between the virtual port and the GEM port is 1:1.

5. In ONU remote management mode, configure the service channel.

```

ZXAN(config)#pon-onu-mng gpon-onu_1/5/1:1

```

```
ZXAN(gpon-onu-mng)#service HSI gempport 1 cos 0 vlan 100
```

6. Configure the user port VLAN.

```
ZXAN(gpon-onu-mng)#vlan port eth_0/1 mode tag vlan 100 pri 0
ZXAN(gpon-onu-mng)#end
```

7. Save the configuration data.

```
ZXAN#write
```

– End of Steps –

2.12 Configuring the GPON Multicast Service

After you configure the GPON multicast service, subscribers can receive multicast service streams.

Prerequisite

- The GPON ONU has been authenticated.
- The T-CONT bandwidth profile has been configured.

Configuration Data

Table 2-13 lists the configuration data of the GPON multicast service.

Table 2-13 Configuration Data of the GPON Multicast Service

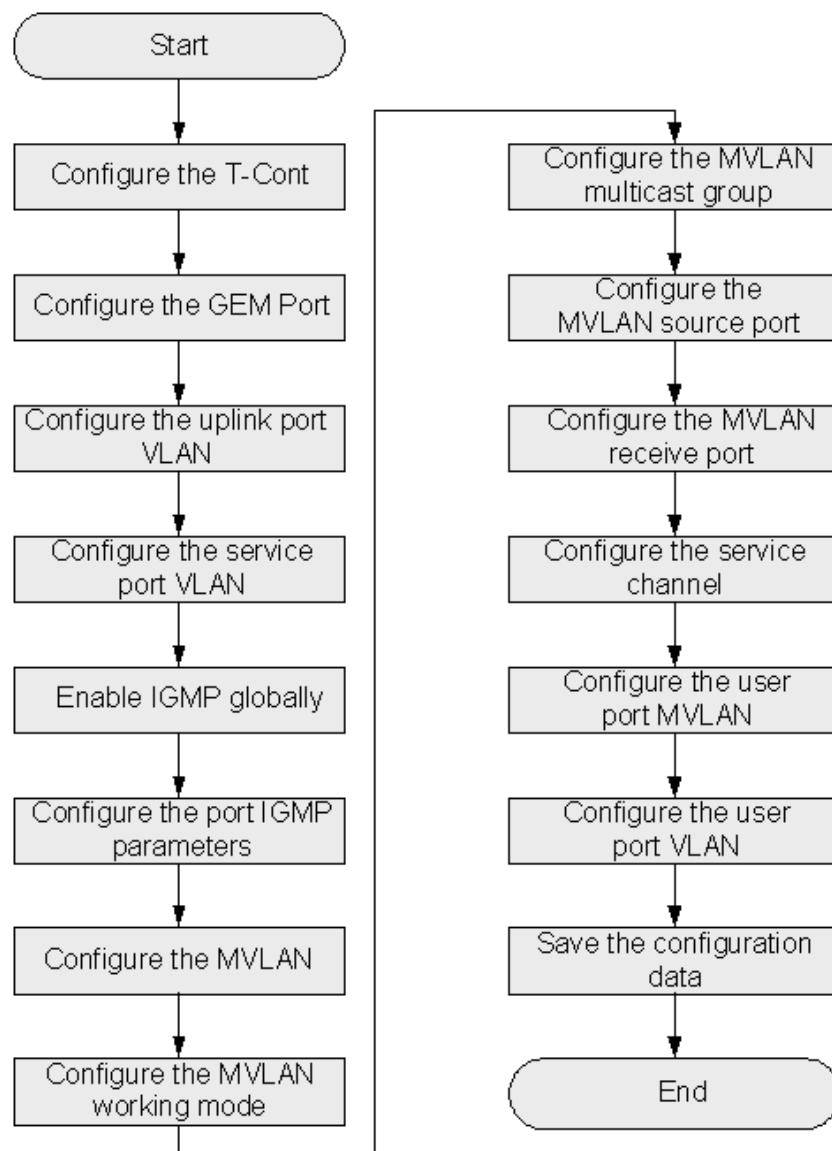
Item	Data
Multicast VLAN (MVLAN) ID	200
Service priority	5
MVLAN working mode	Proxy
MVLAN multicast group	224.1.1.1 – 224.1.1.3
Uplink port	gei_1/22/1
Service port	ONU interface: gpon-onu_1/5/1:1 Service-port ID: 2 Virtual port ID: 2
T-CONT	Index: 2 Name: T2 T-CONT bandwidth profile: 5M
GEM Port	Index: 2 Name: gempport2 T-CONT index: 2

Item	Data
Service channel	Name: mulitcast GEM port index: 2 Priority: 5 VLAN ID: 200
User port VLAN	MVLAN ID: 200 MVLAN tag stripping: enable
User port VLAN	Port: eth_0/2 VLAN mode: tag (The untagged upstream packet is tagged with PVID.) VLAN ID: 200 Priority: 5

Configuration Flowchart

Figure 2-3 shows the configuration flowchart of the GPON multicast service.

Figure 2-3 Configuration Flowchart of the GPON Multicast Service



Steps

1. In ONU interface mode, configure the T-CONT.

```

ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#tcont 2 name T2 profile 5M

```

2. Configure the GEM port.

```

ZXAN(config-if)#gemport 2 name gemport2 tcont 2
ZXAN(config-if)#exit

```

3. In uplink interface configuration mode, configure the uplink port VLAN.

```

ZXAN(config)#interface gei_1/22/1
ZXAN(config-if)#switchport vlan 200 tag
ZXAN(config-if)#exit

```

4. In ONU interface mode, configure the service port VLAN.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#service-port 2 vport 2 user-vlan 200 vlan 200
ZXAN(config-if)#exit
```



Note:

By default, the mapping mode between the virtual port and the GEM port is 1:1.

5. (Optional) Enable IGMP globally.

```
ZXAN(config)#igmp enable
```

6. Configure the port IGMP parameters.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#igmp fast-leave enable vport 2
ZXAN(config-if)#exit
```

7. Configure the MVLAN.

```
ZXAN(config)#igmp mvlan 200
```

8. (Optional) Configure the MVLAN working mode.

```
ZXAN(config)#igmp mvlan 200 work-mode proxy
```

9. Configure the MVLAN multicast group.

```
ZXAN(config)#igmp mvlan 200 group 224.1.1.1 to 224.1.1.3
```

10. Configure MVLAN source port.

```
ZXAN(config)#igmp mvlan 200 source-port gei_1/22/1
```

11. Configure the MVLAN receive port.

```
ZXAN(config)#igmp mvlan 200 receive-port gpon-onu_1/5/1:1 vport 2
```

12. In ONU remote management mode, configure the service channel.

```
ZXAN(config)#pon-onu-mng gpon-onu_1/5/1:1
ZXAN(gpon-onu-mng)#service multicast gemport 2 cos 5 vlan 200
```

13. Configure the user port MVLAN.

```
ZXAN(gpon-onu-mng)#mvlan 200
ZXAN(gpon-onu-mng)#mvlan tag-strip eth_0/2 enable
```

14. Configure the user port VLAN.

```
ZXAN(gpon-onu-mng)#vlan port eth_0/2 mode tag vlan 200 pri 5
ZXAN(gpon-onu-mng)#end
```

15. Save the configuration data.

```
ZXAN#write
```

– End of Steps –

2.13 Configuring the GPON Voice Service (SIP)

After you configure the GPON voice service, subscribers can make and answer phone calls. This section takes the SIP protocol as an example.

Prerequisite

- The GPON ONU has been authenticated.
- The T-CONT bandwidth profile has been configured.
- The GPON VoIP IP profile has been configured.
- The GPON VoIP VLAN profile has been configured.
- The GPON SIP profile has been configured.

Configuration Data

Table 2-14 lists the configuration data of the GPON voice service.

Table 2-14 Configuration Data of the GPON Voice Service

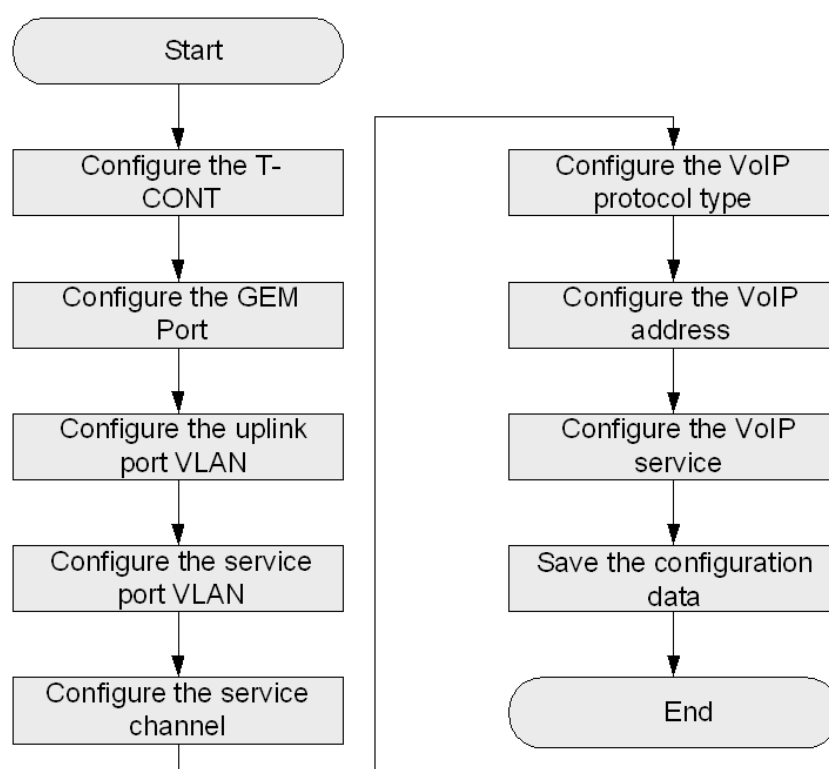
Item	Data
Service VLAN ID	300
Service priority	7
Uplink port	gei_1/22/1
Service port	ONU interface: gpon-onu_1/5/1:1 Service-port ID: 3 Virtual port ID: 3
T-CONT	Index: 3 Name: voip T-CONT bandwidth profile: 2M
GEM Port	Index: 3 Name: gemport3 T-CONT index: 3
Service channel	Name: voip-sip GEM port index: 3 Priority: 7 VLAN ID: 300
VoIP protocol	SIP
VoIP address	IP address allocation mode: static VoIP IP profile: ip-test IP address: 1.2.3.4/24 VoIP VLAN profile: vlan-test

Item	Data
VoIP service	Port: pots_0/1 SIP profile: sip-test User ID: 12345 User name: 12345 Password: 12345

Configuration Flowchart

Figure 2-4 shows the configuration flowchart of the GPON voice service.

Figure 2-4 Configuration Flowchart of the GPON Voice Service



Steps

1. In ONU interface configuration mode, configure the T-CONT.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#tcont 3 name voip profile 2M
```

2. Configure the GEM port.

```
ZXAN(config-if)#gemport 3 name gemport3 tcont 3
ZXAN(config-if)#exit
```

3. In uplink interface configuration mode, configure the uplink port VLAN.

```
ZXAN(config)#interface gei_1/22/1
ZXAN(config-if)#switchport vlan 300 tag
ZXAN(config-if)#exit
```

4. In ONU interface configuration mode, configure the service port VLAN.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#service-port 3 vport 3 user-vlan 300 vlan 300
ZXAN(config-if)#exit
```



Note:

By default, the mapping mode between the virtual port and the GEM port is 1:1.

5. In ONU remote management mode, configure the service channel.

```
ZXAN(config)#pon-onu-mng gpon-onu_1/5/1:1
ZXAN(gpon-onu-mng)#service voip-sip gempport 3 cos 7 vlan 300
```

6. (Optional) Configure the VoIP protocol type.

```
ZXAN(gpon-onu-mng)#voip protocol sip
```

7. Configure the VoIP address.

```
ZXAN(gpon-onu-mng)#voip-ip mode static ip-profile ip-test ip-address 1.2.3.4
mask 255.255.255.0 vlan-profile vlan-test
```

8. Configure the VoIP service.

```
ZXAN(gpon-onu-mng)#sip-service pots_0/1 profile sip-test userid 12345 username
12345 password 12345
ZXAN(gpon-onu-mng)#end
```

9. Save the configuration data.

```
ZXAN#write
```

– End of Steps –

2.14 Configuring the GPON Voice Service (H.248)

After you configure the GPON voice service, subscribers can make and answer phone calls. This section takes the H.248 protocol as an example.

Prerequisite

- The [GPON ONU](#) has been authenticated.
- The T-CONT bandwidth profile has been configured.
- The GPON [VoIP IP](#) profile has been configured.
- The GPON VoIP [VLAN](#) profile has been configured.
- The GPON [MGC](#) profile has been configured.

Configuration Data

[Table 2-15](#) lists the configuration data of the GPON voice service.

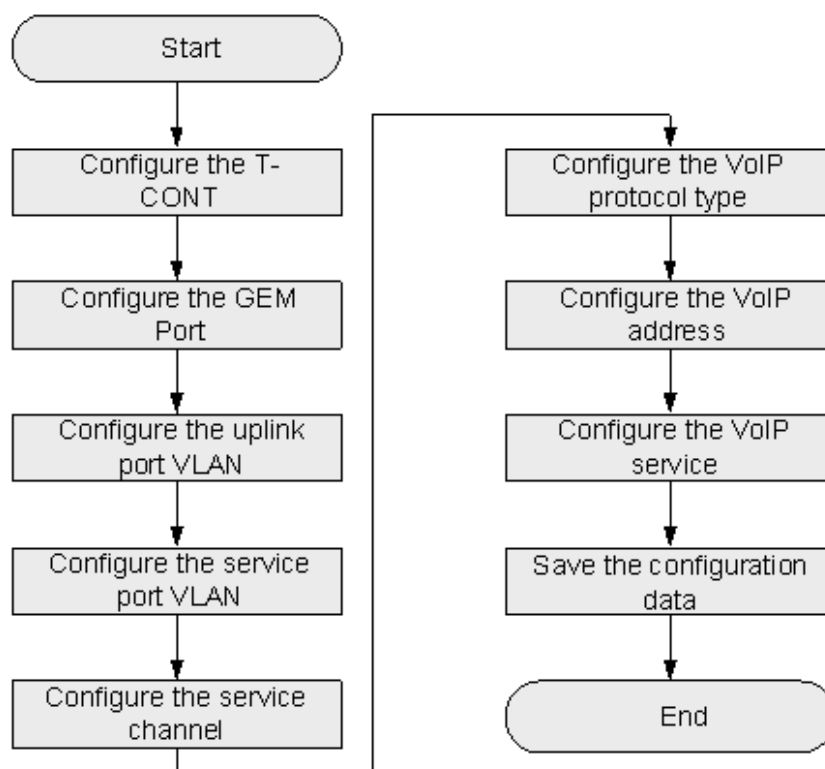
Table 2-15 Configuration Data of the GPON Voice Service

Item	Data
Service VLAN ID	300
Service priority	7
Uplink port	gei_1/22/1
Service port	ONU interface: gpon-onu_1/5/1:1 Service-port ID: 3 Virtual port ID: 3
T-CONT	Index: 3 Name: voip T-CONT bandwidth profile: 2M
GEM Port	Index: 3 Name: gemport3 T-CONT index: 3
Service channel	Name: voip-h248 GEM port index: 3 Priority: 7 VLAN ID: 300
VoIP protocol	H.248
Domain name	iad.zte.com.cn
VoIP address	IP address allocation mode: static VoIP IP profile: ip-test IP address: 1.2.3.4/24 VoIP VLAN profile: vlan-test
VoIP service	Port: pots_0/1 MGC profile: mgc-test

Configuration Flowchart

Figure 2-5 shows the configuration flowchart of the GPON voice service.

Figure 2-5 Configuration Flowchart of the GPON Voice Service



Steps

1. In ONU interface configuration mode, configure the T-CONT.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#tcont 3 name voip profile 2M
```

2. Configure the GEM port.

```
ZXAN(config-if)#gemport 3 name gemport3 tcont 3
ZXAN(config-if)#exit
```

3. In uplink interface configuration mode, configure the uplink port VLAN.

```
ZXAN(config)#interface gei_1/22/1
ZXAN(config-if)#switchport vlan 300 tag
ZXAN(config-if)#exit
```

4. In ONU interface configuration mode, configure the service port VLAN.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#service-port 3 vport 3 user-vlan 300 vlan 300
ZXAN(config-if)#exit
```

**Note:**

By default, the mapping mode between the virtual port and the GEM port is 1:1.

5. In ONU remote management mode, configure the service channel.

```
ZXAN(config)#pon-onu-mng gpon-onu_1/5/1:1
ZXAN(gpon-onu-mng)#service voip-h248 gemport 3 cos 7 vlan 300
```

6. Configure the VoIP protocol type.

```
ZXAN(gpon-onu-mng)#voip protocol h248 domain iad.zte.com.cn
```

7. Configure the VoIP address.

```
ZXAN(gpon-onu-mng)#voip-ip mode static ip-profile ip-test ip-address 1.2.3.4
mask 255.255.255.0 vlan-profile vlan-test
```

8. Configure the VoIP service.

```
ZXAN(gpon-onu-mng)#mgc-service pots_0/1 profile mgc-test
ZXAN(gpon-onu-mng)#end
```

9. Save the configuration data.

```
ZXAN#write
```

– End of Steps –

Chapter 3

CES Service Configuration

The Circuit Emulation Service (CES) provides traditional TDM services based on the Packet Switch Network (PSN), such as E1/T1.

The ZX A10 C300 implements the CES service in two modes that are PWE3 and MEF8.

- The Pseudo Wire Emulation Edge-to-Edge (PWE3) mode of the Internet Engineering Task Force (IETF) implements circuit emulation over the IP network, namely, TDMoIP, based on the IP address.
- The MEF8 mode of the Metro Ethernet Forum (MEF) implements circuit emulation over the Ethernet, namely, TDMoE, based on the MAC address.

The ZX A10 C300 supports the following uplink TDM interfaces:

- E1
- STM-1/STM-4

Table of Contents

Configuring the GPON CES Service.....3-1

3.1 Configuring the GPON CES Service

The ZX A10 C300 connects the ZTE-F621 to implement the E1 uplink MEF8-based CES service.

Context

The ZX A10 C300 supports four types of TDM interface cards, as listed in Table 3-1.

Table 3-1 TDM Interface Cards

Card	Interface	TDM Service Type	TDM Interface Format
CTUB	E1	e1Satop/e1Cetop	tdm_shelf/slot/port
CTLA	STM-1/STM-4	e1Satop	tdm_shelf/slot/port.augno/1/tug3/tug2/e1

The CTLA card supports 126 E1 channels, and the parameter values of each E1 channel must be unique. Table 3-2 provides the parameter value ranges.

Table 3-2 Value Range of CTLA Interface Parameter

Parameter	Value Range
port	1–2 (STM-1 interface) 1 (STM-4 interface)

Parameter	Value Range
augno	1 (STM-1 interface) 1-2 (STM-4 interface)
tug-3	1-3
tug-2	1-7
e1	1-3

Configuration Data

Table 3-3 lists the configuration data of the GPON CES service.

Table 3-3 Configuration Data of the GPON CES Service

Item	Data
NM VLAN	VLAN ID: 40 Priority: 7
CES SVLAN	VLAN ID: 1001 Priority: 7
Uplink port	1/12/1
GPON port	1/5/1
T-CONT profile	Profile name: 6M Bandwidth type: type 1 Assured bandwidth: 6000 kbps
T-CONT	Index: 1 Name: T1 T-CONT profile: 6M
GEM Port	Index: 1 Name: cesport T-CONT index: 1
ONU	ID: 2 Authentication mode: SN authentication Type: ZTE-F621 SN: ZTEG80000001
OLT CES MAC address	0015.EB72.001A
ONU CES MAC address	0015.EB72.0001
OLT PW	Type: E1Satop Inbound ECID: 0x1102 Outbound ECID: 0x1102 Priority: 7 VLAN ID: 1001

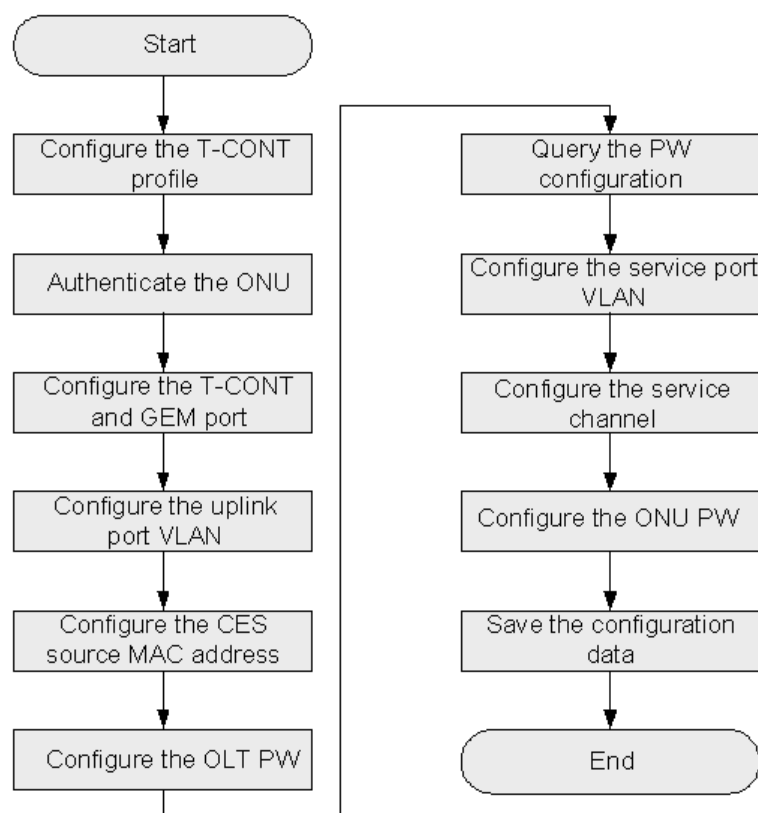
Item	Data
ONU CES service channel	Name: tdm Priority: 7 VLAN ID: 40
ONU PW	Type: E1Satop Tx-ECID: 0x1102 Rx-ECID: 0x1102 Priority: 7 VID: 40

- You need to configure 6 Mbps fixed upstream bandwidth for each channel of E1 service to ensure the quality of service.

Configuration Flowchart

Figure 3-1 shows the configuration flowchart of the GPON CES service.

Figure 3-1 Configuration Flowchart of the GPON CES Service



Steps

- In GPON configuration mode, configure the T-CONT bandwidth profile.

```

ZXAN(config)#gpon
ZXAN(config-gpon)#profile tcont 6M type 1 fixed 6000
[Successful]
ZXAN(config-gpon)#exit

```

2. In OLT interface configuration mode, authenticate the ONU.

```
ZXAN(config)#interface gpon-olt_1/5/1
ZXAN(config-if)#onu 2 type ZTE-F621 sn ZTEG80000001
[Successful]
ZXAN(config-if)#exit
```

3. In ONU interface configuration mode, configure the T-CONT and GEM port.

```
ZXAN(config)#interface gpon-onu_1/5/1:2
ZXAN(config-if)#tcont 1 name T1 profile 6M
ZXAN(config-if)#gemport 1 name cesport tcont 1
ZXAN(config-if)#exit
```

4. In TDM uplink interface configuration mode, configure the uplink port VLAN.

```
ZXAN(config)#interface tdm-gei_1/12/1
ZXAN(config-if)#switchport mode hybrid
ZXAN(config-if)#switchport vlan 1001 tag
ZXAN(config-if)#exit
```

5. Configure the CES source MAC address.

```
ZXAN(config)#ces
ZXAN(config-ces)#mac-address 0015.eb72.001a 12/1
```

6. Configure the OLT PW.

a. Create the PW.

```
ZXAN(config-ces)#pw pw_1/12/1
ZXAN(config-ces-pw)#
```

b. Configure the PW TDM attributes.

```
ZXAN(config-ces-pw)#tdm-service type e1Satop rate 32 tdm_1/12/1
```

c. Configure the PW PSN attributes.

```
ZXAN(config-ces-pw)#psn ethernet 0x1102 0x1102 dst-mac 0015.eb72.0001 vlan 1001
priority 7
```

7. Query the PW configuration.

```
ZXAN(config-ces-pw)#show ces pw prop detail pw_1/12/1
```

```
pw pw_1/12/1
PwType: e1Satop      PsnType: ethernet  Admin-status: enable
OuterInboundLabel:N/A      OuterOutboundLabel:N/A
InboundLable       :0x1102      OutboundLable       :0x1102
```

Service prop:

```
Using tdm interface: tdm_1/12/1
Using TimeSlot      : N/A
TDM-prop profile    : default
```

```

Psn prop:
Destination Mac Address: 0015.EB72.0001
SVlan ID: 1001
CVlan ID: 2304
priority: 7

```

```

Card prop:
Card interface : 12/1
Source Mac Address: 0015.EB72.001A
Source IP Address : 192.192.192.24

```

**Note:**

When the PW CVLAN is not configured, the ZXAN C300 allocates a CVLAN ID to the PW automatically.

8. In ONU interface configuration mode, configure the service port VLAN.

```

ZXAN(config)#interface gpon-onu_1/5/1:2
ZXAN(config-if)#service-port 1 vport 1 user-vlan 40 vlan 768 svlan 1001
ZXAN(config-if)#exit

```

9. In ONU remote management mode, configure the ONU CES service channel.

```

ZXAN(config)#pon-onu-mng gpon-onu_1/5/1:2
ZXAN(gpon-onu-mng)#service tdm gempport 1 cos 7 vlan 40

```

10. Configure the ONU PW.

a. Configure the PW PSN attributes.

```

ZXAN(gpon-onu-mng)#pw psn ethernet ces_0/1 switch switch_0/1 dst-mac 0015.eb72.0
01a loopback disable tx-payload-ecid 0x1102 tx-signalling-ecid 0x0 expected-payl
oad-ecid 0x1102 expected-signalling-ecid 0x0 tag-policy tag pri 7 vlan 40

```

b. Configure the PW TDM attributes.

```

ZXAN(gpon-onu-mng)#pw tdm ces_0/1 payloadsize 32 payload-delay 2 timing-mode
adaptive-timing

```

c. Configure the PW RTP parameters.

```

ZXAN(gpon-onu-mng)#pw rtp ces_0/1 payload-ssrc 3 signalling-ssrc 0 payload-
ptype 97 signalling-ptype 0 expected-payload-ssrc 3 expected-signalling-ssrc 0
expected-payload-ptype 97 expected-signalling-ptype 0 clock-reference 1 rtp-ts-
mode differential

```

11. Save the configuration data.

– End of Steps –

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Chapter 4

P2P Service Configuration

The ZX A10 C300 supports P2P service. P2P interfaces are actually Ethernet interfaces. Using the WDM technology, a P2P interface transmits and receives signals through a single optical fiber, while a traditional Ethernet interface uses two optical fibers to transmit and receiving signals. The P2P service can save a large number of optical fiber resources and thus reduce the network construction cost.

P2P service are suitable for the following scenarios.

- VIP dedicated line
The most popular application. Because each user exclusively possesses an optical fiber, the reliable optical-layer security isolation is provided.
- Base station back-haul
The P2P service provides connection to base stations directly or through P2P ring.
- Device interconnection
When optical fiber resources are limited, the P2P service can be used for device interconnection.

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Configuring the P2P Service.....	4-1
----------------------------------	-----

4.1 Configuring the P2P Service

The ZX A10 C300 are connected ONU s or other Ethernet devices through the P2P interface card. After you configure the P2P service, the subscribers can enjoy the data, multicast, and VoIP service.

Context

The P2P interfaces support smart VLAN configuration based on service-port.

- Adding a SVLAN to user VLANs according to user VLAN range.
- Translating user VALN to SVLAN + VLAN.
- Modifying the 802.1p priority of SVLAN.
- Adding a SVLAN to a user VLAN according to the combination, such as user VLAN + Ethernet type.

Configuration Data

Table 4-1 list the configuration data of the P2P service (take the uplink card GDFO as an example).

Table 4-1 P2P Service Configuration Data

Item	Data
Data service VLAN ID	CVLAN ID: 101 – 124 SVLAN ID: 1001
Multicast service VLAN ID	201
VoIP service VLAN ID	301
Uplink port	gei_1/15/1
Service port	gei_1/5/1
MVLAN working mode	Proxy (default)
MVLAN group	224.1.1.1 – 224.1.1.3

Steps

1. In uplink interface configuration mode, configure uplink port VLAN.

```
ZXAN(config)#interface gei_1/15/1
ZXAN(config-if)#switchport vlan 1001,201,301 tag
ZXAN(config-if)#exit
```

2. In P2P interface configuration mode, configure service port VLAN.

```
ZXAN(config)#interface gei_1/5/1
ZXAN(config-if)#service-port 1 user-vlan 101 to 124 svlan 1001
ZXAN(config-if)#service-port 2 user-vlan 201 vlan 201
ZXAN(config-if)#service-port 3 user-vlan 301 vlan 301
ZXAN(config-if)#exit
```

3. (Optional) Enable global IGMP protocol.

```
ZXAN(config)#igmp enable
```



Note:

By default, the global IGMP protocol is enabled on the ZXA10 C300.

4. Configure IGMP parameters on service port.

```
ZXAN(config)#interface gei_1/5/1
ZXAN(config-if)#igmp fast-leave enable
ZXAN(config-if)#exit
```

5. Configure the MVLAN.

```
ZXAN(config)#igmp mvlan 201
```

6. (Optional) Configure MVLAN working mode.

```
ZXAN(config)#igmp mvlan 201 work-mode proxy
```

7. Configure MVLAN multicast groups.

```
ZXAN(config)#igmp mvlan 201 group 224.1.1.1 to 224.1.1.3
```

8. Configure MVLAN source port.

```
ZXAN(config)#igmp mvlan 201 source-port gei_1/15/1
```

9. Configure MVLAN receiving port.

```
ZXAN(config)#igmp mvlan 201 receive-port gei_1/5/1
```

10. Save configuration data.

– End of Steps –

Follow-Up Action

The P2P interfaces supports the following configuration:

- [QoS](#) (refer to [Chapter 7 QoS Configuration](#))
- [DHCP](#) (refer to [Chapter 11 DHCP Configuration](#))
- Port identification (refer to [14.1 Port Identification Configuration](#))
- Link aggregation (refer to [12.1 Configuring Link Aggregation](#))

The DHCP configuration and port location configuration on the P2P interfaces are similar to the configuration on the PON ONU interfaces.

The QoS configuration and link aggregation configuration on the P2P interfaces are similar to the configuration on the Ethernet interfaces.



Note:

Usually, in the P2P service, the uplink card is the 9U-height Ethernet interface card GDFO. In DHCP snooping configuration, only the 4.5U-height uplink cards can be used for uplink.

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Chapter 5

VLAN Configuration

VLAN is a technology that implements virtual workgroups by dividing the physical equipment in a LAN into several logical network segments. The IEEE issued the IEEE 802.1q standard in 1999 to normalize the VLAN solution.

The ZXAN10 C300 supports 4094 VLANs.

[Table 5-1](#) lists the VLAN specifications.

Table 5-1 VLAN Specifications

VLAN Type	Description
Basic VLAN	Used to isolate ports.
Service port VLAN	Used to implement VLAN translation at the ONU level.
TLS VLAN	Used to add an SVLAN to the packet to implement the Transparent LAN Service (TLS) whatever the user access mode is, or no matter whether the upstream packet has a VLAN tag, or whatever the VLAN tag is.
Cross-connection VLAN	Used to set the special channel for the user port and uplink port. The packets are forwarded in 1:1 mode according to the VLAN ID.

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Configuring the Uplink Port VLAN.....	5-1
Configuring the Service Port VLAN.....	5-2
Configuring the Cross-Connection VLAN.....	5-3

5.1 Configuring the Uplink Port VLAN

By configuring the uplink port VLAN, you can classify ports into different network segments logically to control the communication between ports.

Steps

1. In uplink interface configuration mode, configure the port VLAN.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 2-100 tag
```

**Note:**

When you configure the uplink port VLAN, the system will automatically create the corresponding VLAN.

– End of Steps –

5.2 Configuring the Service Port VLAN

By configuring the service port VLAN on the PON ONU interface, you can implement VLAN translation at the ONU level.

Prerequisite

The ONU has been authenticated.

Context

The service port configuration supports the following applications:

- Add CVLAN + SVLAN to untagged packets
- Add SVLAN to user VLANs according to user VLAN range
- Translate user VLAN to VLAN + SVLAN
- Translate Ethernet protocol type to VLAN + SVLAN
- Translate 802.1p priority to VLAN + SVLAN
- Translate combination (user VLAN, Ethernet protocol type, and 802.1p priority) to VLAN + SVLAN
- Modify SVLAN 802.1p priority
- TLS VLAN

Steps

1. In ONU interface configuration mode, configure the service port VLAN.

```
ZXAN(config)#interface gpon-onu_1/6/1:1
ZXAN(config-if)#service-port 1 vport 1 user-vlan 7 vlan 8 svlan 9
ZXAN(config-if)#service-port 2 vport 1 other-all tls-vlan 501
```

2. (Optional) Query the configured service port VLAN.

```
ZXAN(config-if)#show service-port interface gpon-onu_1/6/1:1
Interface gpon-onu_1/6/1:1
Sport Vport BeginVid EndVid OuterVid InnerVid UserPrio Dscp Etype Filter Vlan C
os SVlan SCos Tls Mode Ingress Egress Queue Status Enable
-----
1 1 7 7 -- -- -- -- -- 8 -
```

```

- 9      --      --      1:1      --      --      --      --      YES
2      1      --      --      --      --      --      --      --      --
- --      --      501      --      --      --      --      --      YES
Sport total number:
2

```

– End of Steps –

5.3 Configuring the Cross-Connection VLAN

By configuring the cross-connection VLAN, you can implement 1:1 VLAN forwarding.

Context

The cross-connection VLAN is a special channel for a user port and an uplink port. When the cross-connection VLAN is configured, packets are forwarded in 1:1 mode according to the VLAN ID but not forwarded in MAC + VLAN mode.

1:1 VLAN exchange is implemented in the following two modes:

- SVLAN
- CVLAN + SVLAN dual tags

Steps

1. In global configuration mode, configure the uplink port VLAN.

```

ZXAN(config)#vlan-translate ingress-port gei_1/21/1 user-outer-vlan 5 user-inner
-vlan 3 vlan 3 svlan 5

```

2. In ONU interface configuration mode, configure the service port VLAN.

```

ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#service-port 1 vport 1 user-vlan 3 svlan 5
ZXAN(config-if)#exit

```

3. In VLAN configuration mode, configure the VLAN to the cross-connection VLAN.

```

ZXAN(config)#vlan 5
ZZXAN(config-vlan5)#xconnect enable cvlan 3

```

4. (Optional) Query the cross-connection VLAN configuration.

```

ZXAN(config-vlan5)#show vlan-xconnect detail
User-Port                Uplink-Port  Svlan  Cvlan  Status
-----
gpon-onu_1/5/1:1 vport 1    gei_1/21/1   5      3      --

```

– End of Steps –

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Chapter 6

IPTV Configuration

The secondary duplication of the Internet Protocol Television (IPTV) layer-2 multicast service is implemented on the OLT and ONU. The related configuration information is as follows:

- Configurations of the basic OLT service parameters

The basic parameters of layer-2 multicast control includes multicast VLAN, source port, receive port, and multicast program address. The multicast VLAN is the VLAN that carries the multicast data. The source port is the uplink port that connects the multicast source. The receive port is the ONU interface that connects the multicast subscriber. The multicast program address consists of the group address and source address.

- Configuration of the OLT multicast protocol mode

The ZXA10 C300 supports the IPv4 and IPv6 multicast dual protocol stack so it can be flexibly configured to accept/drop packets of various protocols. Three working modes (Snooping/Router/Proxy) can be configured based on the multicast VLAN.

- Configuration of ONU user rights

Based on the ITU-T G984.4 standard, the OLT configures the multicast right profile to the ONU via the OMCI interface. The ONU runs the IGMP Snooping protocol, and implements the user right control according to the local multicast right table.

Service Description

As the streaming media such as the multimedia video and data warehouse appear in the IP network, the multicast application gradually becomes the new service demand. The multicast service is applicable to the streaming media, tele-education, video conference, video multicast (Web TV), network game, data copy, and any other point-to-multipoint data transmission application.

Service Specifications

ZXA10 C300 has the carrier-class multicast operation capacity. It supports multicast protocols and controllable multicast and supports a full set of protocols from the subscriber to the network. Hence, it provides a basis for the broadband multicast value-added service and multicast service management. The ZXA10 C300 provides operational and manageable controllable multicast service, supports the Internet Group Management Protocol (IGMP) v1/v2/v3, and supports the IGMP snooping, IGMP proxy, and IGMP router modes.

- Supports the IGMP v1/v2/v3.

- Supports Multicast Listener Discovery (MLD) v1/v2.
- Supports IGMP Snooping/Proxy/Router.
- Supports MLD Snooping/Proxy.
- Supports 8K multicast entries.
- Supports 256 multicast VLANs.
- Supports Channel Access Control (CAC).
- Supports channel preview.
- Supports Call Detail Record (CDR).

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6.1 Configuring the IGMP MVLAN

The IGMP MVLAN is the VLAN that carries the IGMP multicast data, which includes the service VLAN, source port, receive port, and multicast group.

Configuration Data

Table 6-1 lists the configuration data of the IGMP MVLAN.

Table 6-1 Configuration Data of the IGMP MVLAN

Item	Data
IGMP	Enable
Span VLAN function	Enable
MVLAN ID	200
MVLAN working mode	Proxy
MVLAN host version	IGMPv3
MVLAN packet processing mode	IGMPv1: drop IGMPv2: drop IGMPv3: accept
Multicast group IP address	224.1.1.1–224.1.1.3
Multicast source IP address	10.1.1.1
Multicast source port	gei_1/21/1
Multicast receive port	ONU interface: gpon-onu_1/5/1:1 Virtual port ID: 1

Steps

1. In uplink interface configuration mode, configure the uplink port VLAN.


```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 200 tag
ZXAN(config-if)#exit
```

2. In ONU interface configuration mode, configure the service port VLAN.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#service-port 1 vport 1 user-vlan 200 vlan 200
ZXAN(config-if)#exit
```

3. Enable IGMP globally.

```
ZXAN(config)#igmp enable
```

4. Enable IGMP span VLAN function.

```
ZXAN(config)#igmp span-vlan enable
```

5. Configure the MVLAN.

```
ZXAN(config)#igmp mvlan 200
```

6. Configure the MVLAN packet processing mode.

```
ZXAN(config)#igmp mvlan 200 version-mode v1 drop
ZXAN(config)#igmp mvlan 200 version-mode v2 drop
ZXAN(config)#igmp mvlan 200 version-mode v3 accept
```

7. Configure the MVLAN working mode.

```
ZXAN(config)#igmp mvlan 200 work-mode proxy
```

8. Configure the MVLAN host version.

```
ZXAN(config)#igmp mvlan 200 host-version v3
```

9. Configure the MVLAN multicast group.

```
ZXAN(config)#igmp mvlan 200 group 224.1.1.1 to 224.1.1.3 source 10.1.1.1 prejoin
enable
```

10. Configure MVLAN source port.

```
ZXAN(config)#igmp mvlan 200 source-port gei_1/21/1
```

11. Configure the MVLAN receive port.

```
ZXAN(config)#igmp mvlan 200 receive-port gpon-onu_1/5/1:1 vport 1
```

12. (Optional) Query the IGMP global configuration.

```
ZXAN(config)#show igmp
IGMP global parameters:
-----
IGMP is globally enable.
Span vlan is enable.
Host tracking is disable.
IGMP log is disable.
General query gempport mode is unicast.
Prejoin interval is 120(second).
```

13. (Optional) Query the MVLAN.

```
ZXAN(config)#show igmp mvlan
```

```
Total Num is 1.
```

```
VID Status Work-mode GroupFilter Filter-mode ActGroups HostVersion
```

```
-----
```

```
200 enable proxy disable asmssm 0 v3
```

```
ZXAN(config)#show igmp mvlan 200
```

```
Protocol packet's priority is 0 (in proxy/spr mode)
```

```
Act Port is 0.
```

```
Host ip is 192.168.2.14.
```

```
Proxy ip is 192.168.2.14.
```

```
Igmp v1 mode is drop.
```

```
Igmp v2 mode is drop.
```

```
Igmp v3 mode is accept.
```

```
Robustness variable is 2.
```

```
General query interval is 125(second).
```

```
Query max response time is 100(0.1second).
```

```
Last member query interval is 10(0.1second).
```

```
Last member query count is 2.
```

```
Unsolicited report interval is 1(second).
```

```
Startup query interval is 30(second).
```

```
Startup query count is 2.
```

```
Snooping aging time is 300(second).
```

```
-----
```

```
Source Port HostCompatibleMode HostConfigMode V1TimeOut V2TimeOut
```

```
-----
```

```
gei_1/21/1 v3 v3 0 0
```

```
Receive Port
```

```
-----
```

```
gpon-onu_1/5/1:1:1
```

```
SSM Group Range
```

```
-----
```

```
232.0.0.0 mask 255.0.0.0
```

– End of Steps –

6.2 Configuring the MLD MVLAN

The MLD MVLAN is the VLAN that carries the MLD multicast data, which includes the service VLAN, source port, receive port, and multicast group.

Configuration Data

Table 6-2 lists the configuration data of the MLD MVLAN.

Table 6-2 Configuration Data of the MLD MVLAN

Item	Data
MLD	Enable
Span VLAN function	Enable
MVLAN ID	200
MVLAN working mode	Proxy
MVLAN host version	MLDv1
Multicast group IP address	ff1e::0101–ff1e::0103
Multicast source port	gei_1/21/1
Multicast receive port	ONU interface: gpon-onu_1/5/1:1 Virtual port ID: 1

Steps

1. In uplink interface configuration mode, configure the uplink port VLAN.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 200 tag
ZXAN(config-if)#exit
```

2. In ONU interface mode, configure the service port VLAN.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#service-port 1 vport 1 user-vlan 200 vlan 200
ZXAN(config-if)#exit
```

3. Enable MLD globally.

```
ZXAN(config)#mld enable
```

4. Enable MLD span VLAN function.

```
ZXAN(config)#mld span-vlan enable
```

5. Configure the MVLAN.

```
ZXAN(config)#mld mvlan 200
```

6. Configure the MVLAN working mode.

```
ZXAN(config)#mld mvlan 200 work-mode proxy
```

7. Configure the MVLAN host version.

```
ZXAN(config)#mld mvlan 200 host-version vl
```

8. Configure the MVLAN multicast group.

```
ZXAN(config)#mld mvlan 200 group ffile::0101 to ffile::0103
```

9. Configure MVLAN source port.

```
ZXAN(config)#mld mvlan 200 source-port gei_1/21/1
```

10. Configure the MVLAN receive port.

```
ZXAN(config)#mld mvlan 200 receive-port gpon-onu_1/5/1:1 vport 1
```

11. (Optional) Query the MLD global configuration.

```
ZXAN(config)#show mld
MLD global parameters:
-----
MLD is globally enable.
Span vlan is disable.
Host tracking is disable.
MLD log is disable.
General query gempport mode is unicast.
Prejoin interval is 120(second).
```

12. (Optional) Query the MVLAN.

```
ZXAN(config)#show mld mvlan

Total Num is 1.
VID   Status Work-mode   GroupFilter Filter-mode ActGroups HostVersion
-----
200   enable proxy         disable    asmssm      0          v1

ZXAN(config)#show mld mvlan 200
Protocol packet's priority is 0 (in proxy/spr mode)
Act Port is 0.
Host ip is fe80::c0a8:20e.
Proxy ip is fe80::c0a8:20e.
mld v1 mode   is accept.
mld v2 mode   is accept.
Robustness variable is 2.
General query interval is 125(second).
Query max response time is 100(0.1second).
Last member query interval is 10(0.1second).
Last member query count is 2.
Unsolicited report interval is 1(second).
Startup query interval is 30(second).
Startup query count is 2.
Snooping aging time is 300(second).
-----
Source Port           HostCompatibleMode   HostConfigMode   VlTimeOut
```

```
-----  
gei_1/21/1          v1          v1          0
```

```
Receive Port  
-----
```

```
gpon-onu_1/5/1:1:1
```

```
SSM Group Range  
-----
```

```
FF3x:: mask fff0:ffff:ffff:ffff:ffff:ffff::
```

– End of Steps –

6.3 Configuring the IPTV Package

By configuring the IPTV package, you can manage the access right of the IPTV channel.

Prerequisite

The MVLAN has been configured.

Configuration Data

Table 6-3 lists the configuration data of the IPTV package.

Table 6-3 Configuration Data of the IPTV Package

Item	Data
IPTV channel	Name prefix: stv Group IP address: 224.1.1.1–224.1.1.3 Source IP address: 10.1.1.1
IPTV package	Name: pkg1 Channel 0: stv001 (watch) Channel 1: stv002 (watch) Channel 2: stv003 (preview)

Steps

1. Configure the IPTV channel.

```
ZXAN(config)#iptv channel mvlan 200 group 224.1.1.1 to 224.1.1.3 source-address  
10.1.1.1 prename stv
```

2. Create the IPTV package.

```
ZXAN(config)#iptv package name pkg1
```

3. (Optional) Query the IPTV channel.

```
ZXAN(config)#show iptv channel
```

```
Total channel number :3
```

ID	mvlan	group	source	name
0	200	224.1.1.1	10.1.1.1	STV001
1	200	224.1.1.2	10.1.1.1	STV002
2	200	224.1.1.3	10.1.1.1	STV003

4. Configure the channel in the IPTV package.

```
ZXAN(config)#iptv package pkg1 channel stv001 watch
```

```
ZXAN(config)#iptv package pkg1 channel stv002 watch
```

```
ZXAN(config)#iptv package pkg1 channel stv003 preview
```

5. (Optional) Query the IPTV package.

```
ZXAN(config)#show iptv package pkg1
```

```
Package name: PKG1
```

```
Total channel number: 3
```

Group	Source	Mvlan	Right	Id	Name
224.1.1.1	10.1.1.1	200	watch	0	STV001
224.1.1.2	10.1.1.1	200	watch	1	STV002
224.1.1.3	10.1.1.1	200	preview	2	STV003

– End of Steps –

6.4 Configuring the Port IPTV Right

By configuring the IPTV right for port, you can apply the IPTV package to the subscriber port to implement the access control of the IPTV channel.

Prerequisite

- The MVLAN has been configured.
- The IPTV package has been configured.

Context

The ZXA10 C300 supports 2-level [CAC](#).

- When the CAC function is enabled globally, the subscriber port IPTV right takes effect and only the subscriber who subscribes the package can access the channel in the package.
- When the global CAC function is disabled, the subscriber port IPTV right does not take effect and subscribers in the MVLAN can access the channel in the MVLAN.

By default, the global CAC function is disabled.

Steps

1. Enable the CAC function globally.

```
ZXAN(config)#iptv cac enable
```

2. In ONU interface configuration mode, configure the port right.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#iptv package pkg1
ZXAN(config-if)#exit
```

3. (Optional) Query the IPTV global configuration.

```
ZXAN(config)#show iptv control
CAC      : enable
SMS      : 192.168.0.119
```

4. (Optional) Query the IPTV port configuration.

```
ZXAN(config)#show iptv interface gpon-onu_1/5/1:1
auth-mode : auth
right-mode: package
cdrstatus : enable
service   : IN_SERVICE
```

– End of Steps –

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Chapter 7

QoS Configuration

Service Description

Quality of Service (QoS) provides different service qualities to meet different requirements of various applications, for example, providing dedicated bandwidth, reducing the packet loss ratio and reducing packet transmission delay/jitter. Via flexible configuration and application of the QoS attributes, the carrier can provide effective differentiated services and implement and assure the committed service quality.

Service Specifications

The ZXAN C300 supports the following QoS operations:

- Precedence remarking
- Queue scheduling
- Queue mapping
- Traffic shaping

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7.1 Ethernet Interface QoS Configuration

7.1.1 Configuring the Default CoS

When the default CoS is configured, the Ethernet interface adds the default CoS to the untagged packet.

Steps

1. In Ethernet interface mode, configure the default Class of Service (CoS).

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#qos cos default-cos 5
```

2. (Optional) Query the QoS configuration on the interface.

```
ZXAN(config-if)#show qos interface gei_1/21/1
qos cos default-cos 5
```

– End of Steps –

7.1.2 Configuring DSCP-CoS Remarking

Using the DSCP-to-CoS remarking profile, you can remark the packet CoS priority according to the its DSCP value.

Steps

1. In global configuration mode, configure the DSCP-to-CoS remarking profile.

```
ZXAN(config)#qos dscp-to-cos-profile test 3 to 6
```

2. In Ethernet interface mode, apply the DSCP-to-CoS remarking profile.

```
ZXAN(config)#interface gei_1/21/1
```

```
ZXAN(config-if)#qos cos dscp-remark test
```

3. Configure trust Differentiated Services Code Point (DSCP) on the interface.

```
ZXAN(config-if)#qos trust dscp
```

4. (Optional) Query the QoS configuration on the interface.

```
ZXAN(config-if)#show qos interface gei_1/21/1
```

```
qos cos dscp-remark TEST
```

```
qos trust dscp
```

5. (Optional) Query the CoS remarking profile.

```
ZXAN(config-if)#show qos dscp-to-cos-profile test
```

```
-----
profile name      : TEST
```

```
profile detail :
```

```
-----
dscp list  0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15
cos value  0  0  0  6  0  0  0  0  1  1  1  1  1  1  1  1
```

```
-----
dscp list 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
cos value  2  2  2  2  2  2  2  2  3  3  3  3  3  3  3  3
```

```
-----
dscp list 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47
cos value  4  4  4  4  4  4  4  4  5  5  5  5  5  5  5  5
```

```
-----
dscp list 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63
cos value  6  6  6  6  6  6  6  6  7  7  7  7  7  7  7  7
```

```
profile used by:
```

```
gei_1/21/1
```

– End of Steps –

7.1.3 Configuring the Drop Precedence

Using the DSCP-to-drop profile, you can remark the packet drop precedence according to the its DSCP value.

Steps

1. In global configuration mode, configure the drop precedence profile.

```
ZXAN(config)#qos dscp-to-drop-profile test 3 to 2
```

2. In Ethernet interface mode, apply the drop precedence profile.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#qos drop-procedence test
```

3. Configure trust DSCP on the interface.

```
ZXAN(config-if)#qos trust dscp
```

4. (Optional) Query the QoS configuration on the interface.

```
ZXAN(config-if)#show qos interface gei_1/21/1
qos drop-procedence TEST
qos trust dscp
```

5. (Optional) Query the drop precedence profile.

```
ZXAN(config-if)#show qos dscp-to-drop-profile test
-----
profile name      : TEST
profile detail :
-----
dscp list  0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15
drop value 0  0  0  2  0  0  0  0  0  0  0  0  0  0  0  0
-----
dscp list  16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
drop value 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
-----
dscp list  32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47
drop value 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
-----
dscp list  48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63
drop value 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
profile used by:
gei_1/21/1
```

– End of Steps –

7.1.4 Configuring DSCP Remarking

Using the DSCP remark profile, you can remark the packet DSCP priority according to the its original DSCP value.

Steps

1. In global configuration mode, configure the **DSCP** remarking profile.

```
ZXAN(config)#qos dscp-to-dscp-profile test 3 to 13
```

2. In Ethernet interface mode, apply the DSCP remarking profile.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#qos dscp dscp-remark test
```

3. Configure trust DSCP on the interface.

```
ZXAN(config-if)#qos trust dscp
```

4. (Optional) Query the QoS configuration on the interface.

```
ZXAN(config-if)#show qos interface gei_1/21/1
qos dscp dscp-remark TEST
qos trust dscp
```

5. (Optional) Query the DSCP remarking profile.

```
ZXAN(config)#show qos dscp-to-dscp-profile test
-----
profile name      : TEST
profile detail :
-----
dscp list   0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15
dscp value  0  1  2 13 4  5  6  7  8  9 10 11 12 13 14 15
-----
dscp list   16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
dscp value  16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
-----
dscp list   32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47
dscp value  32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47
-----
dscp list   48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63
dscp value  48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63
profile used by:
gei_1/21/1
```

– End of Steps –

7.1.5 Configuring Queue Scheduling

Using the profile, you can implement queue scheduling on the Ethernet interface.

Steps

1. In global configuration mode, configure the queue scheduling profile.

```
ZXAN(config)#qos queue-block-profile test queue0 2 0 queue1 3 0
```

**Note:**

In a queue scheduling profile, the queue of which the queue weight is 0 should be configured at the end.

2. In Ethernet interface mode, apply the queue scheduling profile.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#qos queue-block-profile test
```

3. (Optional) Query the QoS configuration on the interface.

```
ZXAN(config)#show qos interface gei_1/21/1
qos queue-block-profile TEST
```

4. (Optional) Query the QoS queue scheduling profile.

```
ZXAN(config-if)#show qos queue-block-profile test
```

```
-----
profile name      : TEST
profile detail    :
-----
```

```
queue-number      : 8
queue-weight      : 2 3 0 0 0 0 0 0
queue-depth       : 0 0 0 0 0 0 0 0
-----
```

```
profile used by :
gei_1/21/1
```

– End of Steps –

7.1.6 Configuring Traffic Shaping

By implementing traffic shaping, you can set the packet rate to match that of the receiving device, to avoid congestion or packet discarding.

Context

Traffic shaping controls the rate of the output packets so that the packets are sent at a constant rate.

By default, traffic shaping is disabled.

Steps

1. In Ethernet interface mode, configure traffic shaping.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#qos traffic-shape rate-limit 1280 bucket-size 512
```

2. (Optional) Query the QoS configuration on the interface.

```
ZXAN(config)#show qos interface gei_1/21/1
qos traffic-shape rate-limit 1280 bucket-size 512
```

– End of Steps –

7.1.7 Configuring the Mapping Relation From CoS to Local Queues

This section describes how to configure the mapping relation from the Ethernet packet CoS to local queues.

Steps

1. In global configuration mode, configure the mapping relation from CoS to local queues.

```
ZXAN(config)#qos eth-cos-local-map cos0 7 cos1 5 cos2 2 cos3 3 cos4 4 cos5 5
cos6 6 cos7 7
```

2. (Optional) Query the mapping relation from CoS to local queues.

```
ZXAN(config)#show qos eth-cos-local-map
-----
cos value  0  1  2  3  4  5  6  7
queue ID   7  5  2  3  4  5  6  7
```

– End of Steps –

7.2 OLT Interface QoS Configuration

7.2.1 Configuring Queue Scheduling

Using the profile, you can implement queue scheduling on the OLT interface.

Steps

1. In global configuration mode, configure the queue scheduling profile.

```
ZXAN(config)#qos queue-block-profile test queue0 2 12 queue1 3 12
```

2. In OLT interface configuration mode, apply the queue scheduling profile.

```
ZXAN(config)#interface gpon-olt_1/5/1
ZXAN(config-if)#qos queue-block-profile test
```

3. (Optional) Query the QoS queue scheduling profile.

```
ZXAN(config)#show qos queue-block-profile test
-----
profile name      : TEST
profile detail    :
-----
queue-number      : 8
queue-weight      : 2 3 0 0 0 0 0 0
queue-depth       : 12 12 0 0 0 0 0 0
```

```

-----
profile used by:
gpon-olt_1/5/1

```

– End of Steps –

7.2.2 Configuring Queue Mapping

Using the profile, you can implement queue mapping on the OLT interface.

Steps

1. In global configuration mode, configure the queue map profile.

```
ZXAN(config)#qos queue-map-profile test cos-queue-type cos0 2
```

2. In OLT interface configuration mode, apply the queue map profile.

```
ZXAN(config)#interface gpon-olt_1/5/1
ZXAN(config-if)#qos queue-map-profile test
```

3. (Optional) Query the QoS queue mapping profile.

```
ZXAN(config-if)#show qos queue-map-profile test
```

```

-----
profile name      : TEST
profile detail :
-----
queue-map       : cos-queue-type
queue-number    : 8
cos-value       : 0  1  2  3  4  5  6  7
queue-map:      2  1  2  3  4  5  6  7
-----
profile used by:
gpon-olt_1/5/1

```

– End of Steps –

7.2.3 Configuring the Traffic Profile

Using the profile, you can limit the traffic on the GPON OLT interface.

Steps

1. In global configuration mode, configure the traffic profile.

```
ZXAN(config)#traffic-profile test ip cir 10240 cbs 1000 pir 20480 pbs 1000
```

2. In OLT interface configuration mode, apply the traffic profile.

```
ZXAN(config)#interface gpon-olt_1/5/1
ZXAN(config-if)#traffic-profile test direction egress
```

3. (Optional) Query the QoS traffic profile.

```

ZXAN(config-if)#show traffic-profile test
-----
profile name           : TEST
profile detail         :
-----
basic traffic type     : ip
committed information rate : 10240 kbps
committed burst size   : 1000 kbytes
peak information rate   : 20480 kbps
peak burst size        : 1000 kbytes
discard mode           : low priority first
color mode             : blind
-----
profile used by :
gpon-olt_1/5/1

```

– End of Steps –

7.3 ONU Interface QoS Configuration

7.3.1 Configuring the Trust Precedence

This section describes how to configure the ONU virtual port (vport) to trust CoS or DSCP priority of packets.

Context

When the vport trusts [CoS](#) or [DSCP](#) priority, there are two cases:

- When the vport trusts CoS priority, the CoS in packets is marked in the override > cos-remark > trust order based on the ingress CoS.
- When the vport trusts DSCP priority, the CoS is marked according to the configured DSCP-to-CoS mapping relation.

Steps

1. In ONU interface configuration mode, configure the trust precedence of the vport.

```

ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#qos trust dscp vport 1
ZXAN(config-if)#qos trust cos vport 2

```

2. (Optional) Query the QoS configuration on the interface.

```

ZXAN(config-if)#show qos interface gpon-onu_1/5/1:1
qos trust dscp vport 1

```


**Note:**

CoS is the default configuration of the interface and is not displayed.

– End of Steps –

7.3.2 Configuring the Default CoS

When the default CoS is configured, the ONU virtual port (vport) adds the default CoS to the untagged packet.

Context

When the default CoS is configured on a virtual port, the override operation is optional.

- With the override operation: the CoS in all packets (including untagged packets) is modified to the default CoS.
- Without the override operation: Only the CoS in untagged packets is modified to the default CoS.

Steps

1. In ONU interface configuration mode, configure the default CoS.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#qos cos default-cos 5 override vport 1
ZXAN(config-if)#qos cos default-cos 5 vport 2
```

2. (Optional) Query the QoS configuration on the interface.

```
ZXAN(config-if)#show qos interface gpon-onu_1/5/1:1
qos cos default-cos 5 override vport 1
qos cos default-cos 5 vport 2
```

– End of Steps –

7.3.3 Configuring CoS Remarking

Using the CoS remark profile, you can remark packet's CoS priority according to its CoS value on the ONU virtual port (vport).

Context

When the vport trusts [CoS](#) and the default CoS is not configured with the override operation, the CoS in packets is modified according to the mapping relation in the profile after the CoS remarking profile is configured.

Steps

1. In global configuration mode, configure the CoS remarking profile.

```
ZXAN(config)#qos cos-to-cos-profile test cos0 3
```

2. In ONU interface configuration mode, apply the CoS remarking profile.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
```

```
ZXAN(config-if)#qos cos cos-remark test vport 1
```

```
ZXAN(config-if)#qos trust cos vport 1
```

3. (Optional) Query the QoS configuration on the interface.

```
ZXAN(config-if)#show qos interface gpon-onu_1/5/1:1
```

```
qos cos cos-remark TEST vport 1
```

– End of Steps –

7.3.4 Configuring DSCP to CoS Remarking

Using the DSCP-to-CoS remarking profile, the ONU virtual port (vport) modifies packet's CoS priority according to its DSCP value.

Context

When the vport trusts **DSCP**, the **CoS** in packets is modified according to the mapping relation in the profile after the DSCP remarking profile is configured.

Steps

1. In global configuration mode, configure the DSCP-to-CoS remarking profile.

```
ZXAN(config)#qos dscp-to-cos-profile test 12 to 3
```

2. In ONU interface configuration mode, apply the DSCP-to-CoS remarking profile.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
```

```
ZXAN(config-if)#qos cos dscp-remark test vport 1
```

```
ZXAN(config-if)#qos trust dscp vport 1
```

3. (Optional) Query the QoS configuration on the interface.

```
ZXAN(config-if)#show qos interface gpon-onu_1/5/1:1
```

```
qos trust dscp vport 1
```

```
qos cos dscp-remark TEST vport 1
```

– End of Steps –

7.3.5 Configuring the Default Egress CoS

When the default CoS is configured, the ONU virtual port (vport) adds the default egress CoS to the untagged packet.

Context

When the default CoS is configured on a vport, the override operation is optional.

- With the override operation: the CoS in all packets on the vport is modified to the default egress CoS.
- Without the override operation: the vport transparently transmits all packets.

Steps

1. In ONU interface configuration mode, configure the default CoS.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#qos egress-cos default-cos 5 override vport 1
```

2. (Optional) Query the QoS configuration on the interface.

```
ZXAN(config-if)#show qos interface gpon-onu_1/5/1:1
qos egress-cos default-cos 5 override vport 1
```

– End of Steps –

7.3.6 Configuring Egress CoS Remarking

Using the CoS remark profile, you can remark packet's egress CoS priority according to its CoS value on the ONU virtual port (vport).

Context

When the vport trusts CoS and the default CoS is not configured with the override operation, the CoS in packets is modified according to the mapping relation in the profile after the egress CoS remarking profile is configured.

Steps

1. In global configuration mode, configure the CoS remarking profile.

```
ZXAN(config)#qos cos-to-cos-profile test cos0 3
```

2. In ONU interface configuration mode, apply the CoS remarking profile.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#qos egress-cos cos-remark test vport 1
ZXAN(config-if)#qos trust cos vport 1
```

3. (Optional) Query the QoS configuration on the interface.

```
ZXAN(config-if)#show qos interface gpon-onu_1/5/1:1
qos egress-cos cos-remark TEST vport 1
```

– End of Steps –

7.3.7 Configuring Egress DSCP to CoS Remarking

Using the DSCP-to-CoS remarking profile, you can remark packet's egress CoS priority according to its DSCP value on the ONU virtual port (vport).

Context

When the vport trusts DSCP, the CoS in packets is modified according to the mapping relation in the profile after the egress DSCP remarking profile is configured.

Steps

1. In global configuration mode, configure the DSCP-to-CoS remarking profile.

```
ZXAN(config)#qos dscp-to-cos-profile test 12 to 3
```

2. In ONU interface configuration mode, apply the DSCP-to-CoS remarking profile.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#qos egress-cos dscp-remark test vport 1
ZXAN(config-if)#qos trust dscp vport 1
```

3. (Optional) Query the QoS configuration on the interface.

```
ZXAN(config-if)#show qos interface gpon-onu_1/5/1:1
qos trust dscp vport 1
qos egress-cos dscp-remark TEST vport 1
```

– End of Steps –

7.3.8 Configuring CoS Filtering

When CoS filtering is configured on the ONU virtual port (vport), only those packets are forwarded whose CoS is the same as the default CoS of the vport.

Steps

1. In ONU interface configuration mode, configure the default CoS.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#qos cos default-cos 5 vport 1
ZXAN(config-if)#qos cos-filter enable vport 1
```

2. (Optional) Query the QoS configuration on the interface.

```
ZXAN(config-if)#show qos interface gpon-onu_1/5/1:1
qos cos-filter enable vport 1
qos cos default-cos 5 vport 1
```

– End of Steps –

7.3.9 Configuring Queue Scheduling

Using the queue block profile, you can implement queue scheduling on the ONU virtual port (vport).

Steps

1. In global configuration mode, configure the queue block profile.

```
ZXAN(config)#qos queue-block-profile test queue0 2 12 queue1 3 12
```

2. In ONU interface configuration mode, apply the queue block profile.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#qos queue-block-profile test vport 1
```

3. (Optional) Query the QoS queue block profile.

```
ZXAN(config-if)#show qos queue-block-profile test
```

```
-----
profile name      : TEST
profile detail   :
-----
```

```
queue-number      : 8
queue-weight      : 2  3  0  0  0  0  0  0
queue-depth       : 12 12  0  0  0  0  0  0
-----
```

```
profile used by :
gpon-onu_1/5/1:1
```

– End of Steps –

7.3.10 Configuring Queue Mapping

Using the queue map profile, you can implement queue mapping on the ONU virtual port (vport).

Steps

1. In global configuration mode, configure the queue map profile.

```
ZXAN(config)#qos queue-map-profile test cos-queue-type cos0 2
```

2. In ONU interface configuration mode, apply the queue map profile.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#qos queue-map-profile test vport 1
```

3. (Optional) Query the QoS queue map profile.

```
ZXAN(config-if)#show qos queue-map-profile test
```

```
-----
profile name      : TEST
profile detail   :
-----
```

```
queue-map        : cos-queue-type
queue-number     : 8
cos-value        : 0  1  2  3  4  5  6  7
cos-queue-map    : 2  1  2  3  4  5  6  7
-----
```

```
profile used by:
gpon-onu_1/5/1:1
```

– End of Steps –

7.3.11 Configuring the Traffic Profile

Using the traffic profile, you can limit the traffic of the GPON ONU virtual port (vport).

Steps

1. In global configuration mode, configure the traffic profile.

```
ZXAN(config)#traffic-profile test ip cir 10240 cbs 1000 pir 20480 pbs 1000
```

2. In ONU interface configuration mode, apply the traffic profile.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
```

```
ZXAN(config-if)#traffic-profile test vport 1 direction egress
```

3. (Optional) Query the QoS traffic profile.

```
ZXAN(config-if)#show traffic-profile test
```

```
-----  
profile name          : TEST
```

```
profile detail        :  
-----
```

```
basic traffic type    : ip
```

```
committed information rate : 10240 kbps
```

```
committed burst size   : 1000 kbytes
```

```
peak information rate   : 20480 kbps
```

```
peak burst size        : 1000 kbytes
```

```
discard mode           : low priority first
```

```
color mode             : blind  
-----
```

```
profile used by :
```

```
gpon-olt_1/5/1
```

```
gpon-onu_1/5/1:1:1
```

– End of Steps –

Chapter 8

ACL Configuration

The network devices use the Access Control List (ACL) to filter the data packets and control the policy routes and special flows. ACL sets a series of matching rules to identify the objects to be filtered, and permits or denies the corresponding data packet to pass through according to the preset policies.

An ACL can contain one or more rules. These rules enable the device to permit or deny the matching traffic according to specific parameters. An ACL compares the traffic with each rule till it finds a matched rule. The last rule in an ACL is an implicit deny rule.

One interface supports only one ACL.

The ZX10 C300 supports the following four types of ACLs:

- Standard ACL

The standard ACL is only matched by the source IP address.

- Extended ACL

The extended ACL is matched by the source IP address, destination IP address, IP protocol type, TCP/UDP source/destination port number, ICMP type, IGMP type, DSCP, ToS, and IP priority.

- Layer-2 ACL

The layer-2 ACL is matched by the source MAC address, destination MAC address, source VLAN ID, layer-2 Ethernet protocol type, and 802.1p priority value.

- Hybrid ACL

The hybrid ACL is matched by the source MAC address, destination MAC address, source VLAN ID, source IP address, destination IP address, TCP/UDP source/destination port number, including all the matching fields of the preceding three types.

- IPv6 hybrid ACL

It is the IPv6-based hybrid ACL.

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8.1 Configuring a Standard ACL

This section describes how to configure a standard ACL and apply it to an Ethernet interface.

Configuration Data

Table 8-1 lists the configuration data of the standard ACL.

Table 8-1 Configuration Data of the Standard ACL

Item	Data
Time range	Name: worktime Range: 09:00:00–17:00:00 Day: working-day
ACL number	3
Rule 1	Action: deny Source address: 168.1.1.1/24 Time range: worktime
Rule 2	Permit any traffic
Interface	gei_1/21/1

Steps

1. (Optional) In global configuration mode, configure the ACL time range.

```
ZXAN(config)#time-range worktime 09:00:00 to 17:00:00 working-day
```

2. Create a standard ACL.

```
ZXAN(config)#acl standard number 3  
ZXAN(config-std-acl)#
```



Note:

The standard ACL number ranges from 1 to 99. The standard ACL can be applied to the Ethernet interface only.

3. Configure the ACL rules.

```
ZXAN(config-std-acl)#rule 1 deny 168.1.1.1 0.0.0.255 time-range worktime  
ZXAN(config-std-acl)#rule 2 permit any  
ZXAN(config-std-acl)#exit
```


**Note:**

Each standard ACL supports up to 127 rules.
If the time range is not configured, the rule is always effective.

4. In Ethernet interface configuration mode, apply the ACL.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#ip access-group 3 in
```

5. (Optional) Query the ACL configuration.

```
ZXAN(config-if)#show acl 3
acl standard number 3
  rule 1 deny 168.1.1.0 0.0.0.255 time-range worktime
  rule 2 permit any
```

6. (Optional) Query the interface bound with the ACL.

```
ZXAN(config-if)#show access-list bound
```

Interface	Direction	Type	Status	Acl number/name
gei_1/21/1	in	V4STD	successful	3

– End of Steps –

8.2 Configuring an Extended ACL

This section describes how to configure an extended ACL and apply it to an Ethernet interface.

Configuration Data

Table 8-2 lists the configuration data of the extended ACL.

Table 8-2 Configuration Data of the Extended ACL

Item	Data
ACL number	101
Rule 1	Action: deny Source address: 192.168.1.0/24 Protocol type: TCP, Telnet
Rule 2	Permit any TCP and telnet traffic
Interface	gei_1/21/1

Steps

1. In global configuration mode, create an extended ACL.

```
ZXAN(config)#acl extended number 101
ZXAN(config-ext-acl)#
```

**Note:**

The extended ACL number ranges from 100 to 199. An extended ACL can be applied to an Ethernet interface only.

2. Configure the ACL rules.

```
ZXAN(config-ext-acl)#rule 1 deny tcp 192.168.1.0 0.0.0.255 eq telnet any
ZXAN(config-ext-acl)#rule 2 permit tcp any eq telnet any
ZXAN(config-ext-acl)#exit
```

**Note:**

Each extended ACL supports up to 1024 rules.

If the time range is not configured, the rule is always effective.

3. In Ethernet interface configuration mode, apply the ACL.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#ip access-group 101 in
```

4. (Optional) Query the ACL configuration.

```
ZXAN(config-if)#show acl 101
acl extend number 101
rule 1 deny tcp 192.168.1.0 0.0.0.255 eq telnet any
rule 2 permit tcp any eq telnet any
```

5. (Optional) Query the interface bound with the ACL.

```
ZXAN(config-if)#show access-list bound
```

Interface	Direction	Type	Status	Acl number/name
gei_1/21/1	in	V4EXT	successful	101

– End of Steps –

8.3 Configuring a Layer-2 ACL

This section describes how to configure a layer-2 ACL and apply it to an Ethernet interface.

Configuration Data

Table 8-3 lists the configuration data of the layer-2 ACL.

Table 8-3 Configuration Data of the Layer-2 ACL

Item	Data
ACL number	200
Rule 1	Action: deny Source MAC address: 0000.0000.0001 Protocol type: any
Rule 2	Permit any traffic
Interface	gei_1/21/1

Steps

1. In global configuration mode, create a layer-2 ACL.

```
ZXAN(config)#acl link number 200
ZXAN(config-link-acl)#
```



Note:

The layer ACL number ranges from 200 to 299. A layer-2 ACL can be applied to the Ethernet interface and EPON-OLT interface.

2. Configure the ACL rules.

```
ZXAN(config-link-acl)#rule 1 deny any ingress 0000.0000.0001 0000.0000.0000
egress any
ZXAN(config-link-acl)#rule 2 permit any
ZXAN(config-link-acl)#exit
```



Note:

Each layer-2 ACL supports up to 4096 rules.

If the time range is not configured, the rule is always effective.

3. In Ethernet interface configuration mode, apply the ACL.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#ip access-group 200 in
```

4. (Optional) Query the ACL configuration.

```
ZXAN(config-if)#show acl 200
```

```
acl link number 200
  rule 1 deny any ingress 0000.0000.0001 0000.0000.0000 egress any
  rule 2 permit any ingress any egress any
```

5. (Optional) Query the interface bound with the ACL.

```
ZXAN(config-if)#show access-list bound
```

Interface	Direction	Type	Status	Acl number/name
gei_1/21/1	in	V4LVL2	successful	200

– End of Steps –

8.4 Configuring a Hybrid ACL

This section describes how to configure a hybrid ACL and apply it to an Ethernet interface.

Configuration Data

Table 8-4 lists the configuration data of the hybrid ACL.

Table 8-4 Configuration Data of the Hybrid ACL

Item	Data
ACL number	300
Rule 1	Action: deny IP protocol type: any Source address: any Destination address: any Ethernet protocol type: ARP
Rule 2	Action: deny IP protocol type: any Source MAC address: 0000.0000.0001 Destination IP address 192.168.1.0/24 Ethernet protocol type: any
Rule 3	Permit any traffic
Interface	gei_1/21/1

Steps

1. In global configuration mode, create a hybrid ACL.

```
ZXAN(config)#acl hybrid number 300
ZXAN(config-hybd-acl)#
```

**Note:**

The hybrid ACL number ranges from 300 to 399. A hybrid ACL is applied to the Ethernet interface and PON-ONU interface.

2. Configure the ACL rules.

```
ZXAN(config-hybd-acl)#rule 1 deny any any any arp
ZXAN(config-hybd-acl)#rule 2 deny any any 192.168.1.0 0.0.0.255 ip ingress 0000.
0000.0001 0000.0000.0000 egress any
ZXAN(config-hybd-acl)#rule 3 permit any any any any
ZXAN(config-hybd-acl)#exit
```

**Note:**

Each hybrid ACL supports up to 127 rules.

If the time range is not configured, the rule is always effective.

3. In Ethernet interface configuration mode, apply the ACL.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#ip access-group 300 in
```

4. (Optional) Query the ACL configuration.

```
ZXAN(config-if)#show acl 300
acl hybrid number 300
rule 1 deny any any any arp ingress any egress any
rule 2 deny any any 192.168.1.0 0.0.0.255
ip ingress 0000.0000.0001 0000.0000.0000 egress any
rule 3 permit any any any any ingress any egress any
```

5. (Optional) Query the interface bound with the ACL.

```
ZXAN(config-if)#show access-list bound
```

Interface	Direction	Type	Status	Acl number/name
gei_1/21/1	in	V4HYBD	successful	300

– End of Steps –

8.5 Configuring an IPv6 Hybrid ACL

This section describes how to configure an IPv6 hybrid ACL and apply it to an Ethernet interface.

Configuration Data

Table 8-5 lists the configuration data of the IPv6 hybrid ACL.

Table 8-5 Configuration Data of the IPv6 Hybrid ACL

Item	Data
ACL number	600
Rule 1	Action: deny IP protocol type: TCP Source address: any Destination address: any Traffic class: 7 Ethernet protocol type: any
Rule 2	Action: deny Protocol type: any Source address: 00:00::00:22/128 Destination address: any Ethernet protocol type: any CoS priority: 3
Rule 3	Permit any traffic
Interface	gei_1/21/1

Steps

1. In global configuration mode, create an IPv6 hybrid ACL.

```
ZXAN(config)#acl6 hybrid number 600
ZXAN(config-hybd-acl6)#
```



Note:

The IPv6 hybrid ACL number ranges from 600 to 699. An IPv6 hybrid ACL can be applied to an Ethernet interface and PON-OLT interface.

2. Configure the ACL rules.

```
ZXAN(config-hybd-acl6)#rule 1 deny tcp any any traffic-class 7 any
ZXAN(config-hybd-acl6)#rule 2 deny any 00:00::00:22/128 any any cos 3
ZXAN(config-hybd-acl6)#rule 3 permit any any any any
ZXAN(config-hybd-acl6)#exit
```

**Note:**

Each IPv6 hybrid ACL supports up to 127 rules.

If the time range is not configured, the rule is always effective.

3. In Ethernet interface configuration mode, apply the ACL.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#ip access-group 600 in
```

4. (Optional) Query the ACL configuration.

```
ZXAN(config-if)#show acl 600
Acl6 hybrid number 600
rule 1 deny tcp any any traffic-class 7 any
rule 2 deny any ::22/128 any any cos 3 ingress any egress any
rule 3 permit any any any any ingress any egress any
```

5. (Optional) Query the interface bound with the ACL.

```
ZXAN(config-if)#show access-list bound
```

Interface	Direction	Type	Status	Acl number/name
gei_1/21/1	in	V6HYBD	successful	600

– End of Steps –

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Chapter 9

NTP Configuration

The Network Time Protocol (NTP) is a protocol for synchronizing the time of different network members. The devices that support NTP periodically exchange NTP packets to synchronize their clocks.

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9.1 Configuring NTP

The ZXAN C300 works in NTP client mode and synchronizes its time with the NTP server.

Steps

1. In global configuration mode, enable NTP.

```
ZXAN(config)#ntp enable
```

2. Configure the NTP server.

```
ZXAN(config)#ntp server 1.2.1.1 priority 1
```

3. Configure the source IP address of NTP packets for the time synchronization request.

```
ZXAN(config)#ntp source mng1
```

4. Configure the alarm threshold of the NTP time synchronization offset value.

```
ZXAN(config)#ntp alarm-threshold 10
```

5. Configure the NTP synchronization poll interval.

```
ZXAN(config)#ntp poll-interval 5
```

6. (Optional) Query the NTP running status.

```
ZXAN(config)#show ntp status
Clock is unsynchronized , stratum 16, no reference clock
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**16
reference time is 0.0 (0)
clock offset is 0.00 msec, root delay is 0.00 msec
root dispersion is 0.00 msec, peer dispersion is 0.00 msec
server in use is 0:1.2.1.1
```

– End of Steps –

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Chapter 10

STP Configuration

The ZXA10 C300 supports the following three Spanning Tree Protocol (STP) modes:

- Single Spanning Tree Protocol (SSTP)
- Rapid Spanning Tree Protocol (RSTP)
- Multiple Spanning Tree Protocol (MSTP)

SSTP Mode

SSTP complies with the IEEE 802.1d standard. It is compatible with STP, RSTP and MSTP. The bridge in SSTP mode can interwork with the bridge in RSTP and MSTP modes.

RSTP Mode

RSTP complies with the IEEE 802.1w standard. RSTP provides faster convergence than SSTP. When the network topology changes, the port state of the redundant switch port can be quickly changed from Discard to Forward in a point-to-point connection condition.

MSTP Mode

MSTP complies with the IEEE 802.1s standard. MSTP is added with the concepts of instance and VLAN mapping. SSTP and RSTP modes can be considered as a special MSTP instance, in which case, the instance is 0. The MSTP mode provides fast convergence and load balancing for VLAN.

In SSTP and RSTP modes, the VLAN concept does not exist, and each port has only one state. Namely, the port has the same forwarding state in different VLANs.

In MSTP mode, multiple spanning-tree instances can exist, and a port has different forwarding states in different VLANs. Multiple sub-tree instances can be generated in the Multiple Spanning Tree (MST) region to realize load balancing.

MSTP is applied to the redundant network. MSTP can not only provide fast convergence but also distribute flows of different VLANs to the respective paths, which provides a good load sharing mechanism for redundant links.

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10.1 Configuring STP

The ZXA10 C300 supports MSTP and is compatible with SSTP and RSTP. It also supports MSTP ring networking. By default, the ZXA10 C300 uses the MSTP mode. Any one of

the modes is compatible and interconnected with the other two modes. This topic takes MSTP as an example.

Steps

1. In global configuration mode, enable STP.

```
ZXAN(config)#spanning-tree enable
```

2. Configure STP protocol mode.

```
ZXAN(config)#spanning-tree mode mstp
```

3. (Optional) Configure the MST key and digest.

```
ZXAN(config)#spanning-tree mst hmd5-key cisco 0x13ac06a62e47fd51f95d2ba243cd0346
```

```
ZXAN(config)#spanning-tree mst hmd5-digest cisco 0x13ac06a62e47fd51f95d2ba243cd0346
```



Note:

The MSTP packet formats of the Cisco/Huawei devices may not follow the IEEE standard strictly. When the ZXA10 C300 interworks with the Cisco/Huawei devices in the same region, the KEY and DIGEST values are mandatory.

4. In MST configuration mode, configure the MST version number and name.

```
ZXAN(config)#spanning-tree mst configuration
```

```
ZXAN(config-mstp)#revision 10
```

```
ZXAN(config-mstp)#name zte
```

5. Create the MSTP instance.

The ZXA10 C300 has only instance 0 that is the common and internal spanning tree (CIST) in SSTP and RSTP modes. In MSTP mode, instance 0 exists by default and cannot be deleted.

The devices in the same MST region should meet all the following four requirements:

- The MST names are the same.
- The MST version numbers are the same.
- The INS-VLAN mapping tables are the same.
- The devices are connected physically.

```
ZXAN(config-mstp)#instance 1 vlans 10-20
```

```
ZXAN(config-mstp)#exit
```

6. Configure the priority of the local bridge.

```
ZXAN(config)#spanning-tree mst instance 1 priority 4096
```

7. In uplink interface configuration mode, configure the port VLAN.

```
ZXAN(config)#interface gei_1/22/1
```

```
ZXAN(config-if)#switchport vlan 10 tag
```

8. (Optional) Query the MSTP configuration.

```

ZXAN(config-if)#show spanning-tree mst configuration
spanning-tree           : [enable]
mode                    : [MSTP]
CISCO    Hmd5-key       : 0x13ac06a62e47fd51f95d2ba243cd0346
CISCO    Hmd5-digest    : 0x13ac06a62e47fd51f95d2ba243cd0346
HUAWEI   Hmd5-key       : 0x00000000000000000000000000000000
HUAWEI   Hmd5-digest    : 0x00000000000000000000000000000000
BPDU     Hmd5-digest    : 0x6cab52e9278d2d221c83bdfdf1a4da72
Name      : [zte]
Revision  : 10
Instance  Vlans mapped
-----
0          1-9,21-4094

1          10-20

```

9. (Optional) Query the instance configuration.

```

ZXAN(config-if)#show spanning-tree instance 1

MST01
Spanning tree enabled protocol MSTP
RegRootID: Priority      4097;   Address 00d0.d043.3832
           Hello-Time    2 sec;  Max-Age 20 sec
           Forward-Delay 15 sec;

BridgeID: Priority      4097;   Address 00d0.d043.3832
           Hello-Time    2 sec;  Max-Age 20 sec
           Forward-Delay 15 sec;  Max-Hops 20
           Message-Age   0 sec;   RemainHops 20

Interface  Prio.Nbr
Name       Port ID   Cost    Sts      Role      LinkType Bound
-----
gei_1/22/1 128.42   20000   Discard   Designated p2p      MSTP

```

– End of Steps –

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Chapter 11

DHCP Configuration

DHCP

DHCP enables a host on the network to obtain an **IP** address that ensures its proper communication and the relevant configuration information from a DHCP server.

IPv6 DHCP

Dynamic Host Configuration Protocol for IPv6 (IPv6 DHCP) assigns address parameters to hosts, which include IPv6 prefix, IPv6 addresses, and other network configuration parameters.

DHCP Applications

The ZXA10 C300 supports the following DHCP applications:

- DHCP snooping (including IPv6 DHCP snooping)

The ZXA10 C300 snoop on the DHCP communication process of the specified ONU in the specified VLAN to record the user IP/MAC relationship of the specified ONU.

Through DHCP snooping, the administrator can implement IP source-guard according to the IP/MAC binding table.
- DHCP server

The ZXA10 C300 works as a DHCP server to allocate IP addresses for users.
- DHCP client (including IPv6 DHCP client)

The ZXA10 C300 works as a DHCP client. It requires an IP address from the specified DHCP server, so that users can access it through SNMP.

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11.1 Configuring DHCP Snooping

After you configure DHCP snooping, the ZXA10 C300 will intercept the DHCP interaction process on the specified user port, extract the IP address and MAC address, and set up the DHCP snooping binding table that is the basis of IP source guard.

Configuration Data

Table 11-1 lists the configuration data of an DHCP snooping.

Table 11-1 Configuration Data of DHCP Snooping

Item	Data
Global DHCP	● Status: enable ● Option 82: enable
DHCP VLAN ID	200
Uplink interface:	gei_1/21/1
Service interface	gpon-onu_1/5/1:1 (virtual port 1)

Steps

1. In Ethernet interface configuration mode, configure the uplink port VLAN.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 200 tag
ZXAN(config-if)#exit
```

2. In ONU interface configuration mode, configure the VLAN on the virtual port.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#service-port 1 vport 1 user-vlan 200 vlan 200
ZXAN(config-if)#exit
```

3. Enable DHCP snooping on the VLAN.

```
ZXAN(config)#ip dhcp snooping vlan 200
```

4. Enable the global DHCP Option 82 processing.

```
ZXAN(config)#dhcpv4-l2-relay-agent vlan 200 enable
```

5. (Optional) Query the DHCP snooping VLAN configuration.

```
ZXAN(config)#show ip dhcp snooping vlan
DHCP snooping state on vlans
Vlan      State
-----
200       enable
```

– End of Steps –

11.2 Configuring DHCP Server

After you configure the DHCP server function, the ZXA10 C300 can work as a DHCP server to allocate IP addresses to subscribers.

Configuration Data

Table 11-2 lists the configuration data of the DHCP server.

Table 11-2 Configuration Data of DHCP Server

Item	Data
Global DHCP status	enable
IP address pool	● Name: ippool1 ● Range: 10.10.1.3–10.10.1.254
DHCP IP address pool	● Name: dhcp pool1 ● IP pool: ippool1 ● DNS IP address: 10.10.1.2 ● IP address lease period: infinite ● Default route: 10.10.1.254
DHCP policy	● Name: dhcp policy ● Priority: 1 ● DHCP IP address pool: dhcp pool1
DHCP server	● IP address: 10.10.1.1 ● Mode: server ● Policy: dhcp policy

Steps

1. Enable the global DHCP function.

```
ZXAN(config)#ip dhcp enable
```

2. Configure the IP address pool for DHCP clients.

```
ZXAN(config)#ip pool ippool1
ZXAN(config-ip-pool)#range 10.10.1.3 10.10.1.254 255.255.255.0
ZXAN(config-ip-pool)#exit
```

3. Apply the IP address pool to the DHCP IP address pool.

```
ZXAN(config)#ip dhcp pool dhcp pool1
ZXAN(config-dhcp-pool)#ip-pool ippool1
```

4. Configure the lease time of the IP addresses.

```
ZXAN(config-dhcp-pool)#lease-time infinite
```

5. Configure the DHCP DNS server.

```
ZXAN(config-dhcp-pool)#dns-server 10.10.1.2
```

6. Configure the default route.

```
ZXAN(config-dhcp-pool)#default-router 10.10.1.254
ZXAN(config-dhcp-pool)#exit
```

7. Configure the DHCP policy.

```
ZXAN(config)#ip dhcp policy dhcp policy 1
ZXAN(config-dhcp-policy)#dhcp-pool dhcp pool1
ZXAN(config-dhcp-policy)#exit
```

8. In management interface mode, configure the DHCP mode and policy.

```

ZXAN(config)#interface mng1
ZXAN(config-if)#ip address 10.10.1.1 255.255.255.0
ZXAN(config-if)#ip dhcp mode server
ZXAN(config-if)#ip dhcp policy dhcp
ZXAN(config-if)#exit

```

9. Configure the route.

```

ZXAN(config)#ip route mng 0.0.0.0 0.0.0.0 10.10.1.254

```

10. Query the DHCP server clients.

```

ZXAN(config)#show ip dhcp server user
Current online users are 0
Index MAC addr          IP addr          State      Expiration

```

– End of Steps –

11.3 Configuring DHCP Client

The ZXA10 C300 can work as a client to acquire an IP address from a DHCP server.

Configuration Data

Table 11-3 lists the configuration data of the DHCP client.

Table 11-3 Configuration Data of DHCP Client

Item	Data
Global DHCP	Enable
DHCP client	• VLAN ID: 100 • IP address mode: DHCP • Client ID: vlan100 • Class ID: c300 • Hostname: zxan
Uplink interface	gei_1/21/1

Steps

1. Enable the global DHCP function.

```

ZXAN(config)#ip dhcp enable

```

2. Configure the response packet type that is requested by the DHCP client.

```

ZXAN(config)#ip dhcp client broadcast-flag

```

3. In the interface configuration mode, configure the interface VLAN.

```

ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 100 tag
ZXAN(config-if)#exit

```

4. Enter the VLAN interface mode, enable the DHCP client in the VLAN.

```
ZXAN(config)#interface vlan 100
ZXAN(config-if-vlan100)#ip address dhcp
```

5. Configure the DHCP client parameters in the VLAN.

```
ZXAN(config-if-vlan100)#ip dhcp client client-id vlan100
ZXAN(config-if-vlan100)#ip dhcp client class-id c300
ZXAN(config-if-vlan100)#ip dhcp client hostname zxan
ZXAN(config-if-vlan100)#end
ZXAN#
```

6. (Optional) In the administrator mode, configure the DHCP client to obtain IP address again.

```
ZXAN#renew dhcp vlan 100
```

7. (Optional) In the administrator mode, configure the DHCP client to release addresses.

```
ZXAN#release dhcp vlan 100
```

– End of Steps –

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Chapter 12

Uplink Protection Configuration

The ZXA10 C300 adopts the dual uplink protection mechanism to ensure the service stability. When the physical connection between the ZXA10 C300 and upper-layer equipment is broken and the services are interrupted, the device will automatically switch the services to the standby line to restore the services quickly.

The ZXA10 C300 supports the following uplink protection modes:

- Link aggregation
- [UAPS](#)
- CTLA 1+1

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12.1 Configuring Link Aggregation

This section describes how to configure link aggregation to implement load balancing and protection on the uplink port.

Prerequisite

Before this operation, make sure that:

- Link aggregation has been configured on the opposite end.
- Port rate and VLAN properties on the opposite end are the same as that on the ZXA10 C300.

Context

The ZXA10 C300 supports two link aggregation modes.

- Static aggregation

In static aggregation mode, multiple physical ports are directly added to a trunk group to form a logical port. This mode is simple but not suitable for observing the status of the link aggregation port.

- Link Aggregation Control Protocol ([LACP](#))

In LACP mode, multiple physical ports are dynamically aggregated into a trunk group to form a logical port, thus to balance the load of the egress/ingress flow among the member ports. Aggregation is automatically generated to obtain the maximum bandwidth.

The ZXA10 C300 link aggregation function complies with the following rules:

- The link aggregation function supports up to eight trunk groups, and each trunk group contains up to eight member ports.
- The inter-interface card aggregation is supported, and the member ports can be located on any interface card.
- Member ports must operate in full duplex mode, and the working rates and VLAN attributes must be consistent.

The logical port formed by link aggregation on the ZXA10 C300 is called smartgroup. Smartgroup has the same default VLAN attributes as a common Ethernet port.

Steps

1. In global configuration mode, create a smartgroup.

```
ZXAN(config)#interface smartgroup1
ZXAN(config-smartgroup1)#
```

2. Configure load balancing mode.

```
ZXAN(config-smartgroup1)#smartgroup load-balance src-dst-mac
```



Note:

The ZXA10 C300 supports six load balancing modes that are based on source IP, destination IP, source/destination IPs, source MAC, destination MAC, and source/destination MACs respectively. The default mode is based on source/destination MACs.

3. Configure LACP mode.

```
ZXAN(config-smartgroup1)#smartgroup mode 802.3ad
```



Note:

The ZXA10 C300 supports two LACP modes:

- On: LACP static aggregation mode.
- 802.3ad: LACP dynamic negotiation mode

4. Configure the VLAN for the smartgroup.

```
ZXAN(config-smartgroup1)#switchport vlan 100 tag
```

```
ZXAN(config-smartgroup1)#exit
```

5. Configure the VLAN for uplink ports.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 100 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 100 tag
ZXAN(config-if)#exit
```

6. (Optional) Query VLAN properties of the smartgroup and uplink ports.

```
ZXAN(config)#show vlan port smartgroup1
```

PortMode	Pvid	CPvid	Tpid/mode	TLSStatus	TLSVlan	ProtEn	PrioEn
hybrid>=0	1	0	0x8100/PORT	disable	0	disable	disable

```
UntaggedVlan:
1
TaggedVlan:
100
```

```
ZXAN(config)#show vlan port gei_1/21/1
```

PortMode	Pvid	CPvid	Tpid/mode	TLSStatus	TLSVlan	ProtEn	PrioEn
hybrid>=0	1	0	0x8100/PORT	disable	0	disable	disable

```
UntaggedVlan:

TaggedVlan:
100
```

```
ZXAN(config)#show vlan port gei_1/21/2
```

PortMode	Pvid	CPvid	Tpid/mode	TLSStatus	TLSVlan	ProtEn	PrioEn
hybrid>=0	1	0	0x8100/PORT	disable	0	disable	disable

```
UntaggedVlan:

TaggedVlan:
100
```



Note:

Before adding ports to a smartgroup, you need to make sure that the VLAN configuration and switchport mode of member ports should be consistent with the that of the smartgroup.

7. In uplink interface configuration mode, add the port to the aggregation group and set port aggregation mode to active.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#smartgroup 1 mode active
```

**Note:**

The ZXA10 C300 supports the following three port aggregation modes:

- On: Static aggregation trunk. The two ends of the aggregation need to be set to the on mode.
- Active: LACP active negotiation mode
- Passive: LACP passive negotiation mode

It is recommended to set the port at one end to the active aggregation mode, and set the port at the other end to the passive aggregation mode, or set ports at both ends to the active aggregation mode.

8. Configure the timeout mode of the port.

```
ZXAN(config-if)#lacp timeout long
```

**Note:**

The ZXA10 C300 supports the following two LACP timeout modes:

- Long (default): The adjacent port sends a LACPDU packet every 30s.
- Short: The adjacent port sends a LACPDU packet every second.

The LACP timeout mode is valid only when the port is in active or passive aggregation mode.

9. Configure other port in the aggregation group.

```
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#smartgroup 1 mode active
ZXAN(config-if)#lacp timeout long
```

10. (Optional) Query the smartgroup status.

```
ZXAN(config-if)#show lacp internal
Smartgroup:1      Switch attribute:TRUE      Mode:802.3ad
Flag *--Loop is TRUE
Actor      Agg      LACPDUs  Port      Oper      Port  RX      Mux
Port      State      Interval Priority Key      State Machine      Machine
-----
gei_1/21/1  inactive  30      32768     0x100     0x45  port-disabled defaulted
```



```
gei_1/21/2  inactive  30      32768    0x100  0x45  port-disabled defaulted
```

– End of Steps –

12.2 Configuring UAPS

This section describes how to configure UAPS to implement automatic protection switchover of the uplink port.

Context

The ZXAN C300 supports the uplink automatic protection switching (UAPS) function. The system periodically checks the working status of the uplink port. When the system detects that the link of the working port is disconnected or the link is not available due to link quality degradation, it switches the services to the standby port automatically and without interrupting the services.

Steps

1. In global configuration mode, create a UAPS group.

```
ZXAN(config)#uaps-group 1
ZXAN(cfg-uaps-1)#
```

2. Configure the active/standby ports of the UAPS group.

```
ZXAN(cfg-uaps-1)#port master-port gei_1/21/1 slave-port gei_1/21/2
```



Note:

The configuration data on the active port and standby port should be consistent.

3. Enable active/standby auto-switch for the UAPS group.

```
ZXAN(cfg-uaps-1)#revertive enable
```

4. Configure the UAPS group protection time.

```
ZXAN(cfg-uaps-1)#protect-time 400
```

If the UAPS group implements switchover once, it does not implement switchover again during the protection time.

5. Configure the port attribute of the UAPS group.

```
ZXAN(cfg-uaps-1)#switch-type common-port
```

**Note:**

The ZXA10 C300 supports the following two port attributes:

- Common-port: common port
- Trunking-port: link aggregation port

6. (Optional) Query the UAPS group configuration.

```
ZXAN(cfg-uaps-1)#show uaps groupid 1

Revertive control      : enable
PortLight control     : disable
Protect-time          : 400s
Next-hop              : 0.0.0.0
Bfd next_hop          : 0.0.0.0
Link-type             : normal
Link-detect-retry     : 5
Link-detect-interval  : 3
Link status           : connected or NA
Bfd Link status       : connected or NA
Switch-type           : common port
Master ports status   : forwarding
                      gei_1/21/1 : down

Slave ports status    : block
                      gei_1/21/2 : down
```

– End of Steps –

12.3 Configuring CTLA 1+1 Protection

This section describes how to implement 1+1 protection on the uplink [TDM](#) optical interface.

Prerequisite

The CES service has been configured.

Context

CTLA 1+1 protection requires two CTLA cards. The working card is connected to the peer [SDH](#) as the working channel, and the protection card is connected to the peer SDH as the protection channel.

The work principle of CTLA 1+1 protection is as follows:

- In sending direction (from the CTLA card to the peer SDH device), when the working CTLA card and protection CTLA card send data to the peer SDH device simultaneously, the peer SDH device selects either channel to receive the data.
- In receiving direction (from the peer SDH device to the CTLA card), when the peer SDH device sends data to the CTLA cards in two channels, the switching and control card selects one channel to receive the data and discards the data from the other channel.

After the protection group is created, the services on the working card are synchronized to the protection card automatically.

Steps

1. In TDM uplink interface configuration mode of the working card, configure the uplink port VLAN.

```
ZXAN(config)#interface tdm-gei_1/12/1
ZXAN(config-if)#switchport vlan 1001 tag
```

2. In TDM uplink interface configuration mode of the protection card, configure the uplink port VLAN.

```
ZXAN(config)#interface tdm-gei_1/14/1
ZXAN(config-if)#switchport vlan 1001 tag
```

3. In CES configuration mode, configure a protection group.

```
ZXAN(config)#ces
ZXAN(config-ces)#sdhprot group aaa worksdhport tdm_1/12/1 protectsdhport tdm_1/14/1
1plus1
```

4. (Optional) Configure the protection switchover command.

```
ZXAN(config-ces)#sdhprot switch-command group aaa force w2p
```



Note:

By default, the ZXA10 C300 can switch between the ports automatically according to the optical port alarms (LOS, LOF, and MS-AIS). To restore automatic alarm switchover, you need to delete the current protection switchover command.

5. (Optional) Query the protection group configuration.

```
ZXAN(config-ces)#show ces sdhprot group prop
```

```
Name :          aaa
Protect interface:  tdm_1/14/1
Work interface :   tdm_1/12/1
Active interface:  work-channel
Type :            1+1
```

```
Mode:                non-revertive
Holdoff:              0
Wtr:                  0
```

– End of Steps –

Chapter 13

PON Protection Configuration

The ZXA10 C300 uses the active/standby switchover mechanism and PON port protection mechanism to guarantee stable operation of services. When the backbone fiber connection between the ZXA10 C300 and ONU is broken and the services are interrupted, the device will automatically switch the services to the standby PON port to restore the services quickly.

The ZXA10 C300 supports the following four types of PON protection:

- Type A
Type B is the backbone fiber redundancy protection. It backs up the backbone fiber between the PON port and splitter.
- Type B
Type B is the OLT-side redundancy protection. It backs up the OLT PON ports and the backbone fiber between the PON port and splitter. The splitter OLT-side has two input ports and two output ports. This protection mode can recover the service on the OLT side only.
- Type C
Type C is the OLT-side and ONU-side redundancy protection, It backs up the OLT PON port, ONU (dual optical modules), splitter, and all the fibers. In this mode, the fault at any point can be rectified via the active/standby switchover.
- Type D
Type D is the OLT-side and ONU-side redundancy protection, also known as full duplex protection. It backs up the OLT PON port, ONU (dual PON ports), splitter, and all the fibers. In this mode, the fault at any point can be rectified via the active/standby switchover.

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13.1 Configuring PON Port Protection

This section describes how to configure type-B PON protection to implement dual PON port backup protection.

Context

The ZXA10 C300 supports the following three PON port switchover modes:

- Force

The service is switched to the specified PON port unconditionally. The service can be switched from the protection port to the working port (p2w) or from the working port to the protection port (w2p) forcedly.

- Alarm-triggered (default)

- Manual

The service needs to be switched manually. Switchover in p2w or w2p mode is supported.

The priorities of the three modes in descending order are force, alarm-triggered, and then manual.

Configuration Data

Table 13-1 lists the PON protection configuration data.

Table 13-1 PON Protection Configuration Data

Item	Data
PON protection group	zte
Working PON port	1/5/1
Protection PON port	1/5/2
Protection type	Type B
Protection mode	revertive
Restoring time	120s

Steps

1. In PON configuration mode, clear the configuration data on the protection PON port.

```
ZXAN(config)#pon
ZXAN(config-pon)#clear gpon-olt_1/5/2
```

2. Create a PON protection group.

```
ZXAN(config-pon)#protection group zte workpon gpon-olt_1/5/1 protectpon gpon-olt_1/5/2 typeB
```

3. Configure the attributes of the PON protection group.

```
ZXAN(config-pon)#protection prop group zte mode revertive wtr 120
```

4. (Optional) Switch the PON port by forced.

```
ZXAN(config-pon)#protection switch-command group zte force w2p
```

5. (Optional) Query the PON protection group.

```
ZXAN(config-pon)#show protection group information zte
```

```
Name : zte
System model: self-contained
Peer host IP: N/A
Protection type : typeB
Work channel interface : gpon-olt_1/5/1
Protect channel interface: gpon-olt_1/5/2
Protection mode: revertive
Time to restore(s): 120
Active channel: protect-channel
Alarm request:
  Work channel: OLTSF
  Protect channel: OLTSF
External request: force-switch-to-protection-request
```

– End of Steps –

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Chapter 14

Access Security Configuration

Access security configuration can assure the safety of subscriber accounts, prevent illegal users from accessing the device, and illegal user-side packets from attacking the device.

The ZXN10 C300 supports the following access security features:

- Port identification
- MAC address anti-spoofing
- ARP anti-spoofing
- IP source guard
- Split horizon
- MFF
- ARP proxy

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14.1 Port Identification Configuration

The system provides the port identification mechanism to improve network security and prevent user accounts from being stolen. The system implements port identification through the following techniques:

- DHCPv4 Layer-2 Relay Agent
- PPPoE Intermediate Agent
- DHCPv6 Layer-2 Relay Agent
- NDP LIO

14.1.1 Configuring the Port Identification

Port-identification is to define the format and content of the Circuit ID (CID) and Remote ID (RID).

Context

When subscribers access the Internet in [PPPoE](#) Intermediate Agent, DHCPv4 Layer-2 Relay Agent, DHCPv6 Layer-2 Relay Agent, or [NDP](#) Line Identification Option (LIO) mode, the system uses the corresponding agent to locate port. The system sends the packets with port information to authentication servers to bind subscribers accounts and circuits.

Configuration Data

[Table 14-1](#) lists the configuration data of port identification.

Table 14-1 Configuration Data of Port Identification

Item	Data
Port-Identification global	● Master identifier type: access-node-name ● Access node name: ZXA10-C300 ● Slave identifier: ZTE
Port-Identification interface	● Interface: gpon-onu_1/5/1:1 ● Virtual port: 1 ● Remote ID status: enable ● Remote ID name: REMOTE-ID

Steps

1. Configure access node master identifier type.

```
ZXAN(config)#port-identification access-node-id-type access-node-name
```

2. Configure the access node name.

```
ZXAN(config)#port-identification access-node-name ZXA10-C300
```

3. Configure the access node slave identifier.

```
ZXAN(config)#port-identification access-node-slave-id ZTE
```

4. Enter GPON-ONU interface mode, and configure the remote ID field.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#port-identification sub-option remote-id enable vport 1
ZXAN(config-if)#port-identification sub-option remote-id name REMOTE-ID vport 1
ZXAN(config-if)#exit
```

5. (Optional) Query the global configuration of port identification.

```
ZXAN(config)#show port-identification global
access-node-name      : ZXA10-C300
access-node-id-type   : access-node-name
access-node-slave-id  : ZTE
rackno                : 1
shelfno               : 1
```

6. (Optional) Query the interface configuration of port identification.

```
ZXAN(config)#show port-identification port gpon-onu_1/5/1:1 vport 1
```

```

Port          : gpon-onu_1/5/1:1  vport 1
Cid-Format    : CHINA-TELECOM-PON
Rid-status    : Enable
Rid-name      : REMOTE-ID
Rid-Format    :
Access-Loop-Tag : REMOTE-ID

```

– End of Steps –

14.1.2 Configuring the DHCPv4 Layer-2 Relay Agent (DHCPv4L2RA)

When DHCPv4L2RA is enabled, the ZXA10 C300 adds DHCPv4L2RA Option 82 field to the upstream DHCP packets.

Context

The DHCPv4L2RA Option 82 field contains CID and RID, which includes the shelf number, slot number, and port number.

- Only when [DHCPv4L2RA](#) is enabled, the Option 82 field can be added/stripped to/from the DHCP packets.
- When DHCPv4L2RA is disabled, the ZXA10 C300 transparently transmits or directly forwards the DHCP packets without any processing.

The global DHCPv4L2RA function and VLAN DHCPv4L2RA function are mutually exclusive.

Configuration Data

[Table 14-2](#) lists the configuration data of the DHCPv4L2RA.

Table 14-2 Configuration Data of DHCPv4L2RA

Item	Data
DHCPv4L2RA global	Enable
DHCPv4L2RA interface	● Interface: gpon-onu_1/5/1:1 ● Virtual port: 1 ● DHCPv4L2RA status: enable ● Policy: trust and replace

Steps

1. Enable the global DHCPv4L2RA.

```
ZXAN(config)#dhcpv4-l2-relay-agent enable
```

2. In GPON-ONU interface mode, enable DHCPv4L2RA on the interface.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
```

```
ZXAN(config-if)#dhcpv4-l2-relay-agent enable vport 1
```

3. Configure the DHCPv4L2RA policy on the interface.

```
ZXAN(config-if)#dhcpv4-l2-relay-agent trust true replace vport 1
```

4. (Optional) Query either the global DHCPv4L2RA status or DHCPv4L2RA status on vlan.

```
ZXAN(config)#show dhcpv4-l2-relay-agent global
dhcpv4-l2-relay-agent status : enable
```

5. (Optional) Query the interface DHCPv4L2RA configuration.

```
ZXAN(config)#show dhcpv4-l2-relay-agent port gpon-onu_1/5/1:1 vport 1
Onu          Vport  dhcpv4-l2-relay-agent status  Trust    Policy
gpon-onu_1/5/1:1  1      enable                                true     replace
```

– End of Steps –

Result

When the subscriber sends DHCP protocol packets, the system adds the following fields to the packets:

```
Circuit-id:  ZXA10-C300/ZTE eth 5/1/1/0/1:10
Remote-id :  REMOTE-ID
//where, 10 is the original user VLAN.
```

14.1.3 Configuring the PPPoE Intermediate Agent (PPPoE-IA)

When PPPoE-IA is enabled, the ZXA10 C300 adds port information to the upstream PPPoE-IA packets.

Context

When users access the Internet in [PPPoE](#) mode, the ZXA10 C300 uses PPPoE-IA to locate port. The system carries the user information in the PPPoE-IA discovery packets to report to the [BRAS](#) for user authentication, and thus binding the user account and circuit.

The global PPPoE-IA function and VLAN PPPoE-IA function are mutually exclusive.

Configuration Data

[Table 14-3](#) lists the configuration data of PPPoE-IA.

Table 14-3 Configuration Data of PPPoE-IA

Item	Data
PPPoE-IA global	Enable
PPPoE-IA interface	● Interface: gpon-onu_1/5/1:1 ● Virtual port: 1 ● PPPoE-IA status: enable ● Policy: trust and replace

Steps

1. Enable the global PPPoE-IA.

```
ZXAN(config)#pppoe-intermediate-agent enable
```

2. In GPON-ONU interface mode, enable PPPoE-IA on the interface.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#pppoe-intermediate-agent enable vport 1
```

3. Configure the PPPoE-IA policy on the interface.

```
ZXAN(config-if)#pppoe-intermediate-agent trust true replace vport 1
```

4. (Optional) Query either the global DHCPv4L2RA status or DHCPv4L2RA status on VLAN.

```
ZXAN(config)#show pppoe-intermediate-agent global
pppoe-intermediate-agent status : enable
```

5. (Optional) Query the interface PPPoE-IA configuration.

```
ZXAN(config)#show pppoe-intermediate-agent port gpon-onu_1/5/1:1 vport 1
```

Onu	Vport	Pppoe-intermediate-agent status	Trust	Policy
gpon-onu_1/5/1:1	1	enable	true	replace

– End of Steps –

Result

When the subscriber sends PPPoE protocol packets, the system adds the following fields to the packets:

```
Circuit-id: ZXA10-C300/ZTE eth 5/1/1/0/1:10
Remote-id : REMOTE-ID
//where, 10 is original user VLAN.
```

14.1.4 Configuring the DHCPv6 Layer-2 Relay Agent (DHCPv6L2RA)

When DHCPv6L2RA is enabled, the ZXA10 C300 adds DHCPv6L2RA Option 18 and Option 37 fields to the upstream DHCP packets.

Context

The option 18 field includes CID , and the option 37 field includes RID, which provides the physical information such as the shelf number, slot number, and port number.

- Only when DHCPv6L2RA is enabled, the option 18 field and option 37 field can be added/stripped to/from DHCPv6 packets. For option 37, the remote ID status should be enabled and remote ID name should be configured in addition.
- When DHCPv6L2RA is disabled, the system transparently transmits or directly forwards DHCPv6 packets without any processing.

The global DHCPv6L2RA function and VLAN DHCPv6L2RA function are mutually exclusive.

Configuration Data

Table 14-4 lists the configuration data of DHCPv6L2RA.

Table 14-4 Configuration Data of DHCPv6L2RA

Item	Data
DHCPv6L2RA VLAN	• VLAN: 100 • Status: enable
DHCPv6L2RA interface	• Interface: gpon-onu_1/5/1:1 • Virtual port: 1 • DHCPv6L2RA status: enable

Steps

1. Enable DHCPv6L2RA on VLAN.

```
ZXAN(config)#dhcpv6-l2-relay-agent vlan 100 enable
//VLAN 100 is the service vlan after vlan-translation.
```

2. In GPON-ONU interface mode, configure DHCPv6L2RA on the virtual port.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#dhcpv6-l2-relay-agent enable vport 1
ZXAN(config-if)#exit
```

3. (Optional) Query either the global DHCPv6L2RA status or DHCPv6L2RA status on VLAN (mutually exclusive).

```
ZXAN(config)#show dhcpv6-l2-relay-agent vlan all
vlan total   : 1
vlan list    : 100
```

4. (Optional) Query the interface configuration of DHCPv6L2RA.

```
ZXAN(config)#show dhcpv6-l2-relay-agent port gpon-onu_1/5/1:1 vport 1
Port                Pvc dhcpv6-l2-relay-agent status Trust Policy
gpon-onu_1/5/1:1    1   enable                                false add
```

– End of Steps –

Result

When the subscriber sends DHCPv6 protocol packets, the system adds the following fields to the packets:

```
Option 18: ZXA10-C300/ZTE eth 5/1/1/0/1:10
Option 37: REMOTE-ID
//where, 10 is original user VLAN.
```

14.1.5 Configuring the NDP Line Identification Option (NDP-LIO)

When NDP-LIO is enabled, the ZXA10 C300 adds LIO field to the upstream NDP packets.

Context

The LIO field includes CID, and provides the information such as the shelf number, slot number, and port number.

- Only when NDP-LIO is enabled, the LIO field can be added/stripped to/from the NDP packets.
- When NDP-LIO is disabled, the system transparently transmits or directly forwards the NDP packets without any processing.

The global NDP-LIO function and VLAN NDP-LIO function are mutually exclusive.

Configuration Data

Table 14-5 lists the configuration data of NDP-LIO.

Table 14-5 Configuration Data of NDP-LIO

Item	Data
NDP-LIO VLAN	● VLAN: 100 ● Status: enable
NDP-LIO interface	● Interface: gpon-onu_1/5/1:1 ● Virtual port: 1 ● NDP-LIO status: enable ● Policy: trust and replace

Steps

1. Enable NDP-LIO on VLAN.

```
ZXAN(config)#ndp-lio vlan 100 enable
//VLAN 100 is the service vlan after vlan-translation.
```

2. In GPON-ONU interface mode, configure NDP-LIO on the virtual port.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#ndp-lio enable vport 1
ZXAN(config-if)#ndp-lio trust true replace vport 1
ZXAN(config-if)#exit
```

3. (Optional) Query either the global NDP-LIO status or NDP-LIO status on vlan.

```
ZXAN(config)#show ndp-lio vlan all
vlan total   : 1
vlan list    : 100
```

4. (Optional) Query the interface configuration of NDP-LIO.

```
ZXAN(config)#show ndp-lio port gpon-onu_1/5/1:1 vport 1
Onu          Vport    ndp-lio status   Trust   Policy
gpon-onu_1/5/1:1  1        enable          true    replace
```

– End of Steps –

Result

When the subscriber sends NDP protocol packets, the system adds the following LIO field to the packets:

```
Circuit-id: ZXA10-C300/ZTE eth 5/1/1/0/1:10  
//where, 10 is original user VLAN.
```

14.2 MAC Address Anti-Spoofing Configuration

The ZXA10 C300 supports the MAC address anti-spoofing function to prevent malicious MAC address spoofing, which affects the network security.

The ZXA10 C300 MAC address anti-spoofing function has the following features:

- This function constrains the user port that learns the MAC address. When one MAC address is learnt by one user port, the address cannot be learnt by other user ports. Thus, the same MAC address cannot float between different ports.
- Once a user port is detected trying MAC address spoofing, an alarm message including the port and MAC address will be reported.
- This function supports uplink port protection. A user port MAC address can float to an uplink port, whereas an uplink port address cannot float to a user port. A MAC address can float between uplink ports, thus to protect the gateway MAC address of the uplink ports.

14.2.1 Configuring the User Port MAC Address Anti-Spoofing

User-port MAC address anti-spoofing prevents malicious MAC address spoofing between user ports.

Context

The user-port MAC address anti-spoofing has the following features:

- When one MAC address is learnt by one user port, the address cannot be learnt by other user ports.
- Once there is a MAC move event at the first time, the system will generate a notification including the MAC address, VLAN, move-to-port and move-from-port.
- The notification report interval of the same MAC move events can be configured.

Steps

1. Enable global MAC address anti-spoofing function.

```
ZXAN(config)#security mac-anti-spoofing enable
```

2. Enable MAC move notification control.

```
ZXAN(config)#security mac-move-report enable
```

3. (Optional) Configure the notification report interval of the same MAC move log.

```
ZXAN(config)#security mac-move-report interval 30
```


4. (Optional) Query the configuration of MAC address anti-spoofing.

```
ZXAN(config)#show security mac-anti-spoofing configuration
mac-move-report :enable
mac-move-report interval:30[minutes]
mac-anti-spoofing :enable
uplink-protect :disable
```

5. (Optional) Query the MAC move log.

```
ZXAN#show security mac-move-log
Flag *--macMove is forbidden by system.
the total mac-move-log num:2
-----
mac-address      vlan  cfgMacProtect  moveToPort      moveToIfId      moveCount
index trapFlag detector queryPort  moveFromPort    moveFromIfId    trapCount
-----
0002.0304.0506 100  UNNEED  inner-port_1/12/1  unknown(0)      1

1  SENDED  MP  UNNEED  inner-port_1/5/1  unknown(0)      1
-----
0002.0304.0507 100  UNNEED  inner-port_1/12/2  unknown(0)      1

2  *SENDED  MP  UNNEED  inner-port_1/5/1  unknown(0)      1
```

– End of Steps –

14.2.2 Configuring the Service Gateway MAC Anti-Spoofing

Service gateway MAC address anti-spoofing prevents malicious MAC address spoofing between user ports and permits MAC address learning between uplink ports.

Context

The ZX10 C300 supports the following features by service gateway MAC anti-spoofing:

- A MAC address learnt by a user port can be learnt by an uplink port as well.
- The same MAC address cannot be learnt by two user ports.
- The same MAC address can be learnt by multiple uplink ports.

Steps

1. Enable global MAC address anti-spoofing function.

```
ZXAN(config)#security mac-anti-spoofing enable
```

2. Enable MAC address anti-spoofing function with uplink protection.

```
ZXAN(config)#security mac-anti-spoofing uplink-protect enable
```

3. (Optional) Query the configuration of MAC address anti-spoofing.

```
ZXAN(config)#show security mac-anti-spoofing configuration
```

```

mac-move-report :enable
mac-move-report interval:30[minutes]
mac-anti-spoofing :enable
uplink-protect :enable

```

4. (Optional) Query the MAC move log.

```

ZXAN#show security mac-move-log
Flag *--macMove is forbidden by system.
the total mac-move-log num:2
-----
mac-address      vlan  cfgMacProtect  moveToPort      moveToIfId      moveCount
index trapFlag  detector queryPort      moveFromPort    moveFromIfId trapCount
-----
0002.0304.0506 100  UNNEED  inner-port_1/12/1  unknown(0)      1

1   SENDED    MP  UNNEED  inner-port_1/5/1  unknown(0)      1
-----
0002.0304.0507 100  UNNEED  inner-port_1/12/2  unknown(0)      1

2   *SENDED   MP  UNNEED  inner-port_1/5/1  unknown(0)      1

```

– End of Steps –

14.3 Configuring the ARP Anti-Spoofing

The ARP anti-spoofing prevents the ARP spoofing on user side.

Context

The ZXA10 C300 supports user-side [ARP](#) anti-spoofing function, which is implemented based on the following ARP entries:

- The ARP entries inserted by the [DHCP](#) module
- The ARP entries of DHCP snooping static binding item configured by the IP source Guard module

ARP anti-spoofing function is based on both VLAN and service port. Only when the ARP anti-spoofing functions on both VLAN and service port are enabled, the system can implement ARP anti-spoofing on ARP packets with the specific VLAN tag.

When receiving an ARP packet, the ZXA10 C300 compares the packet with the known ARP entries. If the source IP address of the received ARP packet and the [VLAN](#) exist in the ARP table, the ZXA10 C300 checks whether the [MAC](#) addresses are the same. If they are different, the ZXA10 C300 considers the packet as an ARP spoofing behavior and discards it.

The [ARP](#) anti-spoofing function can be configured with up to 256 [VLANs](#).

Configuration Data

Table 14-6 lists the configuration data of the ARP anti-spoofing.

Table 14-6 Configuration Data of ARP Anti-Spoofing

Item	Data
ARP anti-spoofing status	Enable
VLAN ID	200
Direction	User port

Steps

1. Enable the global ARP anti-spoofing function.

```
ZXAN(config)#ip-service arp-anti-spoofing enable
```

2. Configure the ARP anti-spoofing function on the VLAN.

```
ZXAN(config)#ip-service arp-anti-spoofing vlan 200 direction user-port
```

3. (Optional) Query the configuration of ARP anti-spoofing.

```
ZXAN(config)#show ip-service arp-anti-spoofing
```

```
Arp Anti-Spoofing status: Enabled
```

```
vlan          direction
```

```
200           user-port
```

– End of Steps –

14.4 Configuring the Split Horizon

When user communication control is enabled, only subscribers in the specific VLAN and SVLAN can communicate with each other.

Context

The ZXA10 C300 supports the following split horizon features:

- Subscriber separation/intercommunication
- Subscriber intercommunication based on SVLAN and CVLAN

Steps

1. Enable the user communication control function.

```
ZXAN(config)#security user-communication control enable
```

2. Configure the intercommunication VLANs.

```
ZXAN(config)#security user-communication svlan 300 cvlan 200
```

3. (Optional) Query the configuration of intercommunication VLANs.

```
ZXAN(config)#show security user-communication
```

```
usercommunication item:
```

Svlan Cvlan

300 200

– End of Steps –

14.5 Configuring the IP Source Guard

The IP source guard based on the service port prevents illegal users from accessing the Internet.

Context

IP source guard supports IP/MAC anti-spoofing and access security management based on the service port.

The ZXA10 C300 supports IP source guard on both IPv4 and IPv6.

- The legal IPv4 subscribers are managed through either the DHCP snooping table or static IP addresses.
- The legal IPv6 subscribers are managed through either the NDP snooping/DHCPv6 snooping, or static IP addresses.

Configuration Data

Table 14-7 lists the configuration data of the IP source guard.

Table 14-7 Configuration Data of IP Source Guard

Item	Data
Global IP source guard	Enable
Interface IP source guard	● Interface: gpon-onu_1/5/1:2 (virtual port 1) ● Service port: 1 ● IP source guard: enable
Maximum IP address number	● IPv4: 2 ● IPv6: 4
IPv4 DHCP snooping static binding	IP address: 1.1.1.1
IPv6 DHCP snooping static binding	● IPv6 address: 2001::ff01 ● IPv6-mask: 128 ● MAC address: 2365.1498.2369

Steps

1. Enable the IP source guard function.

```
ZXAN(config)#ip-source-guard enable
```

2. In GPON-ONU interface mode, configure the maximum IPv4 and IPv6 subscriber binding entries on the ONU interface.

```
ZXAN(config)#interface gpon-onu_1/5/1:2
ZXAN(config-if)#ip-source-guard ip-limit ipv4 2 ipv6 4
```

3. Configure the service port VLAN.

```
ZXAN(config-if)#service-port 1 vport 1 user-vlan 100 vlan 200
```

4. Enable the IP source guard on the service port.

```
ZXAN(config-if)#ip-source-guard enable sport 1
```

5. Configure the IPv4 DHCP snooping static binding.

```
ZXAN(config-if)#ip dhcp snooping binding 1.1.1.2 sport 1
```

6. Configure the IPv6 DHCP snooping static binding.

```
ZXAN(config-if)#ipv6 dhcp snooping binding mac-address 2365.1498.2369 2001::ff01
ipv6-mask 128 sport 1
```

7. (Optional) Query the IP source guard status.

```
ZXAN(config)#show ip-source-guard
global ip-source-guard status :enable
```

8. (Optional) Query the IPv4 DHCP snooping static binding.

```
ZXAN(config-if)#show ip dhcp snooping static port gpon-onu_1/5/1:2
```

Port	Sport	IP-addr	MAC-addr
gpon-onu_1/5/1:2	1	1.1.1.2	0000.0000.0000

9. (Optional) Query the IPv6 DHCP snooping static binding.

```
ZXAN(config-if)#show ipv6 dhcp snooping static port gpon-onu_1/5/1:2
```

Port	Sport	IPv6-addr	Mask	MAC-addr
gpon-onu_1/5/1:2	1	2001::ff01	128	2365.1498.2369

– End of Steps –

14.6 Configuring MFF

This section describes how to configure MFF to implement layer-3 interworking between subscribers and prevent malicious attacks.

Context

The MAC forced forwarding ([MFF](#)) function prohibits interworking between two subscribers in the same subnet and forcedly forwards the upstream flows of the subscribers to the gateway. The gateway then forwards the flows to implement layer-3 interworking between subscribers. The gateway can analyze traffic between subscribers to prevent malicious attacks.

Steps

1. Enable MFF.

```
ZXAN(config)#ip-service mac-forced-forwarding enable
```

2. Configure the gateway IP address of the MFF VLAN.

```
ZXAN(config)#ip-service mac-forced-forwarding vlan 100 gateway 10.1.1.1
```

3. (Optional) Query the global MFF configuration.

```
ZXAN(config)#show ip-service mac-forced-forwarding
Mac-Forced Forwarding status:Enabled.
```

4. (Optional) Query the gateway information of the MFF VLAN.

```
ZXAN(config)#show ip-service gateway
```

Vlan	Gateway IP	Gateway MAC	Type
100	10.1.1.1	00d0.d0c7.0561	dynamic-600s

– End of Steps –

14.7 Configuring ARP Proxy

This section describes how to configure ARP proxy to implement interworking between subscribers under the same PON port.

Context

By default, the ZXA10 C300 services on different ONUs under the same PON port are isolated. When a service, such as VoIP, requires interworking between the subscribers under the same PON port, the ZXA10 C300 uses the ARP proxy function to achieve interworking between the subscribers in the same VLAN and same network segment under the same PON port.

Steps

1. In layer-3 VLAN interface mode, configure the layer-3 interface IP address.

```
ZXAN(config)#interface vlan 100
ZXAN(config-if-vlan100)#ip address 10.1.1.1 255.255.255.0
```



Note:

The VLAN is the user VLAN. The IP address should be in the same network segment as that of the interworking device.

2. Enable ARP proxy on the layer-3 interface.

```
ZXAN(config-if-vlan100)#ip proxy-arp
```

– End of Steps –

Chapter 15

System Security Configuration

System security configuration can prevent illegal network-side packets from attacking devices, thus to ensure stable running of the devices.

The ZXN10 C300 supports the following system security features:

- Secure Shell ([SSH](#))
- Terminal Access Controller Access-Control ([TACACS+](#))
- Remote Authentication Dial In User Service ([RADIUS](#))
- Management [ACL](#)
- Control panel safety

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15.1 Configuring SSH

[SSH](#) can replace Telnet to implement secure remote login.

Prerequisite

The SSH client software has been installed.

Context

SSH can encrypt the data during transmission to prevent the "intermediate" attacks. In addition, SSH compresses the data to be transmitted, thus increasing the transmission speed. When the SSH client communicates with the SSH server, the user name and password are encrypted, thus to prevent the password from being intercepted.

The ZXN10 C300 supports the SSH server function.

Steps

1. In global configuration mode, enable SSH server.

```
ZXAN(config)#ssh server enable
```

2. Configure the SSH server protocol version.

```
ZXAN(config)#ssh server version 2
```

3. Configure the SSH server authentication mode.

```
ZXAN(config)#ssh server authentication mode local
```

4. Configure the SSH server authentication type.

```
ZXAN(config)#ssh server authentication type pap
```

5. (Optional) Query the SSH configuration.

```
ZXAN(config)#show ssh
```

SSH configuration:

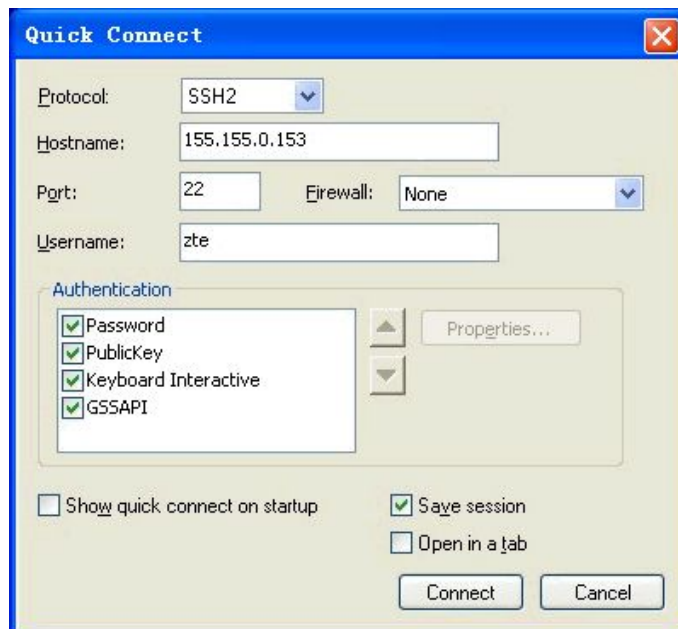
```
SSH enable-flag configuration : enable
SSH version                  : ver2.0
SSH only configuration       : disable
SSH init server key          : not initialized
SSH auth mode                : local
SSH auth type                : pap
```

– End of Steps –

Follow-Up Action

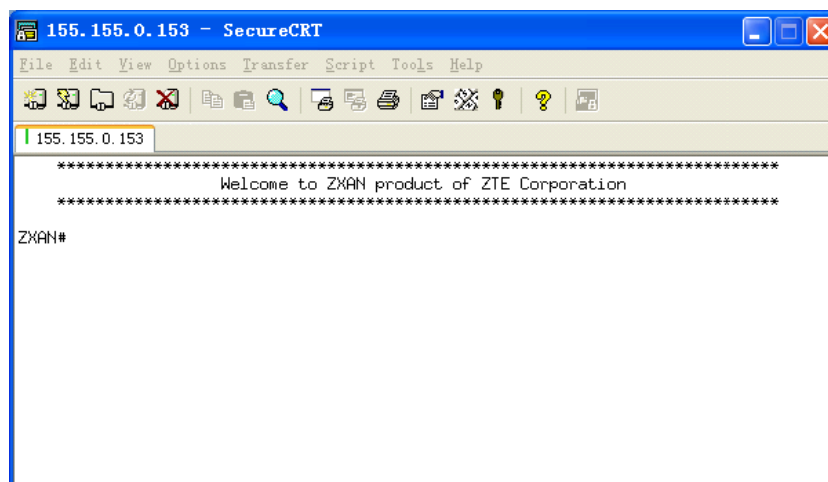
1. In a Windows OS, run the SSH client software (SecureCRT, for example). The **Quick Connect** dialog box opens, as shown in [Figure 15-1](#).

Figure 15-1 Quick Connect Dialog Box



2. In the **Quick Connect** dialog box, select **Protocol**, type **Hostname** and **Username**, and then click **Connect**. The login window opens, as shown in [Figure 15-2](#).

Figure 15-2 SSH Login Window

**Note:**

The hostname is the in-band/out-of-band NM IP address of the ZXA10 C300.

15.2 Configuring TACACS+

TACACS+ ensures data safety of the ZXA10 C300 by implementing safety authentication and authorization for remote subscribers who access the ZXA10 C300.

Context

TACACS+ supports two login modes:

- Telnet
- SSH

Steps

1. Configure telnet user authentication type.

```
ZXAN(config)#user-authentication-type aaa
```

2. Configure telnet user authorization type.

```
ZXAN(config)#user-authorization-type aaa
```

3. Configure **SSH** server authentication mode.

```
ZXAN(config)#ssh server authentication mode aaa
```

4. Enable TACACS+.

```
ZXAN(config)#tacacs enable
```

5. Configure TACACS+ server.

- In-band NM mode (The ZXA10 C300 is connected to the TACACS+ server through the in-band NM channel.)

```
ZXAN(config)#tacacs-server host 1.2.2.3
```

- Out-of-band NM mode (The ZXA10 C300 is connected to the TACACS+ server through the out-of-band NM channel.)

```
ZXAN(config)#tacacs-server host vrf mng 1.2.2.3
```

6. Configure TACACS+ server group.

- In-band NM mode

```
ZXAN(config)#aaa group-server tacacs+ zte
```

```
ZXAN(config-sg)#server 1.2.2.3
```

```
ZXAN(config-sg)#exit
```

- Out-band NM mode

```
ZXAN(config)#aaa group-server tacacs+ zte
```

```
ZXAN(config-sg)#server vrf mng 1.2.2.3
```

```
ZXAN(config-sg)#exit
```

7. Configure the authorization, authentication, and accounting group.

```
ZXAN(config)#aaa authentication login default group zte
```

```
ZXAN(config)#aaa authorization exec default group zte
```

```
ZXAN(config)#aaa accounting commands 10 default stop-only group zte
```

– End of Steps –

15.3 Configuring RADIUS

RADIUS ensures data safety of the ZXA10 C300 by implementing safety authentication and authorization for remote users who access the ZXA10 C300.

Context

RADIUS supports two login modes:

- Telnet
- SSH

Steps

1. Configure telnet user authentication type.

```
ZXAN(config)#user-authentication-type aaa
```

2. Configure telnet user authorization type.

```
ZXAN(config)#user-authorization-type aaa
```

3. Configure SSH server authentication mode.

```
ZXAN(config)#ssh server authentication mode aaa
```

4. Configure the RADIUS server group.

```
ZXAN(config)#aaa group-server radius-authen 1
ZXAN(config-authgrp-1)#
```

5. Configure the RADIUS server.

```
ZXAN(config-authgrp-1)#server 1 2.2.2.3 key zteRad
```

6. (Optional) Configure the route.

```
ZXAN(config-authgrp-1)#ip mng
ZXAN(config-authgrp-1)#exit
```



Note:

When the ZXA10 C300 is connected to the RADIUS server through the in-band NM channel, you can skip this step.

7. Configure the authentication group.

```
ZXAN(config)#aaa authentication login default rds-group 1
```

8. Configure the authorization group.

```
ZXAN(config)#aaa authorization exec default rds-group 1
```

– End of Steps –

15.4 Configuring Management ACL

After you configure the management ACL, accessing the ZXA10 C300 in Telnet/SNMP mode can be restricted.

Context

The management ACL is a standard ACL, which controls the source IP address of the received IP packets. The management ACL restricts users' access to the ZXA10 C300 NM module.

Steps

1. Create a standard ACL.

```
ZXAN(config)#acl standard number 10
ZXAN(config-std-acl)#
```

2. Configure the ACL rules.

```
ZXAN(config-std-acl)#rule 1 deny 1.1.1.10 0.0.0.0
ZXAN(config-std-acl)#rule 2 permit 1.1.1.0 0.0.0.255
ZXAN(config-std-acl)#exit
```

3. Apply the ACL.

```
ZXAN(config)#line telnet access-class 10
```

– End of Steps –

15.5 Configuring Control Panel Safety

After you configure control panel safety, the ZXA10 C300 can limit the protocol packet rate and prevent DoS packet attacks.

Context

Control panel safety includes the following three functions:

- Rate limit of protocol packets

Different rate limits are set for packets of different protocols.

- Rate limit of CPU queue packets

Packet rate limits for eight queues of the exchange chip can be set separately. When the packet rate of a certain queue is too high, a corresponding rate limit can be set to reduce the impact on the CPU.

- Black list

When the number of packets sent to the CPU by a user in one polling period (5s by default) exceeds the threshold, the ZXA10 C300 considers that the user implements a DoS attack on the NE and includes the user into the black list. Then packets sent by the user will be dropped till the user stops the attack.

Steps

1. Enter control panel mode, and configure packet limit.

```
ZXAN(config)#control-panel
ZXAN(control-panel)#packet-limit dhcp 20
ZXAN(control-panel)#packet-limit arp 50
```

2. Configure the rate limit of CPU queue packets.

```
ZXAN(control-panel)#cpu queue 1 25
```

3. Enable anti-DoS.

```
ZXAN(control-panel)#anti-dos enable
```

4. Enable the anti-DoS drop function.

```
ZXAN(control-panel)#anti-dos drop enable
```

5. Configure the threshold of the black list.

```
ZXAN(control-panel)#anti-dos limit-number 20
```

6. Configure the polling time of the black list.

```
ZXAN(control-panel)#anti-dos blocking-time 10
```

7. (Optional) Query the black list.

```
ZXAN(control-panel)#show control-panel anti-dos black-table
-----MP BLACK TABLE-----
mac-address      vlan      port      onu-sn      state      PktIn      Drop

-----NP BLACK-TABLE-----
mac-address      port      onu-sn      state      PktIn      Drop
-----
```

– End of Steps –

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Chapter 16

MPLS Service Configuration

Multiprotocol Label Switching ([MPLS](#)) is originally developed to increase forward speed of routers. Having prominent advantages in the applications of traffic engineering and [VPN](#), the MPLS protocol plays an important role in [IP](#) networks.

The key point of the MPLS protocol is the concept of labels, which are short, easy-to-handle, and without topology information. Labels contain information that only affects in a certain area.

Initially, MPLS focus on IPv4. However, the key techniques of MPLS can be expanded to multiple protocols, such as IPv6, [IPX](#), Apple-talk, and [SNA](#). MPLS has no limitation on the link layer techniques and can transport network groups over any media between network entities.

MPLS packets are forwarded based on labels. When IP packets enter an MPLS network, the [PE](#) routers on the MPLS ingress analyze the contents of the packets and select the proper labels for the packets. The packets are forwarded by nodes in the MPLS network according to these short labels. When the packets leave the MPLS network, the PE routers on the MPLS egress remove the labels.

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16.1 MPLS Configuration

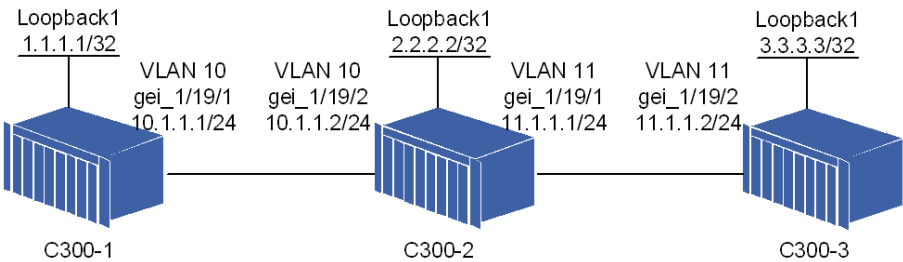
16.1.1 Configuring the Basic MPLS Function

In the basic MPLS application, the MPLS forwarding is based on the LDP sessions between routers.

Networking Diagram

[Figure 16-1](#) shows the networking diagram of the basic MPLS function.

Figure 16-1 Networking Diagram of Basic MPLS Function



Three ZXA10 C300s are in the same OSPF area and interconnected by the OSPF protocol.

Configuration Data

Table 16-1 lists the configuration data of the basic MPLS function.

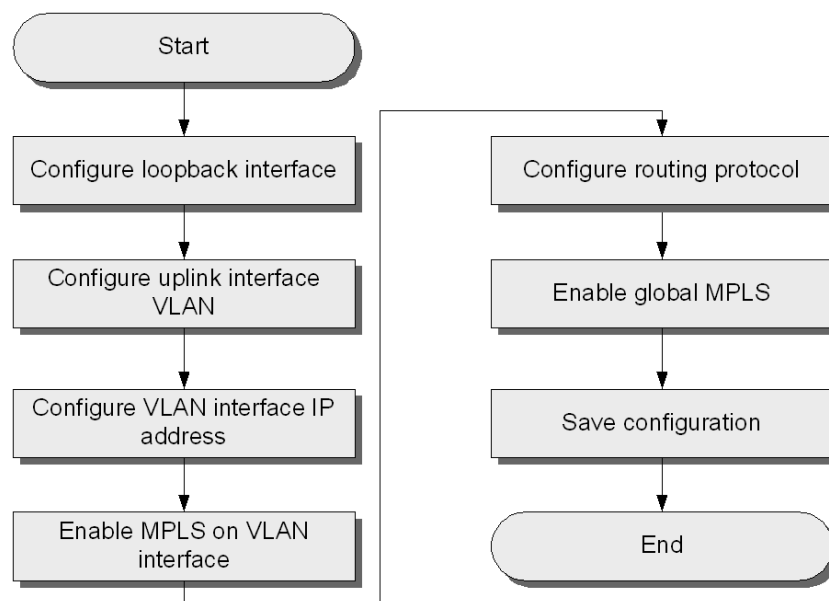
Table 16-1 Configuration Data of Basic MPLS Function

Item		Data
C300-1	Loopback interface	- ID: 1 - IP address: 1.1.1.1/32
	L3 interface	- VLAN ID: 10 - Interface: gei_1/21/1 - IP address: 10.1.1.1/24
	Reserved VPN VLAN	VLAN ID: 4090
C300-2	Loopback interface	- ID: 1 - IP address: 2.2.2.2/32
	L3 interface 1	- VLAN ID: 11 - Interface: gei_1/21/1 - IP address: 11.1.1.1/24
	L3 interface 2	- VLAN ID: 10 - Interface: gei_1/21/2 - IP address: 10.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
C300-3	Loopback interface	- ID: 1 - IP address: 3.3.3.3/32
	L3 interface	- VLAN ID: 11 - Interface: gei_1/21/2 - IP address: 11.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090

Configuration Flow

Figure 16-2 shows the configuration flowchart of the basic MPLS function.

Figure 16-2 Configuration Flowchart of Basic MPLS Function



Steps

- Configuration on C300-1.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 1.1.1.1 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.1 255.255.255.0
```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 1.1.1.1 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#vlan-reserve 4090 vpn
```

```
ZXAN(config)#mpls ip
```

7. Save configuration data.

- Configuration on C300-2.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 2.2.2.2 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.1 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 2.2.2.2 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#mpls ip
```

7. Save configuration data.

- Configuration on C300-3.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 3.3.3.3 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
XAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.2 255.255.255.0
```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 3.3.3.3 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#mpls ip
```

7. Save configuration data.

Result

Execute the command **ping mpls ipv4 3.3.3.3 32 repeat 1000** on C300-1. The ping is successful.

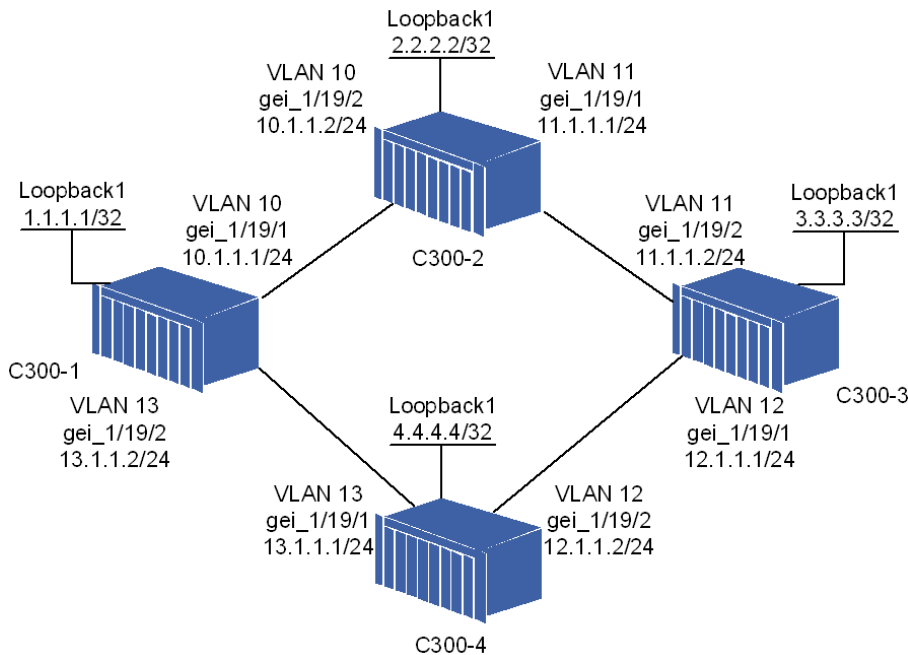
16.1.2 Configuring the Static LSP

The static LSP is used in MPLS L2VPN.

Networking Diagram

Figure 16-3 shows the networking diagram of the static Label Switched Path (LSP) configuration.

Figure 16-3 Networking Diagram of Static LSP Configuration



The Label Distribution Protocol (LDP) is running on the four ZXA10 C300s. Configure two static LSPs between C300-1 and C300-3.

The static LSP from C300-1 to C300-3 is: C300-1 > C300-2 > C300-3.

The static LSP from C300-3 to C300-1 is: C300-3 > C300-4 > C300-1.

Configuration Data

Table 16-2 lists the configuration data of the static LSP.

Table 16-2 Configuration Data of the Static LSP

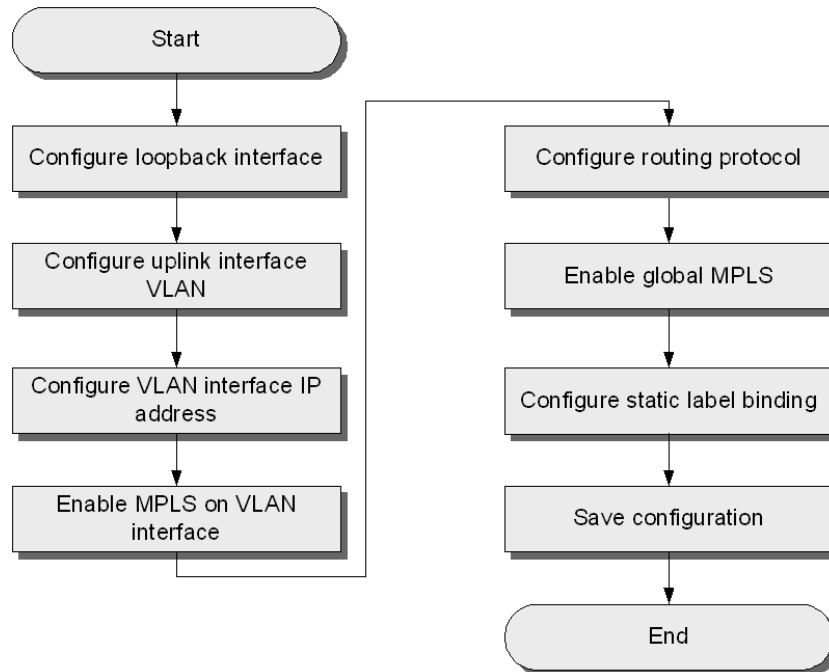
Item		Data
C300-1	Loopback interface	- ID: 1 - IP address: 1.1.1.1/32
	L3 interface 1	- VLAN ID: 10 - Interface: gei_1/21/1 - IP address: 10.1.1.1/24
	L3 interface 2	- VLAN ID: 13 - Interface: gei_1/21/2 - IP address: 13.1.1.12/24
	Static LSP (C300-1 to C300-3)	- Destination IP address: 3.3.3.3/32 - Next hop IP address: 10.1.1.2/24 - Out label: 1040001
	Static LSP (C300-3 to C300-1)	- Destination IP address: 1.1.1.1/32 - In label: 1041004

Item		Data
C300-2	Loopback interface	● ID: 1 ● IP address: 2.2.2.2/32
	L3 interface 1	● VLAN ID: 11 ● Interface: gei_1/21/1 ● IP address: 11.1.1.1/24
	L3 interface 2	● VLAN ID: 10 ● Interface: gei_1/21/2 ● IP address: 10.1.1.2/24
	Static LSP (C300-1 to C300-3)	● Destination IP address: 3.3.3.3/32 ● Next hop IP address: 11.1.1.2/24 ● Out label: 1040002 ● In label: 1040001
C300-3	Loopback interface	● ID: 1 ● IP address: 3.3.3.3/32
	L3 interface 1	● VLAN ID: 12 ● Interface: gei_1/21/1 ● IP address: 12.1.1.1/24
	L3 interface 2	● VLAN ID: 11 ● Interface: gei_1/21/2 ● IP address: 11.1.1.2/24
	Static LSP (C300-1 to C300-3)	● Destination IP address: 3.3.3.3/32 ● In label: 1040002
	Static LSP (C300-3 to C300-1)	● Destination IP address: 1.1.1.1/32 ● Next hop IP address: 10.1.1.2/24 ● Out label: 1041005
C300-4	Loopback interface	● ID: 1 ● IP address: 4.4.4.4/32
	L3 interface 1	● VLAN ID: 13 ● Interface: gei_1/21/1 ● IP address: 13.1.1.1/24
	L3 interface 2	● VLAN ID: 12 ● Interface: gei_1/21/2 ● IP address: 12.1.1.2/24
	Static LSP (C300-3 to C300-1)	● Destination IP address: 1.1.1.1/32 ● Next hop IP address: 13.1.1.2/24 ● Out label: 1040004 ● In label: 1040005

Configuration Flow

Figure 16-4 shows the configuration flowchart of the static LSP.

Figure 16-4 Configuration Flowchart of Static LSP



Steps

- Configuration on C300-1.

1. Configure the loopback interface.

```

ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 1.1.1.1 255.255.255.255
ZXAN(config-loopback1)#exit
  
```

2. Configure the VLAN of the uplink interfaces.

```

ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 13 tag
ZXAN(config-if)#exit
  
```

3. Configure the IP address of the VLAN interfaces.

```

ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.1 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 13
ZXAN(config-if-vlan13)#ip address 13.1.1.2 255.255.255.0
ZXAN(config-if-vlan13)#exit
  
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 13
ZXAN(config-if-vlan13)#mpls ip
ZXAN(config-if-vlan13)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 1.1.1.1 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 13.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the static label binding.

```
ZXAN(config)#mpls static binding ipv4 3.3.3.3 255.255.255.255 outlabel 10.1.1.2
1040001 //static LSP from C300-1 to C300-3
ZXAN(config)#mpls static binding ipv4 1.1.1.1 255.255.255.255 inlabel 1041004
//static LSP from C300-3 to C300-1
```

8. Save the configuration data.

● Configuration on C300-2.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 2.2.2.2 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.1 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 2.2.2.2 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the static label binding.

```
ZXAN(config)#mpls static binding ipv4 3.3.3.3 255.255.255.255 inlabel 1040001
//static LSP from C300-1 to C300-3
ZXAN(config)#mpls static binding ipv4 3.3.3.3 255.255.255.255 outlabel 11.1.1.2
1040002 //static LSP from C300-1 to C300-3
```

8. Save the configuration data.

● Configuration on C300-3.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 3.3.3.3 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 12 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.2 255.255.255.0
ZXAN(config-if-vlan11)#exit
ZXAN(config)#interface vlan 12
ZXAN(config-if-vlan12)#ip address 12.1.1.1 255.255.255.0
ZXAN(config-if-vlan12)#exit
```


4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
ZXAN(config)#interface vlan 12
ZXAN(config-if-vlan12)#mpls ip
ZXAN(config-if-vlan12)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 3.3.3.3 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 12.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the static label binding.

```
ZXAN(config)#mpls static binding ipv4 3.3.3.3 255.255.255.255 inlabel 1040002
//static LSP from C300-1 to C300-3
ZXAN(config)#mpls static binding ipv4 1.1.1.1 255.255.255.255 outlabel 12.1.1.2
1041005 //static LSP from C300-3 to C300-1
```

8. Save the configuration data.

● Configuration on C300-4.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 4.4.4.4 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 13 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 12 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 12
ZXAN(config-if-vlan12)#ip address 12.1.1.2 255.255.255.0
ZXAN(config-if-vlan12)#exit
ZXAN(config)#interface vlan 13
ZXAN(config-if-vlan13)#ip address 13.1.1.1 255.255.255.0
ZXAN(config-if-vlan13)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 12
ZXAN(config-if-vlan12)#mpls ip
ZXAN(config-if-vlan12)#exit
ZXAN(config)#interface vlan 13
ZXAN(config-if-vlan13)#mpls ip
ZXAN(config-if-vlan13)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 4.4.4.4 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 12.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 13.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the static label binding.

```
ZXAN(config)#mpls static binding ipv4 1.1.1.1 255.255.255.255 inlabel 1041005
//static LSP from C300-3 to C300-1
ZXAN(config)#mpls static binding ipv4 1.1.1.1 255.255.255.255 outlabel 13.1.1.2
1041004 //static LSP from C300-3 to C300-1
```

8. Save the configuration data.

Result

Execute the command **ping mpls ipv4 3.3.3.3 32 repeat 1000** on C300-1. The ping is successfully.

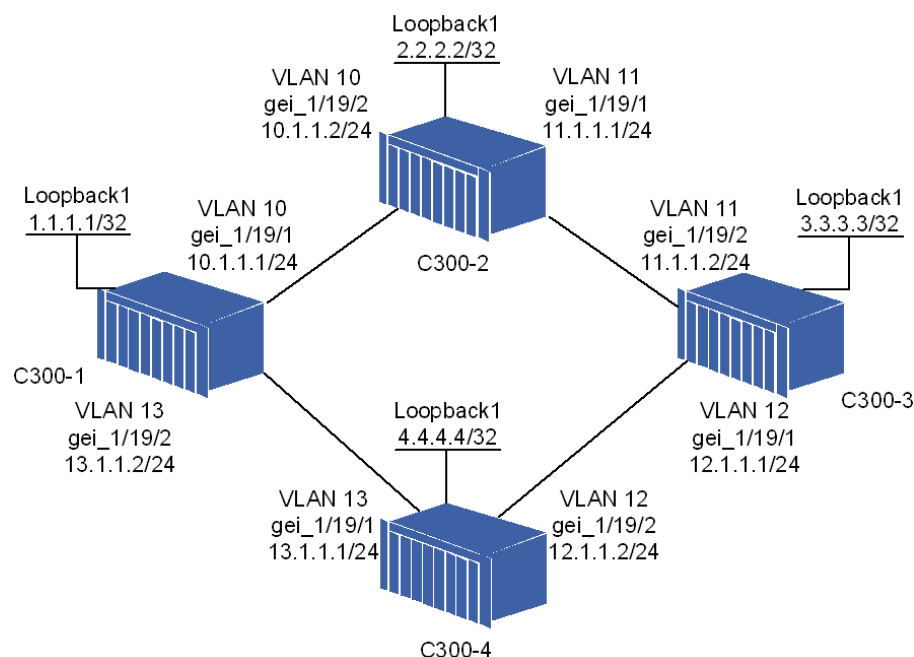
16.1.3 Configuring the LDP FRR Function

The LDP fast reroute is a mechanism that is used to reroute packets around a failed link.

Networking Diagram

Figure 16-5 shows the networking diagram of the Label Distribution Protocol (LDP) Fast Reroute (FRR) function.

Figure 16-5 Networking Diagram of LDP FRR Configuration



The LDP is running on the four ZXA10 C300s. Configure two LSPs between C300-1 and C300-3.

- The active LSP is: C300-1 > C300-2 > C300-3.
- The backup LSP is: C300-1 > C300-4 > C300-3.

Configuration Data

Table 16-3 lists the configuration data of the LDP FRR function.

Table 16-3 Configuration Data of the LDP FRR Function

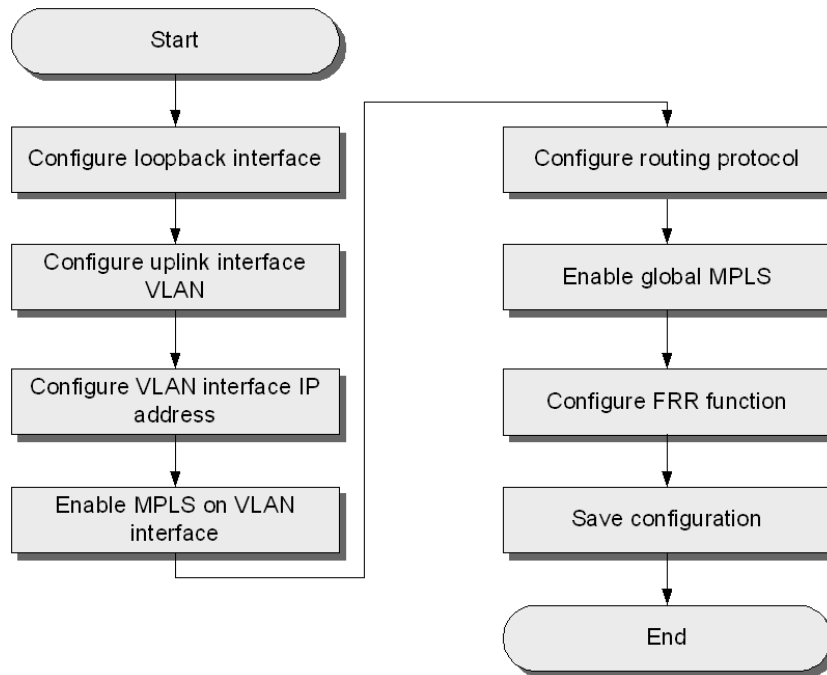
Item		Data
C300-1	Loopback interface	● ID: 1 ● IP address: 1.1.1.1/32
	L3 interface 1	● VLAN ID: 10 ● Interface: gei_1/21/1 ● IP address: 10.1.1.1/24
	L3 interface 2	● VLAN ID: 13 ● Interface: gei_1/21/2 ● IP address: 13.1.1.12/24
	Active route	● Destination IP address: 3.3.3.3/32 ● Next hop IP address: 10.1.1.2/24
	Backup route	● Destination IP address: 3.3.3.3/32 ● Next hop IP address: 13.1.1.1/24 ● Tag value: 200 ● Priority: 60

Item		Data
C300-2	Loopback interface	● ID: 1 ● IP address: 2.2.2.2/32
	L3 interface 1	● VLAN ID: 11 ● Interface: gei_1/21/1 ● IP address: 11.1.1.1/24
	L3 interface 2	● VLAN ID: 10 ● Interface: gei_1/21/2 ● IP address: 10.1.1.2/24
C300-3	Loopback interface	● ID: 1 ● IP address: 3.3.3.3/32
	L3 interface 1	● VLAN ID: 12 ● Interface: gei_1/21/1 ● IP address: 12.1.1.1/24
	L3 interface 2	● VLAN ID: 11 ● Interface: gei_1/21/2 ● IP address: 11.1.1.2/24
C300-4	Loopback interface	● ID: 1 ● IP address: 4.4.4.4/32
	L3 interface 1	● VLAN ID: 13 ● Interface: gei_1/21/1 ● IP address: 13.1.1.1/24
	L3 interface 2	● VLAN ID: 12 ● Interface: gei_1/21/2 ● IP address: 12.1.1.2/24

Configuration Flow

Figure 16-6 shows the configuration flowchart of the LDP FRR function.

Figure 16-6 Configuration Flowchart of LDP FRR Function



Steps

- Configuration on C300-1.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 1.1.1.1 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 13 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.1 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 13
ZXAN(config-if-vlan13)#ip address 13.1.1.2 255.255.255.0
ZXAN(config-if-vlan13)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
```

```
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 13
ZXAN(config-if-vlan13)#mpls ip
ZXAN(config-if-vlan13)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 1.1.1.1 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 13.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the FRR function.

```
ZXAN(config)#mpls ldp lsp-control ordered
ZXAN(config)#ip route 3.3.3.3 255.255.255.255 10.1.1.2
ZXAN(config)#ip route 3.3.3.3 255.255.255.255 13.1.1.1 200 slave 60
```

8. Save the configuration data.

● Configuration on C300-2.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 2.2.2.2 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.1 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
```

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 2.2.2.2 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Save the configuration data.

● Configuration on C300-3.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 3.3.3.3 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 12 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.2 255.255.255.0
ZXAN(config-if-vlan11)#exit
ZXAN(config)#interface vlan 12
ZXAN(config-if-vlan12)#ip address 12.1.1.1 255.255.255.0
ZXAN(config-if-vlan12)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
ZXAN(config)#interface vlan 12
ZXAN(config-if-vlan12)#mpls ip
ZXAN(config-if-vlan12)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
```

```
ZXAN(config-router)#network 3.3.3.3 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 12.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Save the configuration data.

● Configuration on C300-4.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 4.4.4.4 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 13 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 12 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 12
ZXAN(config-if-vlan12)#ip address 12.1.1.2 255.255.255.0
ZXAN(config-if-vlan12)#exit
ZXAN(config)#interface vlan 13
ZXAN(config-if-vlan13)#ip address 13.1.1.1 255.255.255.0
ZXAN(config-if-vlan13)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 12
ZXAN(config-if-vlan12)#mpls ip
ZXAN(config-if-vlan12)#exit
ZXAN(config)#interface vlan 13
ZXAN(config-if-vlan13)#mpls ip
ZXAN(config-if-vlan13)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 4.4.4.4 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 12.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 13.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.


```

ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip

```

7. Save the configuration data.

Result

When you reboot C300-2, the LDP session between two C300-1 and C300-3 is not affected.

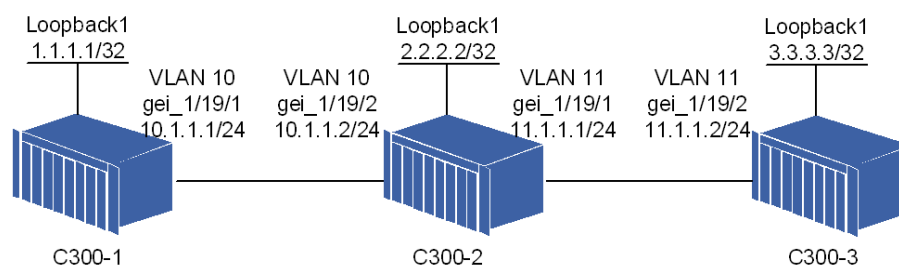
16.1.4 Configuring the LDP GR Function

When a router is configured with LDP GR function, it assists a neighboring router to recover gracefully from an interruption in service.

Networking Diagram

Figure 16-7 shows the networking diagram of the LDP Graceful Restart (GR) function.

Figure 16-7 Networking Diagram of LDP GR



Three ZXA10 C300s are in the same OSPF area and interconnected by the OSPF protocol. C300-2 supports two control and switching cards, which work in active and backup mode. When one of the control switching cards is faulty and two cards are swapped, the LDP session between C300-1 and C300-3 is not interrupted.

Configuration Data

Table 16-4 lists the configuration data of the LDP GR function.

Table 16-4 Configuration Data of LDP GR Function

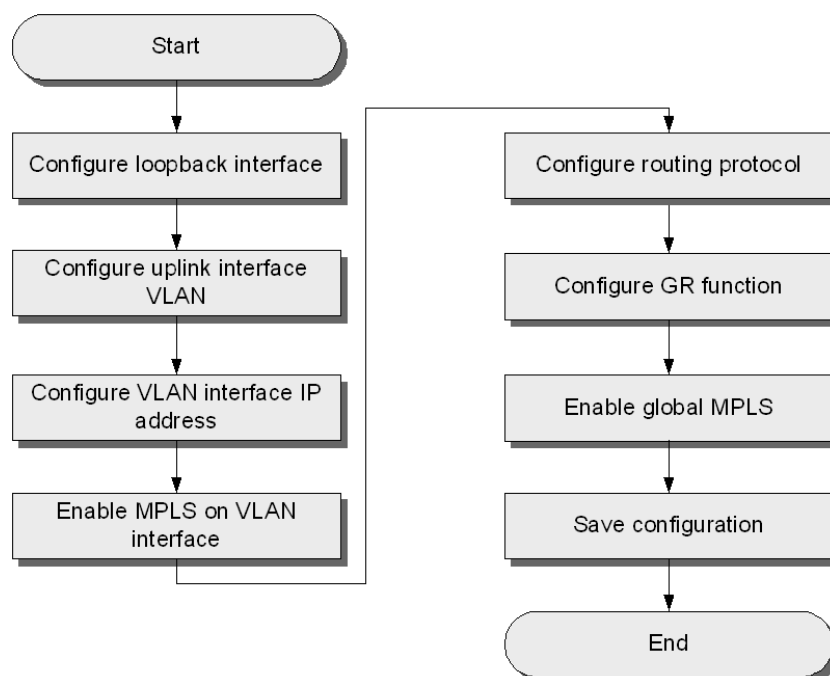
Item		Data
C300-1	Loopback interface	- ID: 1 - IP address: 1.1.1.1/32
	L3 interface	- VLAN ID: 10 - Interface: gei_1/21/1 - IP address: 10.1.1.1/24
	Grace period	180 s

Item		Data
C300-2	Loopback interface	● ID: 1 ● IP address: 2.2.2.2/32
	L3 interface 1	● VLAN ID: 11 ● Interface: gei_1/21/1 ● IP address: 11.1.1.1/24
	L3 interface 2	● VLAN ID: 10 ● Interface: gei_1/21/2 ● IP address: 10.1.1.2/24
	Grace period	180 s
C300-3	Loopback interface	● ID: 1 ● IP address: 3.3.3.3/32
	L3 interface	● VLAN ID: 11 ● Interface: gei_1/21/2 ● IP address: 11.1.1.2/24
	Grace period	180 s

Configuration Flow

Figure 16-8 shows the configuration flowchart of the LDP GR function.

Figure 16-8 Configuration Flowchart of LDP GR Function



Steps

- Configuration on C300-1.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 1.1.1.1 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.1 255.255.255.0
```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 1.1.1.1 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
```

6. Configure the GR function.

```
ZXAN(config-router)#nsf
ZXAN(config-router)#grace-period 180
ZXAN(config-router)#exit
```

7. Enable the global MPLS function.

```
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ldp graceful-restart
ZXAN(config)#mpls ip
```

8. Save the configuration data.

- Configuration on C300-2.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 2.2.2.2 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.1 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 2.2.2.2 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
```

6. Configure the GR function.

```
ZXAN(config-router)#nsf
ZXAN(config-router)#grace-period 180
ZXAN(config-router)#exit
```

7. Enable the global MPLS function.

```
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ldp graceful-restart
ZXAN(config)#mpls ip
```

8. Save the configuration data.

● Configuration on C300-3.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 3.3.3.3 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
XAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.2 255.255.255.0
```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if-vlan11)#mpls ip
```

```
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 3.3.3.3 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
```

6. Configure the GR function.

```
ZXAN(config-router)#nsf
ZXAN(config-router)#grace-period 180
ZXAN(config-router)#exit
```

7. Enable the global MPLS function.

```
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ldp graceful-restart
ZXAN(config)#mpls ip
```

8. Save the configuration data.

Result

The LDP GR function is successfully configured. Execute the command **ping mpls ipv4 3.3.3.3** on C300-1. The ping is successful. When you swap the active and backup control switching cards on C300-2, the LDP session between C300-1 and C300-3 is not interrupted.

16.2 VPLS Configuration

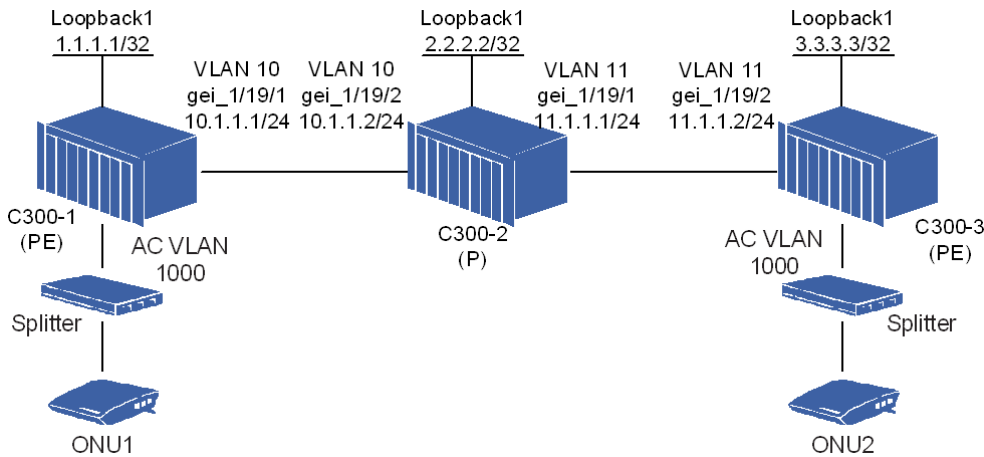
16.2.1 Configuring the Basic VPLS

VPLS provides Ethernet based multipoint to multipoint communication over IP/MPLS networks. It allows geographically dispersed sites to share an Ethernet broadcast domain by connecting sites through PWs.

Networking Diagram

Figure 16-9 shows the networking diagram of the basic Virtual Private LAN Service (VPLS).

Figure 16-9 Networking Diagram of Basic VPLS



ONU1 and ONU2 are in the same VPLS network. These two ONUs communicate with each other through the Pseudo-Wires (PWs) based on the LDP sessions.

Configuration Data

Table 16-5 lists the configuration data of the basic VPLS.

Table 16-5 Configuration Data of Basic VPLS

Item		Data
C300-1	Loopback interface	- ID: 1 - IP address: 1.1.1.1/32
	L3 interface	- VLAN ID: 10 - Interface: gei_1/21/1 - IP address: 10.1.1.1/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW	- Name: 3and1000pw - Mode: dynamic pwe3 - Type: ethernet-vlan - Peer: 3.3.3.3 - Router ID: 3.3.3.3 - VC ID: 1000
	VFI	- Name: vpls-vfi - PW: 3and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000

Item		Data
C300-2	Loopback interface	● ID: 1 ● IP address: 2.2.2.2/32
	L3 interface 1	● VLAN ID: 11 ● Interface: gei_1/21/1 ● IP address: 11.1.1.1/24
	L3 interface 2	● VLAN ID: 10 ● Interface: gei_1/21/2 ● IP address: 10.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
C300-3	Loopback interface	● ID: 1 ● IP address: 3.3.3.3/32
	L3 interface	● VLAN ID: 11 ● Interface: gei_1/21/2 ● IP address: 11.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW	● Name: 1and1000pw ● Mode: dynamic pwe3 ● Type: ethernet-vlan ● Peer: 1.1.1.1 ● Router ID: 1.1.1.1 ● VC ID: 1000
	VFI	● Name: vpls-vfi ● PW: 1and1000pw
	Class map	● Name: classmap1000 ● SVLAN ID: 1000
	CIP	● Index: 1 ● Service type: ethernet class-map ● Class map: classmap1000

**Note:**

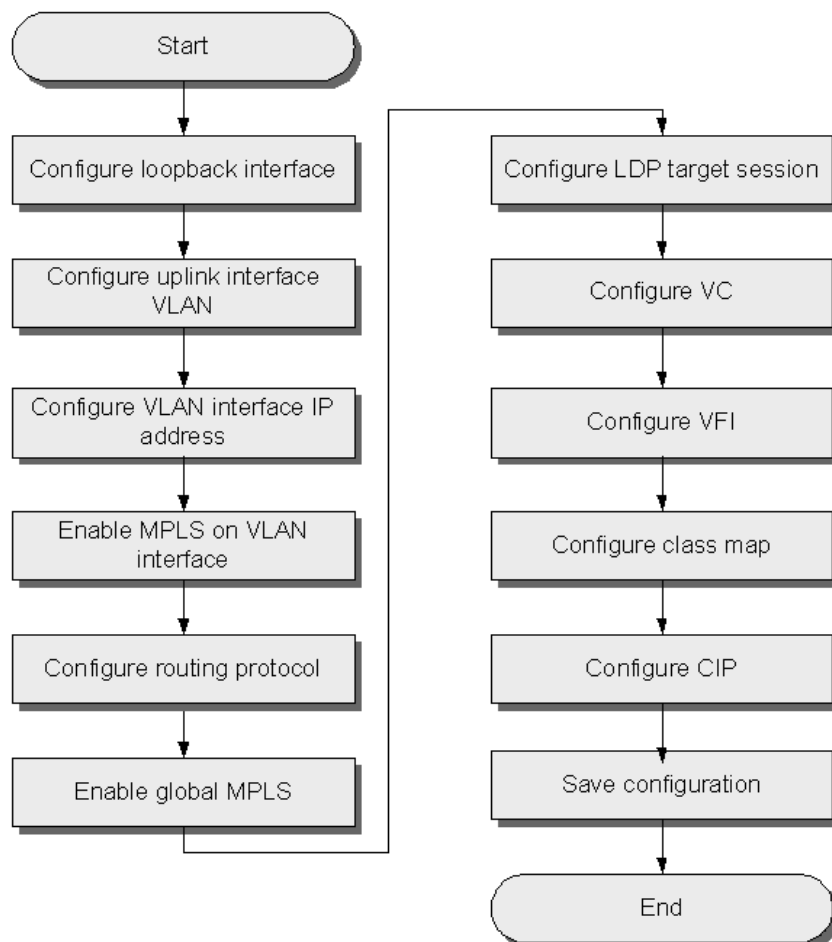
A class map defines the policy of class mapping.

A Customer Instance Port (CIP) is a logical interface that correlates to an L2VPN service instance, that is, an Attachment Circuit (AC). The data on a CIP is a certain service flow, which is specified by a class map, from a [UNI](#).

Configuration Flow

Figure 16-10 shows the configuration flowchart of the basic VPLS.

Figure 16-10 Configuration Flowchart of Basic VPLS



Steps

- Configuration on C300-1.

1. Configure the loopback interface.

```

ZXAN(config)#interface loopback1
ZXAN(config-loopback1)# address 1.1.1.1 255.255.255.255
ZXAN(config-loopback1)#exit

```

2. Configure the VLAN of the uplink interface.

```

ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit

```

3. Configure the IP address of the VLAN interface.

```

ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.1 255.255.255.0

```


4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 1.1.1.1 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 3.3.3.3
```

8. Configure the VC.

```
ZXAN(config)#pw 3and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwtype ethernet-vlan
ZXAN(config-pw)#peer 3.3.3.3 router-id 3.3.3.3 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the VFI.

```
ZXAN(config)#vfi vpls-vfi
ZXAN(config-vfi)#mpls xconnect pw 3and1000pw
ZXAN(config-vfi)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect vpls-vfi
ZXAN(config-cip1)#exit
```

12. Save the configuration data.**● Configuration on C300-2.****1. Configure the loopback interface.**

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 2.2.2.2 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.1 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 2.2.2.2 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Save the configuration data.

● Configuration on C300-3.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 3.3.3.3 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interface.

```
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 11 tag
```

```
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 11  
ZXAN(config-if-vlan11)#ip address 11.1.1.2 255.255.255.0
```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if-vlan11)#mpls ip  
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1  
ZXAN(config-router)#network 3.3.3.3 0.0.0.0 area 0.0.0.0  
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0  
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn  
ZXAN(config)#ip mpls-forwarding enable  
ZXAN(config)#mpls ldp router-id loopback1  
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 1.1.1.1
```

8. Configure the VC.

```
ZXAN(config)#pw 1and1000pw  
ZXAN(config-pw)#mode dynamic pwe3 fec128  
ZXAN(config-pw)#pwttype ethernet-vlan  
ZXAN(config-pw)#peer 1.1.1.1 router-id 1.1.1.1 vcid 1000  
ZXAN(config-pw)#exit
```

9. Configure the VFI.

```
ZXAN(config)#vfi vpls-vfi  
ZXAN(config-vfi)#mpls xconnect pw 1and1000pw  
ZXAN(config-vfi)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap1000  
ZXAN(config-cmap)#match svlan-id 1000  
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1  
ZXAN(config-cip1)#service-type ethernet class-map classmap1000  
ZXAN(config-cip1)#xconnect vpls-vfi  
ZXAN(config-cip1)#exit
```

12. Save the configuration data.

Result

Two ONUs can ping each other successfully.

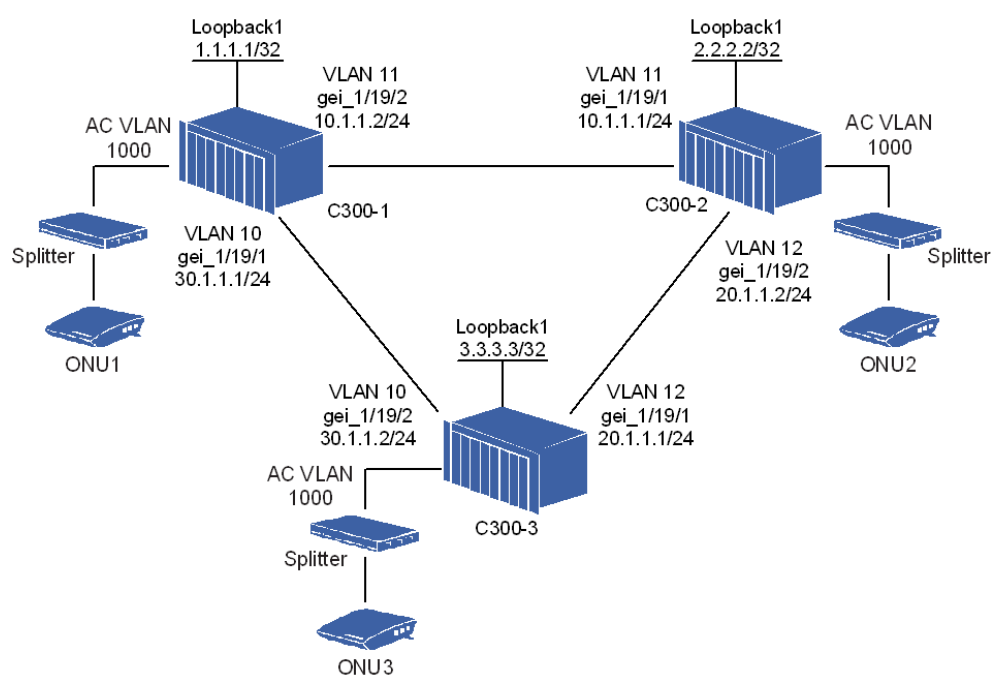
16.2.2 Configuring the Full-Mesh VPLS

To connect all of the customer LANs, full mesh connectivity is required. LDP is used for full mesh establishment for VPLS.

Networking Diagram

Figure 16-11 shows the networking diagram of the full-mesh VPLS.

Figure 16-11 Networking Diagram of Full-Mesh VPLS



ONU1, ONU2, and ONU2 are in the same VPLS network. These three ONUs communicate with each other through the PWs based on the LDP sessions.

Configuration Data

Table 16-6 lists the configuration data of the full-mesh VPLS.

Table 16-6 Configuration Data of Full-Mesh VPLS

Item		Data
C300-1	Loopback interface	● ID: 1 ● IP address: 1.1.1.1/32
	L3 interface 1	● VLAN ID: 10 ● Interface: gei_1/21/1 ● IP address: 30.1.1.1/24
	L3 interface 2	● VLAN ID: 11 ● Interface: gei_1/21/2 ● IP address: 10.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW 1	● Name: 2and1000pw ● Mode: dynamic pwe3 ● Type: ethernet-vlan ● Peer: 2.2.2.2 ● Router ID: 2.2.2.2 ● VC ID: 1000
	PW 2	● Name: 3and1000pw ● Mode: dynamic pwe3 ● Type: ethernet-vlan ● Peer: 3.3.3.3 ● Router ID: 3.3.3.3 ● VC ID: 1000
	VFI	● Name: fullmesh-vfi ● PW1: 2and1000pw ● PW2: 3and1000pw
	Class map	● Name: classmap1000 ● SVLAN ID: 1000
	CIP	● Index: 1 ● Service type: ethernet class-map ● Class map: classmap1000
C300-2	Loopback interface ID	● ID: 1 ● IP address: 2.2.2.2/32
	L3 interface 1	● VLAN ID: 11 ● Interface: gei_1/21/1 ● IP address: 10.1.1.1/24
	L3 interface 2	● VLAN ID: 12 ● Interface: gei_1/21/2 ● IP address: 20.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090

Item		Data
	PW 1	● Name: 1and1000pw ● Mode: dynamic pwe3 ● Type: ethernet-vlan ● Peer: 1.1.1.1 ● Router ID: 1.1.1.1 ● VC ID: 1000
	PW 2	● Name: 3and1000pw ● Mode: dynamic pwe3 ● Type: ethernet-vlan ● Peer: 3.3.3.3 ● Router ID: 3.3.3.3 ● VC ID: 1000
	VFI	● Name: fullmesh-vfi ● PW1: 1and1000pw ● PW2: 3and1000pw
	Class map	● Name: classmap1000 ● SVLAN ID: 1000
	CIP	● Index: 1 ● Service type: ethernet class-map ● Class map: classmap1000
C300-3	Loopback interface	● ID: 1 ● IP address: 3.3.3.3/32
	L3 interface 1	● VLAN ID: 12 ● Interface: gei_1/21/1 ● IP address: 20.1.1.1/24
	L3 interface 2	● VLAN ID: 10 ● Interface: gei_1/21/2 ● IP address: 30.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW 1	● Name: 1and1000pw ● Mode: dynamic pwe3 ● Type: ethernet-vlan ● Peer: 1.1.1.1 ● Router ID: 1.1.1.1 ● VC ID: 1000
	PW 2	● Name: 2and1000pw ● Mode: dynamic pwe3 ● Type: ethernet-vlan ● Peer: 2.2.2.2 ● Router ID: 2.2.2.2

Item		Data
		VC ID: 1000
	VFI	- Name: fullmesh-vfi - PW1: 1and1000pw - PW2: 2and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000

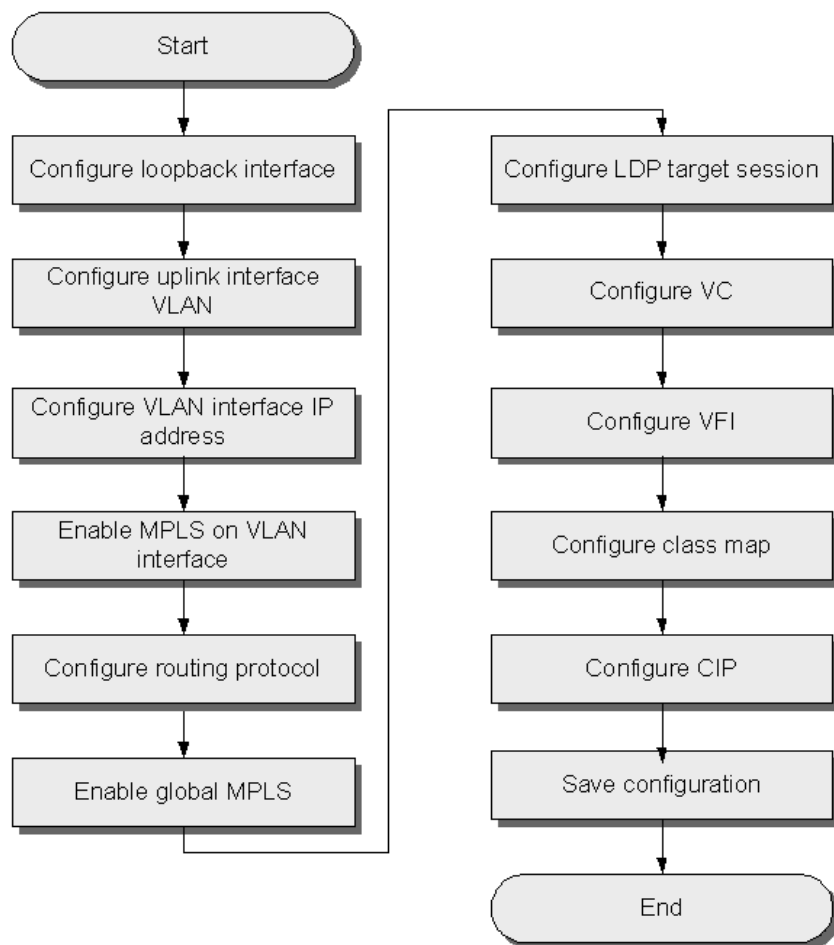
**Note:**

A class map defines the policy of class mapping.

A CIP is a logical interface that correlates to an L2VPN service instance, that is, AC. The data on a CIP is a certain service flow, which is specified by a class map, from a UNI.

Configuration Flow

Figure 16-12 shows the configuration flowchart of the full-mesh VPLS.

Figure 16-12 Configuration Flowchart of Full-Mesh VPLS

Steps

- Configuration on C300-1.

1. Configure the loopback interface.

```

ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 1.1.1.1 255.255.255.255
ZXAN(config-loopback1)#exit

```

2. Configure the VLAN of the uplink interfaces.

```

ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit

```

3. Configure the IP address of the VLAN interfaces.

```

ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 30.1.1.1 255.255.255.0
ZXAN(config-if-vlan10)#exit

```



```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 1.1.1.1 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 30.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 2.2.2.2
ZXAN(config)#mpls ldp target-session 3.3.3.3
```

8. Configure the VC.

```
ZXAN(config)#pw 2and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwtpe ethernet-vlan
ZXAN(config-pw)#peer 2.2.2.2 router-id 2.2.2.2 vcid 1000
ZXAN(config-pw)#exit
ZXAN(config)#pw 3and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwtpe ethernet-vlan
ZXAN(config-pw)#peer 3.3.3.3 router-id 3.3.3.3 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the VFI.

```
ZXAN(config)#vfi fullmesh-vfi
ZXAN(config-vfi)#mpls xconnect pw 2and1000pw
ZXAN(config-vfi)#mpls xconnect pw 3and1000pw
ZXAN(config-vfi)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect fullmesh-vfi
ZXAN(config-cip1)#exit
```

12. Save the configuration data.

● Configuration on C300-2.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 2.2.2.2 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interface.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 12 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 10.1.1.1 255.255.255.0
ZXAN(config-if-vlan11)#exit
ZXAN(config)#interface vlan 12
ZXAN(config-if-vlan12)#ip address 20.1.1.2 255.255.255.0
ZXAN(config-if-vlan12)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
ZXAN(config)#interface vlan 12
ZXAN(config-if-vlan12)#mpls ip
ZXAN(config-if-vlan12)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 2.2.2.2 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 20.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 1.1.1.1
ZXAN(config)#mpls ldp target-session 3.3.3.3
```

8. Configure the VC.

```
ZXAN(config)#pw 1and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet-vlan
ZXAN(config-pw)#peer 1.1.1.1 router-id 1.1.1.1 vcid 1000
ZXAN(config)#pw 3and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet-vlan
ZXAN(config-pw)#peer 3.3.3.3 router-id 3.3.3.3 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the VFI.

```
ZXAN(config)#vfi fullmesh-vfi
ZXAN(config-vfi)#mpls xconnect pw 1and1000pw
ZXAN(config-vfi)#mpls xconnect pw 3and1000pw
ZXAN(config-vfi)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect fullmesh-vfi
ZXAN(config-cip1)#exit
```

12. Save the configuration data.**● Configuration on C300-3.****1. Configure the loopback interface.**

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 3.3.3.3 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interface.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 12 tag
```

```
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 30.1.1.2 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 12
ZXAN(config-if-vlan12)#ip address 20.1.1.1 255.255.255.0
ZXAN(config-if-vlan12)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 12
ZXAN(config-if-vlan12)#mpls ip
ZXAN(config-if-vlan12)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 3.3.3.3 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 20.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 30.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 1.1.1.1
ZXAN(config)#mpls ldp target-session 2.2.2.2
```

8. Configure the VC.

```
ZXAN(config)#pw 1and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwtype ethernet-vlan
ZXAN(config-pw)#peer 1.1.1.1 router-id 1.1.1.1 vcid 1000
ZXAN(config)#pw 2and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwtype ethernet-vlan
ZXAN(config-pw)#peer 2.2.2.2 router-id 2.2.2.2 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the VFI.

```
ZXAN(config)#vfi fullmesh-vfi
ZXAN(config-vfi)#mpls xconnect pw 1and1000pw
ZXAN(config-vfi)#mpls xconnect pw 2and1000pw
ZXAN(config-vfi)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect fullmesh-vfi
ZXAN(config-cip1)#exit
```

12. Save the configuration data.

Result

Three ONUs can ping each other successfully.

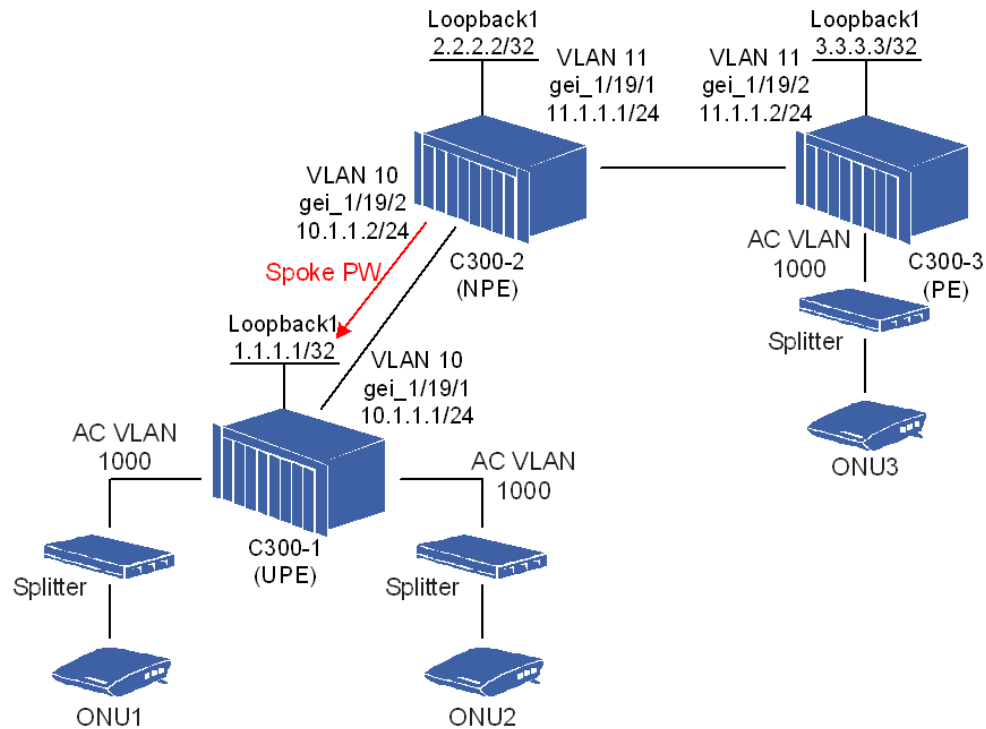
16.2.3 Configuring the Hierarchical VPLS

HVPLS is a method of subdividing a VPLS VPN into two or three tiered hierarchical networks. This can significantly reduce the number of LDP sessions and LSPs, and thus unburden the core network, by concentrating customers in edge devices. HVPLS may also be used to join two VPLS mesh structures together.

Networking Diagram

Figure 16-13 shows the networking diagram of the Hierarchical VPLS (HVPLS).

Figure 16-13 HPVLS Networking Diagram



ONU1, ONU2, and ONU3 are in the same VPLS network. ONU1 and ONU2 are connected to the VPLS network through the User-facing PE (UPE), while ONU3 is connected to the VPLS through the PE. The Network PE (NPE) communicates with the UPE through a spoke PW sessions.

Configuration Data

Table 16-7 lists the configuration data of the HVPLS.

Table 16-7 HVPLS Configuration Data

Item		Data
C300-1	Loopback interface	- ID: 1 - IP address: 1.1.1.1/32
	L3 interface	- VLAN ID: 10 - Interface: gei_1/21/1 - IP address: 10.1.1.1/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW	- Name: 2and1000pw - Mode: dynamic pwe3 - Type: ethernet-vlan - Peer: 2.2.2.2 - Router ID: 2.2.2.2 - VC ID: 1000

Item		Data
	VFI	- Name: hvpls-vfi - PW: 2and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000
C300-2	Loopback interface	- ID: 1 - IP address: 2.2.2.2/32
	L3 interface 1	- VLAN ID: 11 - Interface: gei_1/21/1 - IP address: 11.1.1.1/24
	L3 interface 2	- VLAN ID: 10 - Interface: gei_1/21/2 - IP address: 10.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW 1	- Name: 1and1000pw - Mode: dynamic pwe3 - Type: ethernet-vlan - Peer: 1.1.1.1 - Router ID: 1.1.1.1 - VC ID: 1000
	PW 2	- Name: 3and1000pw - Mode: dynamic pwe3 - Type: ethernet-vlan - Peer: 3.3.3.3 - Router ID: 3.3.3.3 - VC ID: 1000
	VFI	- Name: hvpls-vfi - PW1: 1and1000pw (spoke) - PW2: 3and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000

Item		Data
C300-3	Loopback interface	● ID: 1 ● IP address: 3.3.3.3/32
	L3 interface	● VLAN ID: 11 ● Interface: gei_1/21/2 ● IP address: 11.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW	● Name: 2and1000pw ● Mode: dynamic pwe3 ● Type: ethernet-vlan ● Peer: 2.2.2.2 ● Router ID: 2.2.2.2 ● VC ID: 1000
	VFI	● Name: hvpls-vfi ● PW: 2and1000pw
	Class map	● Name: classmap1000 ● SVLAN ID: 1000
	CIP	● Index: 1 ● Service type: ethernet class-map ● Class map: classmap1000

**Note:**

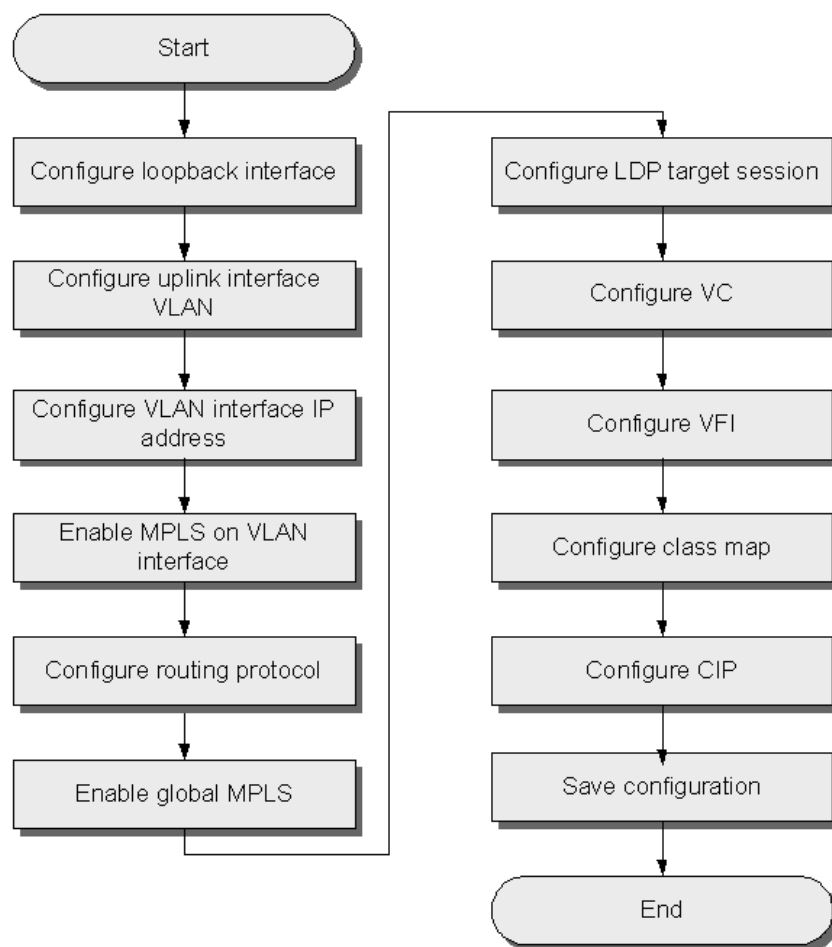
A class map defines the policy of class mapping.

A CIP is a logical interface that correlates to an L2VPN service instance, that is, an AC. The data on a CIP is a certain service flow, which is specified by a class map, from a UNI.

Configuration Flow

Figure 16-14 shows the configuration flowchart of the HVPLS.

Figure 16-14 HVPLS Configuration Flowchart



Steps

- Configuration on C300-1.

1. Configure the loopback interface.

```

ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 1.1.1.1 255.255.255.255
ZXAN(config-loopback1)#exit

```

2. Configure the VLAN of the uplink interface.

```

ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit

```

3. Configure the IP address of the VLAN interface.

```

ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.1 255.255.255.0

```

4. Enable the MPLS on the VLAN interface.

```

ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit

```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 1.1.1.1 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 2.2.2.2
```

8. Configure the VC.

```
ZXAN(config)#pw 2and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet-vlan
ZXAN(config-pw)#peer 2.2.2.2 router-id 2.2.2.2 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the VFI.

```
ZXAN(config)#vfi hvpls-vfi
ZXAN(config-vfi)#mpls xconnect pw 2and1000pw
ZXAN(config-vfi)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect hvpls-vfi
ZXAN(config-cip1)#exit
```

12. Save the configuration data.**● Configuration on C300-2.****1. Configure the loopback interface.**

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 2.2.2.2 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
```

```
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.1 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 2.2.2.2 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 1.1.1.1
ZXAN(config)#mpls ldp target-session 3.3.3.3
```

8. Configure the VC.

```
ZXAN(config)#pw 1and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet-vlan
ZXAN(config-pw)#peer 1.1.1.1 router-id 1.1.1.1 vcid 1000
ZXAN(config)#pw 3and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet-vlan
ZXAN(config-pw)#peer 3.3.3.3 router-id 3.3.3.3 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the VFI.

```
ZXAN(config)#vfi hvpls-vfi
ZXAN(config-vfi)#mpls xconnect pw 1and1000pw spoke
ZXAN(config-vfi)#mpls xconnect pw 3and1000pw
ZXAN(config-vfi)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect hvpls-vfi
ZXAN(config-cip1)#exit
```

12. Save the configuration data.

- Configuration on C300-3.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 3.3.3.3 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interface.

```
XAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.2 255.255.255.0
```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 3.3.3.3 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 2.2.2.2
```

8. Configure the VC.

```
ZXAN(config)#pw 2and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwtpe ethernet-vlan
ZXAN(config-pw)#peer 2.2.2.2 router-id 2.2.2.2 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the VFI.

```
ZXAN(config)#vfi hvpls-vfi
ZXAN(config-vfi)#mpls xconnect pw 2and1000pw
ZXAN(config-vfi)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect hvpls-vfi
ZXAN(config-cip1)#exit
```

12. Save the configuration data.

Result

These three ONUs can ping each other successfully.

16.3 Configuring the VPWS

VPWS is suitable for applications based on ATM or FR connection.

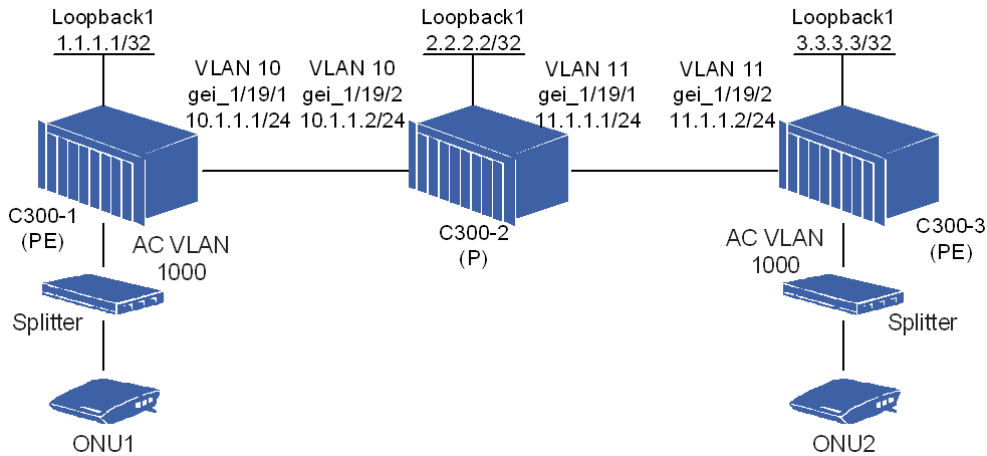
Background Information

The **VPWS** is one of the L2VPN application. VPWS uses **VLLs** to provide Ethernet based point to point communication over **IP/MPLS** networks. VLL uses the **PW** encapsulation for transporting Ethernet traffic over an MPLS tunnel across an IP/MPLS backbone.

Networking Diagram

Figure 16-15 shows the networking diagram of the **VPWS**.

Figure 16-15 VPWS Networking Diagram



Two ZXA10 C300s and a router are in the same **OSPF** area.

ONU1 and **ONU2** access the two **PEs** (ZXA10 C300s) through the **AC VLAN**. The two **ONUs** communicate with each other through the **VPWS** based on the **LDP** sessions.

Configuration Data

Table 16-8 lists the configuration data of the **VPWS**.

Table 16-8 VPWS Configuration Data

Item		Data
C300-1	Loopback interface	- ID: 1 - IP address: 1.1.1.1/32
	L3 interface	- VLAN ID: 10 - Interface: gei_1/21/1 - IP address: 10.1.1.1/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW	- Name: 3and1000pw - Mode: dynamic pwe3 - Type: ethernet - Peer: 3.3.3.3 - Router ID: 3.3.3.3 - VC ID: 1000
	VLL	- Name: zte-vpws - Type: ethernet - PW: 3and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000

Item		Data
C300-2	Loopback interface	● ID: 1 ● IP address: 2.2.2.2/32
	L3 interface 1	● VLAN ID: 11 ● Interface: gei_1/21/1 ● IP address: 11.1.1.1/24
	L3 interface 2	● VLAN ID: 10 ● Interface: gei_1/21/2 ● IP address: 10.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
C300-3	Loopback interface	● ID: 1 ● IP address: 3.3.3.3/32
	L3 interface	● VLAN ID: 1 ● Interface: gei_1/21/2 ● IP address: 11.1.1.2/24
	PW	● Name: 1and1000pw ● Mode: dynamic pwe3 ● Type: ethernet ● Peer: 1.1.1.1 ● Router ID: 1.1.1.1 ● VC ID: 1000
	VLL	● Name: zte-vpws ● Type: ethernet ● PW: 1and1000pw
	Class map	● Name: classmap1000 ● SVLAN ID: 1000
	CIP	● Index: 1 ● Service type: ethernet class-map ● Class map: classmap1000

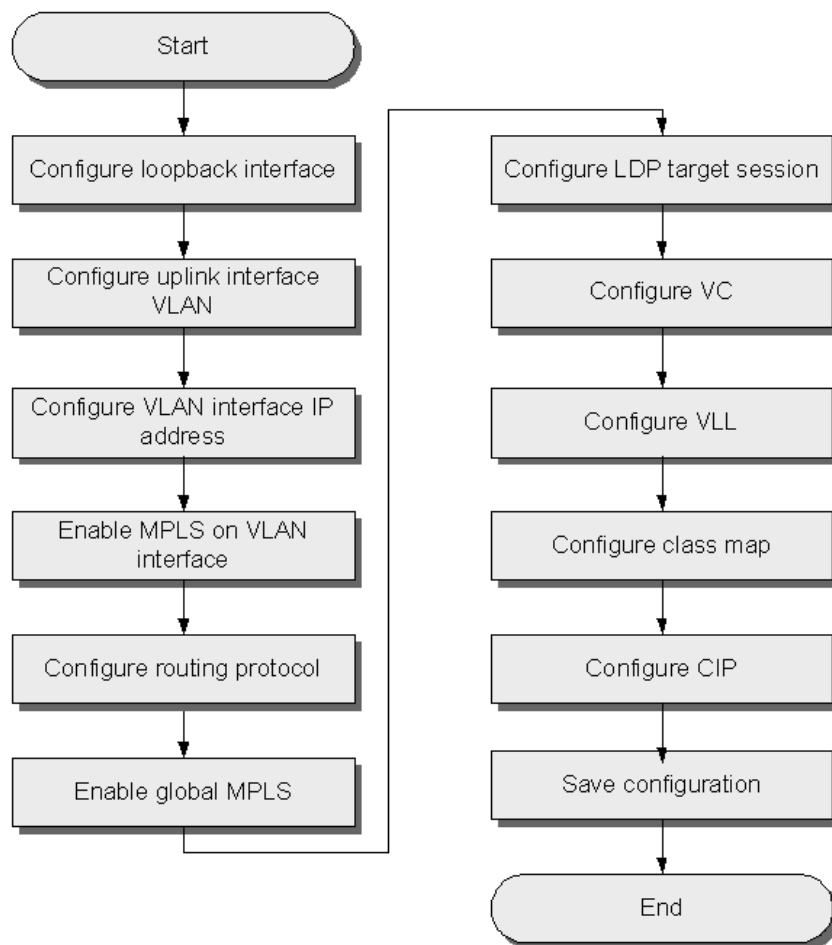
**Note:**

A class map defines the policy of class mapping.

A CIP is a logical interface that correlates to an L2VPN service instance, that is, an AC. The data on a CIP is a certain service flow, which is specified by a class map, from a UNI.

Configuration Flow

Figure 16-16 shows the configuration flowchart of the VPWS.

Figure 16-16 Configuration Flowchart of VPWS

Steps

- Configuration on C300-1.

1. Configure the loopback interface.

```

ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 1.1.1.1 255.255.255.255
ZXAN(config-loopback1)#exit

```

2. Configure the VLAN of the uplink interface.

```

ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit

```

3. Configure the IP address of the VLAN interface.

```

ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.1 255.255.255.0

```

4. Enable the MPLS on the VLAN interface.

```

ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit

```


5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 1.1.1.1 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 3.3.3.3
```

8. Configure the VC.

```
ZXAN(config)#pw 3and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwtpe ethernet
ZXAN(config-pw)#peer 3.3.3.3 router-id 3.3.3.3 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure VLL.

```
ZXAN(config)#vll zte-vpws
ZXAN(config-vll)#service-type ethernet
ZXAN(config-vll)#mpls xconnect pw 3and1000pw
ZXAN(config-vll)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect zte-vpws
ZXAN(config-cip1)#exit
```

12. Save the configuration data.

● Configuration on C300-2.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 2.2.2.2 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
```

```
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.1 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 2.2.2.2 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Save the configuration data.

● Configuration on C300-3.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 3.3.3.3 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interface.

```
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.2 255.255.255.0
```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 3.3.3.3 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 1.1.1.1
```

8. Configure the VC.

```
ZXAN(config)#pw 1and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#peer 1.1.1.1 router-id 1.1.1.1 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the VLL.

```
ZXAN(config)#vll zte-vpws
ZXAN(config-vll)#service-type ethernet
ZXAN(config-vll)#mpls xconnect pw 1and1000pw
ZXAN(config-vll)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect zte-vpws
ZXAN(config-cip1)#exit
```

12. Save the configuration data.

Result

The two ONUs can ping each other successfully.

16.4 PWE3 Configuration

16.4.1 Configuring the Basic PWE3 Service

In the basic PWE3 application, the PEs transparently transmit the services of CEs through the PWs.

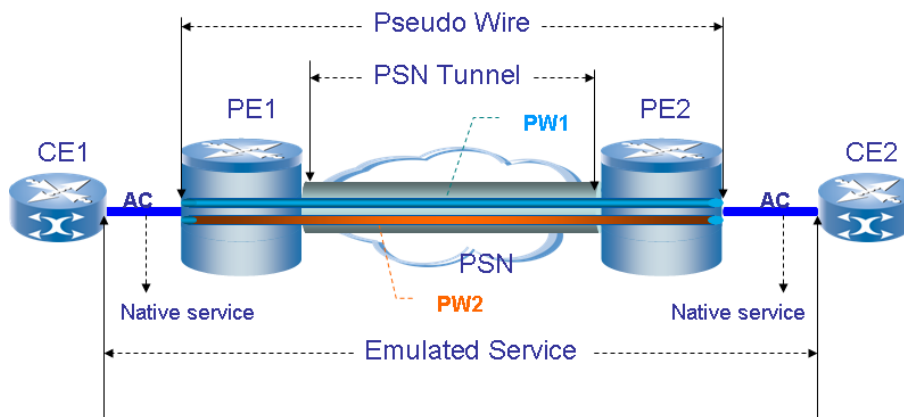
Background Information

The IETF working group develops PWE3, an architecture for service providers to emulate the native services over PSNs.

The native service may be ATM, FR, Ethernet, HDLC, TDM, or SONET/SDH, while the PSN may be MPLS, IP (either IPv4 or IPv6), or L2TPv3.

As shown in Figure 16-17, PEs transparently transmit native services of CEs.

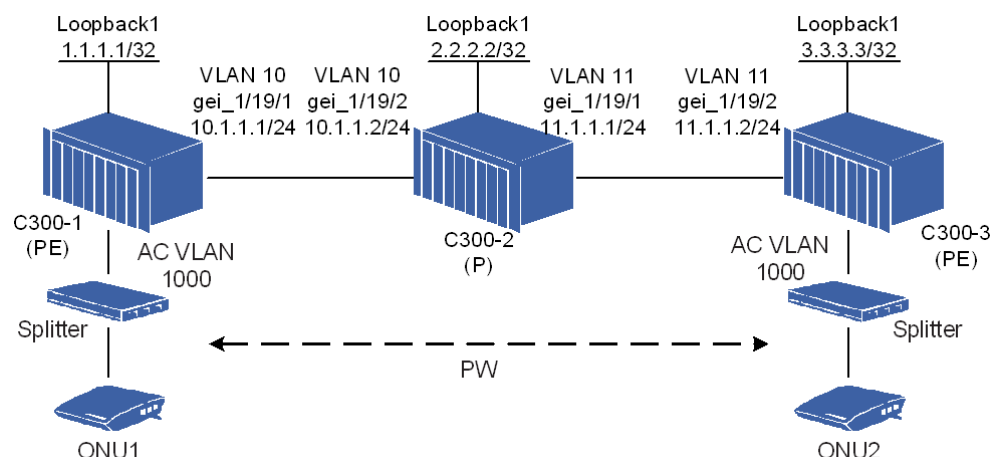
Figure 16-17 PWE3 Network Reference Model



Networking Diagram

Figure 16-18 shows the networking diagram of the basic PWE3 service.

Figure 16-18 Networking Diagram of Basic PWE3 Service



Three ZXA10 C300s are in the same **OSPF** area. **ONU1** and **ONU2** access the two **PEs** (ZXA10 C300s) through the **AC VLAN**. The two ONUs communicate with each other through the **PWs** based on the **LDP** sessions.

Configuration Data

Table 16-9 lists the configuration data of the basic PWE3 service.

Table 16-9 Configuration Data of Basic PWE3 Service

Item		Data
C300-1	Loopback interface	- ID: 1 - IP address: 1.1.1.1/32
	L3 interface	- VLAN ID: 10 - Interface: gei_1/21/1 - IP address: 10.1.1.1/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW	- Name: 3and1000pw - Mode: dynamic pwe3 - Type: ethernet - Peer: 3.3.3.3 - Router ID: 3.3.3.3 - VC ID: 1000
	VLL	- Name: pwe3-vll - Type: ethernet - PW: 3and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000

Item		Data
C300-2	Loopback interface	- ID: 1 - IP address: 2.2.2.2/32
	L3 interface 1	- VLAN ID: 11 - Interface: gei_1/21/1 - IP address: 11.1.1.1/24
	L3 interface 2	- VLAN ID: 10 - Interface: gei_1/21/2 - IP address: 10.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
C300-3	Loopback interface	- ID: 1 - IP address: 3.3.3.3/32
	L3 interface	- VLAN ID: 11 - Interface: gei_1/21/2 - IP address: 11.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW	- Name: 1and1000pw - Mode: dynamic pwe3 - Type: ethernet - Peer: 1.1.1.1 - Router ID: 1.1.1.1 - VC ID: 1000
	VLL	- Name: pwe3-vll - Type: ethernet - PW: 1and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000

**Note:**

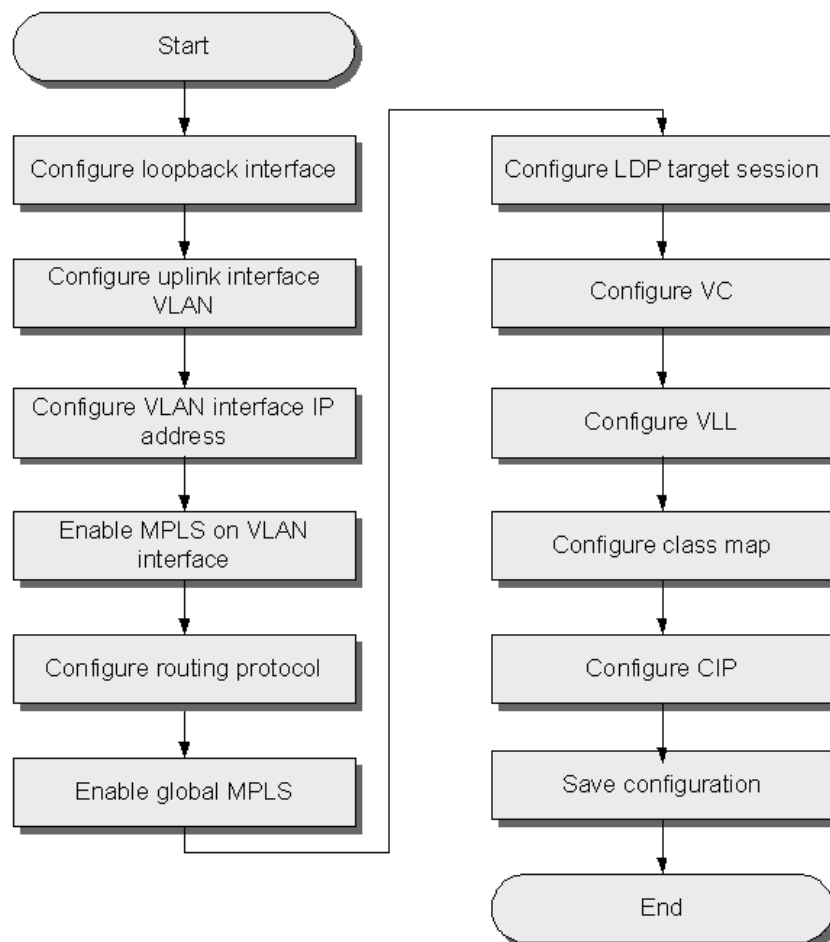
A class map defines the policy of class mapping.

A CIP is a logical interface that correlates to an L2VPN service instance, that is, an AC. The data on a CIP is a certain service flow, which is specified by a class map, from a UNI.

Configuration Flow

Figure 16-19 shows the configuration flowchart of the basic PWE3 service.

Figure 16-19 Configuration Flowchart of PW Redundancy



Steps

- Configuration on C300-1.

1. Configure the loopback interface.

```

ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 1.1.1.1 255.255.255.255
ZXAN(config-loopback1)#exit
  
```

2. Configure the VLAN of the uplink interfaces.

```

ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
  
```

3. Configure the IP address of the VLAN interface.

```

ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.1 255.255.255.0
  
```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 1.1.1.1 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 3.3.3.3
```

8. Configure the VC.

```
ZXAN(config)#pw 3and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#peer 3.3.3.3 router-id 3.3.3.3 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the VLL.

```
ZXAN(config)#vll pwe3-vll
ZXAN(config-vll)#service-type ethernet
ZXAN(config-vll)#mpls xconnect pw 3and1000pw
ZXAN(config-vll)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect pwe3-vll
ZXAN(config-cip1)#exit
```

12. Save the configuration data.**● Configuration on C300-2.****1. Configure the loopback interface.**

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 2.2.2.2 255.255.255.255
```



```
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.1 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 2.2.2.2 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Save the configuration data.

● Configuration on C300-3.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 3.3.3.3 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
XAN(config)#interface gei_1/21/2
```

```
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.2 255.255.255.0
```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 3.3.3.3 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 1.1.1.1
```

8. Configure the VC.

```
ZXAN(config)#pw 1and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#peer 1.1.1.1 router-id 1.1.1.1 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the VLL.

```
ZXAN(config)#vll pwe3-vll
ZXAN(config-vll)#service-type ethernet
ZXAN(config-vll)#mpls xconnect pw 1and1000pw
ZXAN(config-vll)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect pwe3-vll
ZXAN(config-cip1)#exit
```

12. Save the configuration data.

Result

The two ONUs can ping each other successfully.

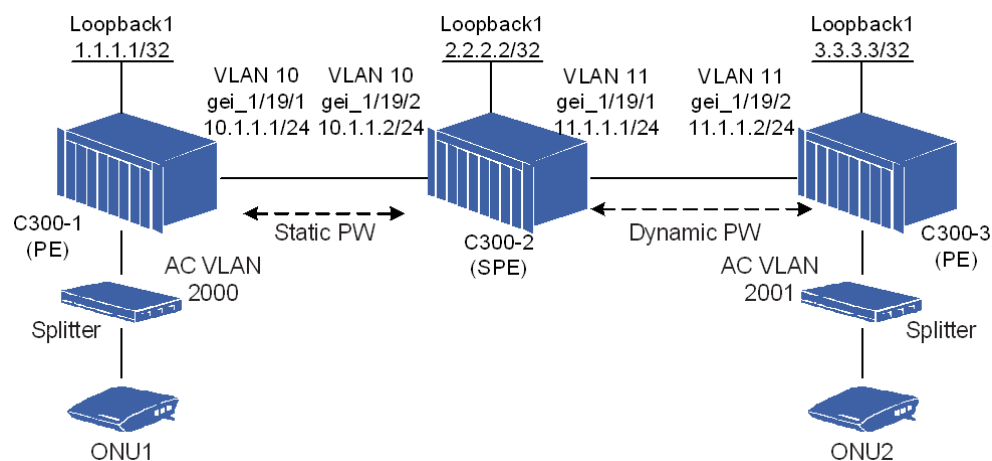
16.4.2 Configuring Multi-Segment PWs

Multi-segment PWs, which consist of static PWs and dynamic PWs, enhance the flexibility and feasibility of PWE3 networking.

Networking Diagram

Figure 16-20 shows the networking diagram of the multi-segment PWs.

Figure 16-20 Networking Diagram of Multi-Segment PWs



ONU1 and ONU2 access the two PEs (ZXA10 C300s) through the AC VLAN. Both ZXA10 C300s are connected to an MPLS network. The two ONUs communicate with each other through the multi-segment PWs between two ZXA10 C300s and the switching PE (SPE).

Configuration Data

Table 16-10 lists the configuration data of the multi-segment PWs.

Table 16-10 Configuration Data of Multi-Segment PWs

Item		Data
C300-1	Loopback interface	- ID: 1 - IP address: 1.1.1.1/32
	L3 interface	- VLAN ID: 10 - Interface: gei_1/21/1 - IP address: 10.1.1.1/24
	Reserved VPN VLAN	VLAN ID: 4090

Item		Data
	PW	● Name: 2and2000pw ● Mode: static ● Type: ethernet ● Peer: 2.2.2.2 ● Router ID: 2.2.2.2 ● VC ID: 2000 ● Local label: 1042431 ● Remote label: 1042432
	VLL	● Name: mspw-vll ● Type: ethernet ● PW: 2and2000pw
	Class map	● Name: classmap2000 ● SVLAN ID: 2000
	CIP	● Index: 1 ● Service type: ethernet class-map ● Class map: classmap2000
C300-2	Loopback interface	● ID: 1 ● IP address: 2.2.2.2/32
	L3 interface 1	● VLAN ID: 11 ● Interface: gei_1/21/1 ● IP address: 11.1.1.1/24
	L3 interface 2	● VLAN ID: 10 ● Interface: gei_1/21/2 ● IP address: 10.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW 1	● Name: 1and2000pw ● Mode: static ● Type: ethernet ● Peer: 1.1.1.1 ● Router ID: 1.1.1.1 ● VC ID: 2000 ● Local label: 1042432 ● Remote label: 1042431
	PW 2	● Name: 3and2001pw ● Mode: dynamic pwe3 ● Type: ethernet ● Peer: 3.3.3.3 ● Router ID: 3.3.3.3 ● VC ID: 2001

Item		Data
	Multi-segment PW	● Name: zte-mspw ● PW 1: 1and2000pw ● PW 2: 3and2001pw
C300-3	Loopback interface	● ID: 1 ● IP address: 3.3.3.3/32
	L3 interface	● VLAN ID: 11 ● Interface: gei_1/21/2 ● IP address: 11.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW	● Name: 2and2001pw ● Mode: dynamic pwe3 ● Type: ethernet ● Peer: 2.2.2.2 ● Router ID: 2.2.2.2 ● VC ID: 2001
	VLL	● Name: mspw-vll ● Type: ethernet ● PW: 2and2001pw
	Class map	● Name: classmap2001 ● SVLAN ID: 2001
	CIP	● Index: 1 ● Service type: ethernet class-map ● Class map: classmap2001

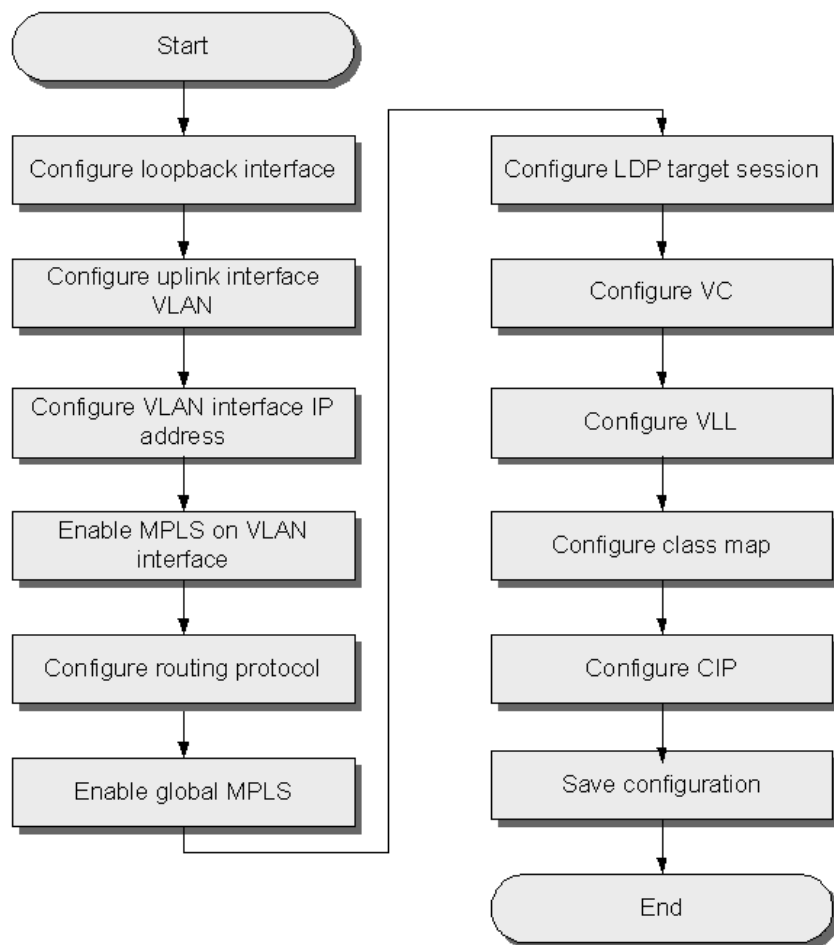
**Note:**

A class map defines the policy of class mapping.

A CIP is a logical interface that correlates to an L2VPN service instance, that is, an AC. The data on a CIP is a certain service flow, which is specified by a class map, from a UNI.

Configuration Flow

Figure 16-21 shows the configuration flowchart of the multi-segment PWs.

Figure 16-21 Configuration Flowchart of Multi-Segment PWs

Steps

- Configuration on C300-1.

1. Configure the loopback interface.

```

ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 1.1.1.1 255.255.255.255
ZXAN(config-loopback1)#exit

```

2. Configure the VLAN of the uplink interface.

```

ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit

```

3. Configure the IP address of the VLAN interface.

```

ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.1 255.255.255.0

```

4. Enable the MPLS on the VLAN interface.

```

ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit

```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 1.1.1.1 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 2.2.2.2
```

8. Configure the VC.

```
ZXAN(config)#pw 2and2000pw
ZXAN(config-pw)#mode static fec128
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#peer 2.2.2.2 router-id 2.2.2.2 vcid 2000 local-label 1042431
remote-label 1042432
ZXAN(config-pw)#exit
```

9. Configure the VLL.

```
ZXAN(config)#vll mspw-vll
ZXAN(config-vll)#service-type ethernet
ZXAN(config-vll)#mpls xconnect pw 2and2000pw
ZXAN(config-vll)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap2000
ZXAN(config-cmap)#match svlan-id 2000
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap2000
ZXAN(config-cip1)#xconnect mspw-vll
ZXAN(config-cip1)#exit
```

12. Save the configuration data.

```
ZXAN(config)#exit
ZXAN#write
Building configuration...
.....[OK]
```

- **Configuration on C300-2.**

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 2.2.2.2 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.1 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 2.2.2.2 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target sessions.

```
ZXAN(config)#mpls ldp target-session 1.1.1.1
ZXAN(config)#mpls ldp target-session 3.3.3.3
```

8. Configure the VCs.

```
ZXAN(config)#pw 1and2000pw
ZXAN(config-pw)#mode static fec128
```



```
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#peer 1.1.1.1 router-id 1.1.1.1 vcid 2000 local-label 1042432
remote-label 1042431
ZXAN(config-pw)#exit
ZXAN(config)#pw 3and2001pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#peer 3.3.3.3 router-id 3.3.3.3 vcid 2001
ZXAN(config-pw)#exit
```

9. Configure the multi-segment PW.

```
ZXAN(config)#mspw zte-mspw
ZXAN(config-mspw)#mpls xconnect pw 1and2000pw
ZXAN(config-mspw)#mpls xconnect pw 3and2001pw
ZXAN(config-mspw)#exit
```

10. Save the configuration data.

```
ZXAN(config)#exit
ZXAN#write
Building configuration...
.....[OK]
```

● Configuration on C300-3.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 3.3.3.3 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
XAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.2 255.255.255.0
```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 3.3.3.3 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
```

```
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 2.2.2.2
```

8. Configure the VC.

```
ZXAN(config)#pw 2and2001pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwtype ethernet
ZXAN(config-pw)#peer 2.2.2.2 router-id 2.2.2.2 vcid 2001
ZXAN(config-pw)#exit
```

9. Configure the VLL.

```
ZXAN(config)#vll mspw-vll
ZXAN(config-vll)#service-type ethernet
ZXAN(config-vll)#mpls xconnect pw 2and2001pw
ZXAN(config-vll)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap2001
ZXAN(config-cmap)#match svlan-id 2001
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap2001
ZXAN(config-cip1)#xconnect mspw-vll
ZXAN(config-cip1)#exit
```

12. Save the configuration data.

```
ZXAN(config)#exit
ZXAN#write
Building configuration...
.....[OK]
```

Result

The two ONUs can ping each other successfully.

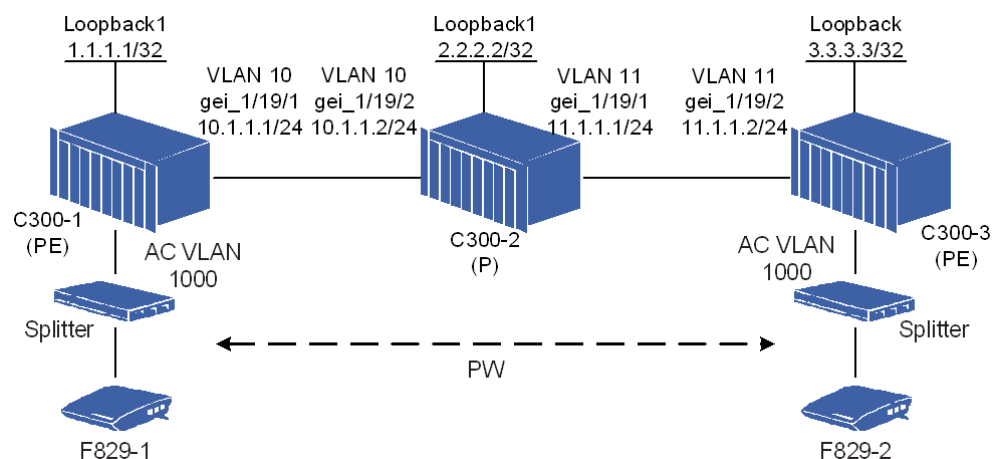
16.4.3 Configuring the TDM Relay Service

Though the PWE3 emulation, the TDM service can be carried over MPLS network.

Networking Diagram

Figure 16-22 shows the networking diagram of the TDM relay service.

Figure 16-22 Networking Diagram of TDM Relay Service



Two ZXA10 C300s and a router are in the same **OSPF** area. Two ZXA10 F829s access the two **PE**s (ZXA10 C300s) through the **AC** VLAN. The **TDM** services on the two ONUs are transported over the **PW**.

Configuration Data

Table 16-11 lists the configuration data of the TDM relay service.

Table 16-11 Configuration Data of TDM Relay Service

Item		Data
C300-1	Loopback interface	- ID: 1 - IP address: 1.1.1.1/32
	L3 interface	- VLAN ID: 10 - Interface: gei_1/21/1 - IP address: 10.1.1.1/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW class	- Name: 3and1000pw - Control-word: preferred
	PW	- Name: 3and1000pw - Mode: dynamic pwe3 - Type: e1 - Peer: 3.3.3.3 - Router ID: 3.3.3.3 VC ID: 1000
	VLL	- Name: tdm-vll - Type: tdm PW: 3and1000pw
	CIP	- Index: 1 - PSN local label: 1046016 - PSN remote label: 6016

Item		Data
		- PW local label: 1046015 - PW remote label: 6015 - Destination MAC address: 0015.eb72.0004 - VLAN ID: 1000
	MAC address	0015.eb72.0002
	RTD slot number	15
F829-1	CES MAC address	0015.eb72.0004
	TDM profile	- Name: f829 - Packet reorder: enable - RTP header: enable - Timestamp mode: differential - Queue size: 6
	TDM service	- Type: e1Satop - Clock source: adaptive - Primary clock port: 1 - Secondary clock port: 2 - Slot: 0/2
	PW TDM properties	- Name: pw_0/2/1 - TDM interface: tdm_0/2/1 - Number of trunk frames per packet: 1 - Time slots: 1-12 - TDM profile: f829
	PW PSN properties	- Type: mpls-staticdouble - Inbound transport label: 0x1780 - Outbound transport label: 0xff600 - Inbound interworking label: 0x177f - Outbound interworking label: 0xff5ff - Destination MAC address: 0015.eb72.0002 - VLAN ID: 1000 - Priority: 7
C300-2	Loopback interface	- ID: 1 - IP address: 2.2.2.2/32
	L3 interface 1	- VLAN ID: 11 - Interface: gei_1/21/1 - IP address: 11.1.1.1/24
	L3 interface 2	- VLAN ID: 10 - Interface: gei_1/21/2 - IP address: 10.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090

Item		Data
C300-3	Loopback interface	- ID: 1 - IP address: 3.3.3.3/32
	L3 interface	- VLAN ID: 11 - Interface: gei_1/21/2 - IP address: 11.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW class	- Name: 1and1000pw - Control-word: preferred
	PW name	- Name: 1and1000pw - Mode: dynamic pwe3 - Type: e1 - Peer: 1.1.1.1 - Router ID: 1.1.1.1 - VC ID: 1000
	VLL	- Name: tdm-vll - Type: tdm - PW: 1and1000pw
	CIP	- Index: 1 - PSN local label: 1047016 - PSN remote label: 7016 - PW local label: 1047015 - PW remote label: 7015 - Destination MAC address: 0015.eb72.0005 - VLAN ID: 1000
	MAC address	0015.eb72.0003
F829-2	RTD slot number	15
	CES MAC address	0015.eb72.0005
	TDM profile	- Name: f829 - Packet reorder: enable - RTP header: enable - Timestamp mode: differential - Queue size: 6
	TDM service	- Type: e1Satop - Clock source: adaptive - Primary clock port: 1 - Secondary clock port: 2 - Slot: 0/2
	PW TDM properties	- Name: pw_0/2/1 - TDM interface: tdm_0/2/1

Item		Data
		● Number of trunk frames per packet: 1 ● Time slots: 1–12 ● TDM profile: f829
	PW PSN properties	● Type: mpls-staticdouble ● Inbound transport label: 0x1b68 ● Outbound transport label: 0xff9e8 ● Inbound interworking label: 0x1b67 ● Outbound interworking label: 0xff9e7 ● Destination MAC address: 0015.eb72.0003 ● VLAN ID: 1000 ● Priority: 7

**Note:**

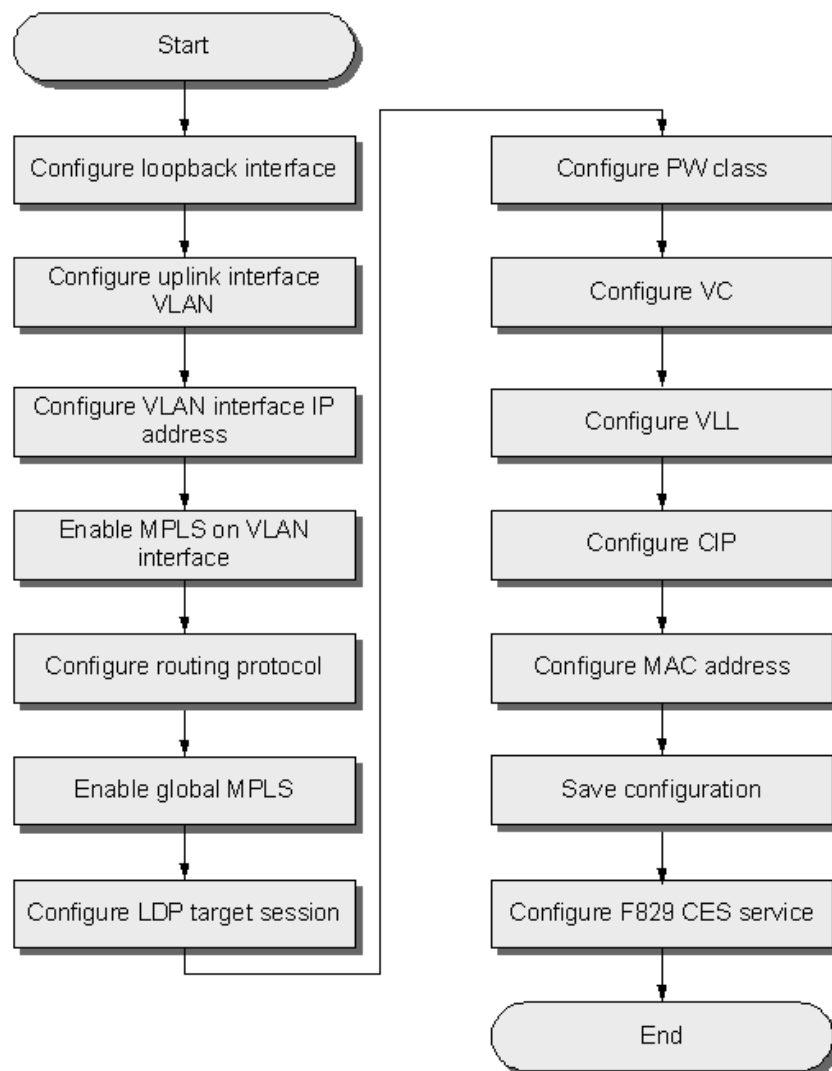
A class map defines the policy of class mapping.

A CIP is a logical interface that correlates to an L2VPN service instance, that is, an AC. The data on a CIP is a certain service flow, which is specified by a class map, from a UNI.

Configuration Flow

Figure 16-23 shows the configuration flowchart of the TDM relay service.

Figure 16-23 Configuration Flowchart of TDM Relay Service



Steps

- Configuration on C300-1.

1. Configure the loopback interface.

```

ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 1.1.1.1 255.255.255.255
ZXAN(config-loopback1)#exit

```

2. Configure the VLAN of the uplink interface.

```

ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit

```

3. Configure the IP address of the VLAN interface.

```

ZXAN(config)#interface vlan 10
ZXAN(config-if)#ip address 10.1.1.1 255.255.255.0

```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if)#mpls ip
ZXAN(config-if)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 1.1.1.1 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 3.3.3.3
```

8. Configure the PW class.

```
ZXAN(config)#pw-class 3and1000pw
ZXAN(config-pw-class)#control-word preferred
ZXAN(config-pw-class)#exit
```

9. Configure the VC.

```
ZXAN(config)#pw 3and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwtype e1
ZXAN(config-pw)#apply pw-class 3and1000pw
ZXAN(config-pw)#peer 3.3.3.3 router-id 3.3.3.3 vcid 1000
ZXAN(config-pw)#exit
```

10. Configure the VLL.

```
ZXAN(config)#vll tdm-vll
ZXAN(config-vll)#service-type tdm
ZXAN(config-vll)#mpls xconnect pw 3and1000pw
ZXAN(config-vll)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type tdm-relay mpls 1046016 6016 1046015 6015 0015.eb72.
0004 1000
ZXAN(config-cip1)#xconnect tdm-vll
ZXAN(config-cip1)#exit
```

12. Configure the MAC address.

```
ZXAN(config)#mac 0015.eb72.0002 15
```

13. Save the configuration data.

- Configuration on F829-1.

1. Configure the CES MAC address.

```
ZXAN(config)#ces
ZXAN(config-ces)# mac-add 0015.eb72.0004
```

2. Configure the TDM profile.

```
ZXAN(config-ces)# tdm-profile f829 reorder en rtpHder en jtrBfr 4000 TSMODE dif 6
```

3. Configure the TDM service.

```
ZXAN(config-ces)# tdm-service type elsatop clock-source adaptive primary-clock
1 secondary-clock 3 0/2
```

4. Configure the PW.

```
ZXAN(config-ces)# pw pw_0/2/1
ZXAN(config-ces-pw)# tdm-relation tdm_0/2/1 frame 1 channellist 1-12 tdm-profile
-name f829
ZXAN(config-ces-pw)# psn mpls-staticdouble 0x1780 0xff600 0x177f 0xff5ff dst-mac
0015.eb72.0002 vlan 1000 priority 7
```

5. Save the configuration data.

```
ZXAN(config-ces-pw)# end
ZXAN# write
```

- Configuration on C300-2.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 2.2.2.2 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.1 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
```

```
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 2.2.2.2 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 11.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Save the configuration data.

● Configuration on C300-3.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 3.3.3.3 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interface.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if)#ip address 10.1.1.1 255.255.255.0
```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if)#mpls ip
ZXAN(config-if)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 3.3.3.3 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 1.1.1.1
```

8. Configure the PW class.

```
ZXAN(config)#pw-class land1000pw
ZXAN(config-pw-class)#control-word preferred
ZXAN(config-pw-class)#exit
```

9. Configure the VC.

```
ZXAN(config)#pw land1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwtype e1
ZXAN(config-pw)#apply pw-class land1000pw
ZXAN(config-pw)#peer 1.1.1.1 router-id 1.1.1.1 vcid 1000
ZXAN(config-pw)#exit
```

10. Configure the VLL.

```
ZXAN(config)#vll tdm-vll
ZXAN(config-vll)#service-type tdm
ZXAN(config-vll)#mpls xconnect pw land1000pw
ZXAN(config-vll)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type tdm-relay mpls 1047016 7016 1047015 7015 0015.eb72.
0005 1000
ZXAN(config-cip1)#xconnect tdm-vll
ZXAN(config-cip1)#exit
```

12. Configure the MAC address.

```
ZXAN(config)#mac 0015.eb72.0003 15
```

13. Save the configuration data.**● Configuration on F829-2.****1. Configure the CES MAC address.**

```
ZXAN(config)#ces
ZXAN(config-ces)# mac-add 0015.eb72.0005
```

2. Configure the TDM profile.

```
ZXAN(config-ces)# tdm-profile f829 reorder en rtpHder en jtrBfr 4000 TSMODE dif 6
```

3. Configure the TDM service.

```
ZXAN(config-ces)# tdm-service type elsatop clock-source adaptive primary-clock
1 secondary-clock 3 0/2
```

4. Configure the PW.

```
ZXAN(config-ces)# pw pw_0/2/1
ZXAN(config-ces-pw)# tdm-relation tdm_0/2/1 frame 1 channellist 1-12 tdm-profile
-name f829
```

```
ZXAN(config-ces-pw)# psn mpls-staticdouble 0x1b68 0xff9e8 0x1b67 0xff9e7 dst-mac
0015.eb72.0003 vlan 1000 priority 7
```

5. Save the configuration data.

```
ZXAN(config-ces-pw)# end
ZXAN# write
```

Result

The TDM relay service is successfully configured on both C300s and F829s.

16.5 PW Redundancy Configuration

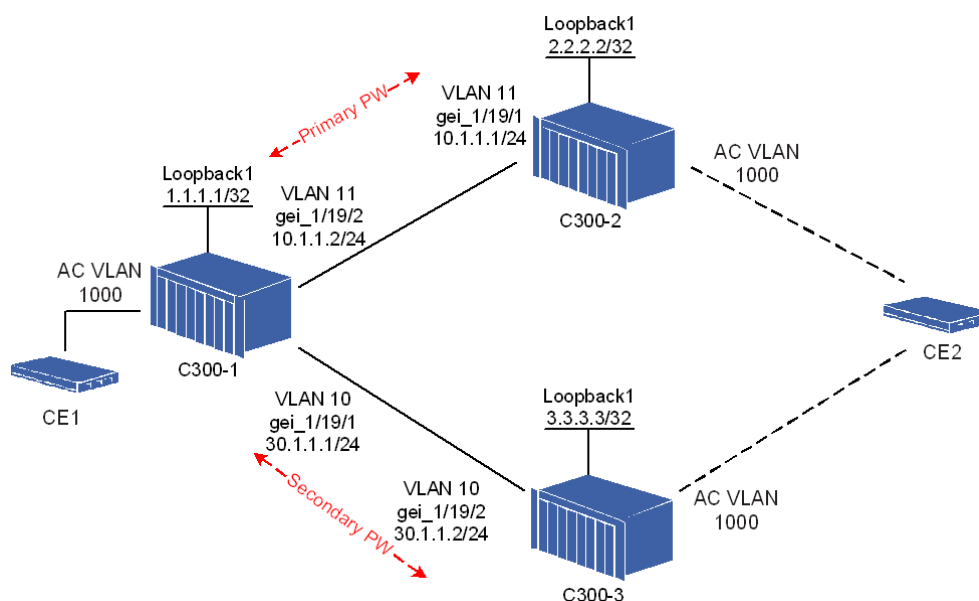
16.5.1 Configuring the VPLS Redundancy

VPLS redundancy enhances the stability and reliability of MPLS network.

Networking Diagram

Figure 16-24 shows the networking diagram of the VPLS redundancy.

Figure 16-24 Networking Diagram of VPLS Redundancy



The PW redundancy group works in either 1:1 or 1+1 mode.

- In 1:1 mode, only the active PW forwards the user traffic.
- In 1+1 mode, both PWs forward the user traffic.

The network supports the [OSPF](#) and [LDP](#) protocols.

Configuration Data

Table 16-12 lists the configuration data of the VPLS redundancy.

Table 16-12 Configuration Data of VPLS Redundancy

Item		Data
C300-1	Loopback interface	- ID: 1 - IP address: 1.1.1.1/32
	L3 interface 1	- VLAN ID: 10 - Interface: gei_1/21/1 - IP address: 30.1.1.1/24
	L3 interface 2	- VLAN ID: 11 - Interface: gei_1/21/2 - IP address: 10.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW 1	- Name: 2and1000pw - Mode: dynamic pwe3 - Peer: 2.2.2.2 - Router ID: 2.2.2.2 - VC ID: 1000
	PW 2	- Name: 3and1000pw - Mode: dynamic pwe3 - Peer: 3.3.3.3 - Router ID: 3.3.3.3 - VC ID: 1000
	PW redundancy group	- Name: 2and3vcid1000 - Primary PW: 2and1000pw - Secondary PW: 3and1000pw
	VFI	- Name: redundancy-vfi - PW: 2and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
C300-2	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000
	Loopback interface ID	- ID: 1 - IP address: 2.2.2.2/32
	L3 interface	- VLAN ID: 11 - Interface: gei_1/21/1 - IP address: 10.1.1.1/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW	- Name: 1and1000pw - Mode: dynamic pwe3 - Peer: 1.1.1.1

Item		Data
		- Router ID: 1.1.1.1 - VC ID: 1000
	PW redundancy group	- Name: 2and3vcid1000 - PW: 1and1000pw
	VFI	- Name: redundancy-vfi - PW: 1and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000
C300-3	Loopback interface	- ID: 1 - IP address: 3.3.3.3/32
	L3 interface 2	- VLAN ID: 10 - Interface: gei_1/21/2 - IP address: 30.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW	- Name: 1and1000pw - Mode: dynamic pwe3 - Peer: 1.1.1.1 - Router ID: 1.1.1.1 - VC ID: 1000
	PW redundancy group	- Name: 2and3vcid1000 - PW: 1and1000pw
	VFI	- Name: redundancy-vfi - PW: 1and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000

**Note:**

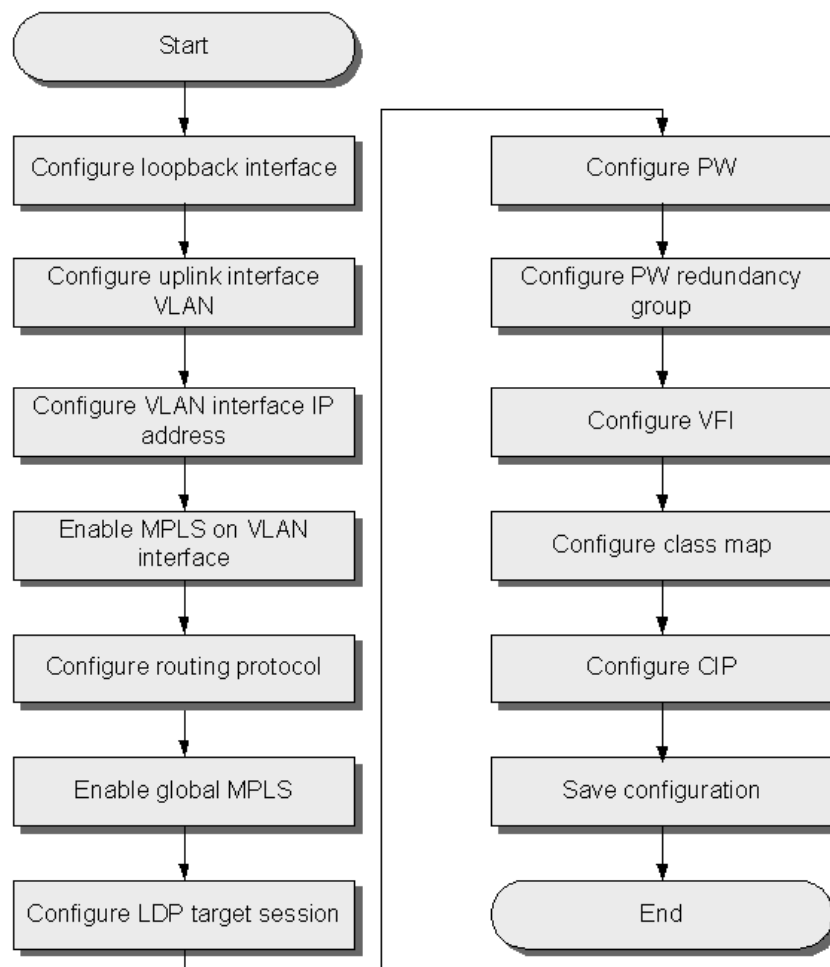
A class map defines the policy of class mapping.

A CIP is a logical interface that correlates to an L2VPN service instance, that is, an AC. The data on a CIP is a certain service flow, which is specified by a class map, from a UNI.

Configuration Flow

Figure 16-25 shows the configuration flowchart of the VPLS redundancy.

Figure 16-25 Configuration Flowchart of VPLS Redundancy



Steps

● Configuration on C300-1.

1. Configure the loopback interface.

```

ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 1.1.1.1 255.255.255.255
ZXAN(config-loopback1)#exit

```

2. Configure the VLAN of the uplink interfaces.

```

ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit

```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 30.1.1.1 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 1.1.1.1 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 30.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 2.2.2.2
ZXAN(config)#mpls ldp target-session 3.3.3.3
```

8. Configure the PWs.

```
ZXAN(config)#pw 2and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#peer 2.2.2.2 router-id 2.2.2.2 vcid 1000
ZXAN(config-pw)#exit
ZXAN(config)#pw 3and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#peer 3.3.3.3 router-id 3.3.3.3 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the PW redundancy group.

→ Non-negotiation, revertive, 1:1 mode


```

ZXAN(config)#pw-redundancy-group 2and3vcid1000
ZXAN(config-pw-group)#negotiation disable
ZXAN(config-pw-group)#protect-type 1by1
ZXAN(config-pw-group)#protect hold-off 0
ZXAN(config-pw-group)#protect mode revertive wtr 1
ZXAN(config-pw-group)#pw 2and1000pw
ZXAN(config-pw-group)#second-pw 3and1000pw
ZXAN(config-pw-group)#exit

```

→ **Non-negotiation, non-revertive, 1+1 mode**

```

ZXAN(config)#pw-redundancy-group 2and3vcid1000
ZXAN(config-pw-group)#negotiation disable
ZXAN(config-pw-group)#protect-type 1plus1 2send2recv
ZXAN(config-pw-group)#protect mode non-revertive
ZXAN(config-pw-group)#pw 2and1000pw
ZXAN(config-pw-group)#second-pw 3and1000pw
ZXAN(config-pw-group)#exit

```

→ **Negotiation, master, non-revertive, 1:1 mode**

```

ZXAN(config)#pw-redundancy-group 2and3vcid1000
ZXAN(config-pw-group)#negotiation enable master
ZXAN(config-pw-group)#protect-type 1by1
ZXAN(config-pw-group)#protect mode non-revertive
ZXAN(config-pw-group)#pw 2and1000pw
ZXAN(config-pw-group)#second-pw 3and1000pw
ZXAN(config-pw-group)#exit

```

10. Configure the VFI.

```

ZXAN(config)#vfi redundancy-vfi
ZXAN(config-vfi)#mpls xconnect pw 2and1000pw spoke
ZXAN(config-vfi)#exit

```

11. Configure the class map.

```

ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit

```

12. Configure the CIP.

```

ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect redundancy-vfi
ZXAN(config-cip1)#exit

```

13. Save the configuration data.

- **Configuration on C300-2.**

1. Configure the loopback interface.

```

ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 2.2.2.2 255.255.255.255

```

```
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interface.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 10.1.1.1 255.255.255.0
```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 2.2.2.2 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 1.1.1.1
```

8. Configure the PW.

```
ZXAN(config)#pw 1and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#peer 1.1.1.1 router-id 1.1.1.1 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the PW redundancy group.

→ Non-negotiation, revertive, 1:1 mode

You may skip this step.

→ Non-negotiation, non-revertive, 1+1 mode

You may skip this step.

→ Negotiation, slave, non-revertive, 1:1 mode

```
ZXAN(config)#pw-redundancy-group 2and3vcid1000
ZXAN(config-pw-group)#negotiation enable slave
ZXAN(config-pw-group)#protect-type 1by1
ZXAN(config-pw-group)#protect mode non-revertive
```

```
ZXAN(config-pw-group) #pw land1000pw
ZXAN(config-pw-group) #exit
```

10. Configure the VFI.

```
ZXAN(config) #vfi redundancy-vfi
ZXAN(config-vfi) #mpls xconnect pw land1000pw spoke
ZXAN(config-vfi) #exit
```

11. Configure the class map.

```
ZXAN(config) #class-map classmap1000
ZXAN(config-cmap) #match svlan-id 1000
ZXAN(config-cmap) #exit
```

12. Configure the CIP.

```
ZXAN(config) #cip 1
ZXAN(config-cip1) #service-type ethernet class-map classmap1000
ZXAN(config-cip1) #xconnect redundancy-vfi
ZXAN(config-cip1) #exit
```

13. Save the configuration data.

- Configuration on C300-3.

1. Configure the loopback interface.

```
ZXAN(config) #interface loopback1
ZXAN(config-loopback1) #ip address 3.3.3.3 255.255.255.255
ZXAN(config-loopback1) #exit
```

2. Configure the VLAN of the uplink interface.

```
ZXAN(config) #interface gei_1/21/2
ZXAN(config-if) #switchport vlan 10 tag
ZXAN(config-if) #exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config) #interface vlan 10
ZXAN(config-if-vlan10) #ip address 30.1.1.2 255.255.255.0
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config-if-vlan10) #mpls ip
ZXAN(config-if-vlan10) #exit
```

5. Configure the routing protocol.

```
ZXAN(config) #router ospf 1
ZXAN(config-router) #network 3.3.3.3 0.0.0.0 area 0.0.0.0
ZXAN(config-router) #network 30.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router) #exit
```

6. Enable the global MPLS function.

```
ZXAN(config) #vlan-reserve 4090 vpn
ZXAN(config) #ip mpls-forwarding enable
ZXAN(config) #mpls ldp router-id loopback1
```

```
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 1.1.1.1
```

8. Configure the PW.

```
ZXAN(config)#pw 1and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#peer 1.1.1.1 router-id 1.1.1.1 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the PW redundancy group.

→ Non-negotiation, revertive, 1:1 mode

You may skip this step.

→ Non-negotiation, non-revertive, 1+1 mode

You may skip this step.

→ Negotiation, slave, non-revertive, 1:1 mode

```
ZXAN(config)#pw-redundancy-group 2and3vcid1000
ZXAN(config-pw-group)#negotiation enable slave
ZXAN(config-pw-group)#protect-type 1by1
ZXAN(config-pw-group)#protect mode non-revertive
ZXAN(config-pw-group)#pw 1and1000pw
ZXAN(config-pw-group)#exit
```

10. Configure the VFI.

```
ZXAN(config)#vfi redundancy-vfi
ZXAN(config-vfi)#mpls xconnect pw 1and1000pw spoke
ZXAN(config-vfi)#exit
```

11. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

12. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect redundancy-vfi
ZXAN(config-cip1)#exit
```

13. Save the configuration data.

Result

When the primary PW is disabled, the secondary PW is activated, and the two CEs can ping each other successfully.

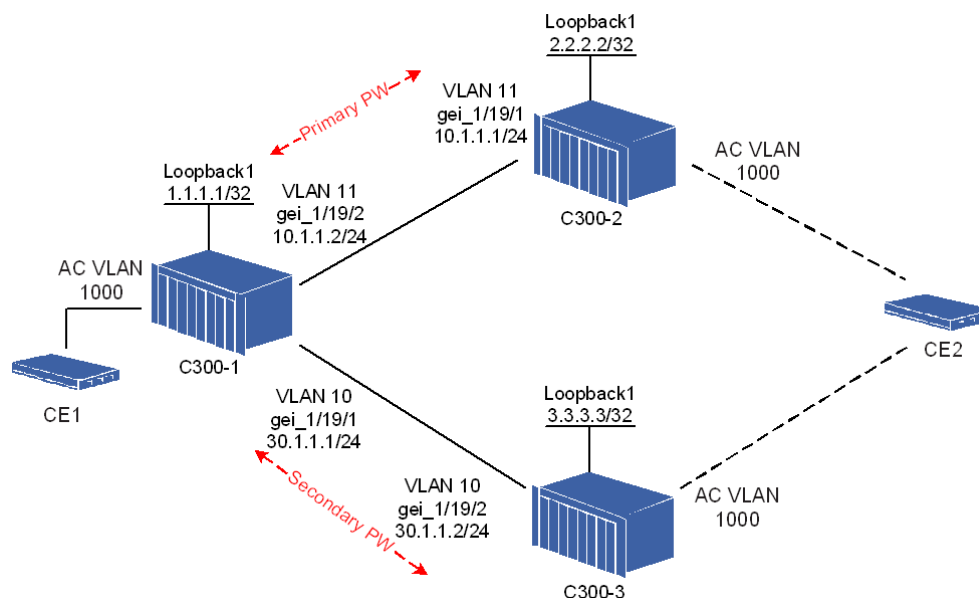
16.5.2 Configuring the VLL Resilience

VLL resilience enhances the stability and reliability of MPLS network.

Networking Diagram

Figure 16-26 shows the networking diagram of the VLL resilience.

Figure 16-26 Networking Diagram of VLL Resilience



The PW redundancy group works in either 1:1 or 1+1 mode.

- In 1:1 mode, only the active PW forwards the user traffic.
- In 1+1 mode, both PWs forward the user traffic.

The network supports the [OSPF](#) and [LDP](#) protocols.

Configuration Data

Table 16-13 lists the configuration data of the VLL resilience.

Table 16-13 Configuration Data of VLL Resilience

Item		Data
C300-1	Loopback interface	● ID: 1 ● IP address: 1.1.1.1/32
	L3 interface 1	● VLAN ID: 10 ● Interface: gei_1/21/1 ● IP address: 30.1.1.1/24
	L3 interface 2	● VLAN ID: 11 ● Interface: gei_1/21/2 ● IP address: 10.1.1.2/24

Item		Data
	Reserved VPN VLAN	VLAN ID: 4090
	PW 1	● Name: 2and1000pw ● Mode: dynamic pwe3 ● Type: ethernet ● Peer: 2.2.2.2 ● Router ID: 2.2.2.2 ● VC ID: 1000
	PW 2	● Name: 3and1000pw ● Mode: dynamic pwe3 ● Type: ethernet ● Peer: 3.3.3.3 ● Router ID: 3.3.3.3 ● VC ID: 1000
	PW redundancy group	● Name: 2and3vcid1000 ● Primary PW: 2and1000pw ● Secondary PW: 3and1000pw
	VLL	● Name: redundancy-vfi ● PW: 2and1000pw
	Class map	● Name: classmap1000 ● SVLAN ID: 1000
	CIP	● Index: 1 ● Service type: ethernet class-map ● Class map: classmap1000
C300-2	Loopback interface ID	● ID: 1 ● IP address: 2.2.2.2/32
	L3 interface	● VLAN ID: 11 ● Interface: gei_1/21/1 ● IP address: 10.1.1.1/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW	● Name: 1and1000pw ● Mode: dynamic pwe3 ● Type: ethernet ● Peer: 1.1.1.1 ● Router ID: 1.1.1.1 ● VC ID: 1000
	PW redundancy group	● Name: 2and3vcid1000 ● PW: 1and1000pw
	VLL	● Name: redundancy-vll ● PW: 1and1000pw

Item		Data
	Class map	● Name: classmap1000 ● SVLAN ID: 1000
	CIP	● Index: 1 ● Service type: ethernet class-map ● Class map: classmap1000
C300-3	Loopback interface	● ID: 1 ● IP address: 3.3.3.3/32
	L3 interface 2	● VLAN ID: 10 ● Interface: gei_1/21/2 ● IP address: 30.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW	● Name: 1and1000pw ● Mode: dynamic pwe3 ● Type: ethernet ● Peer: 1.1.1.1 ● Router ID: 1.1.1.1 ● VC ID: 1000
	PW redundancy group	● Name: 2and3vcid1000 ● PW: 1and1000pw
	VLL	● Name: redundancy-vll ● PW: 1and1000pw
	Class map	● Name: classmap1000 ● SVLAN ID: 1000
	CIP	● Index: 1 ● Service type: ethernet class-map ● Class map: classmap1000

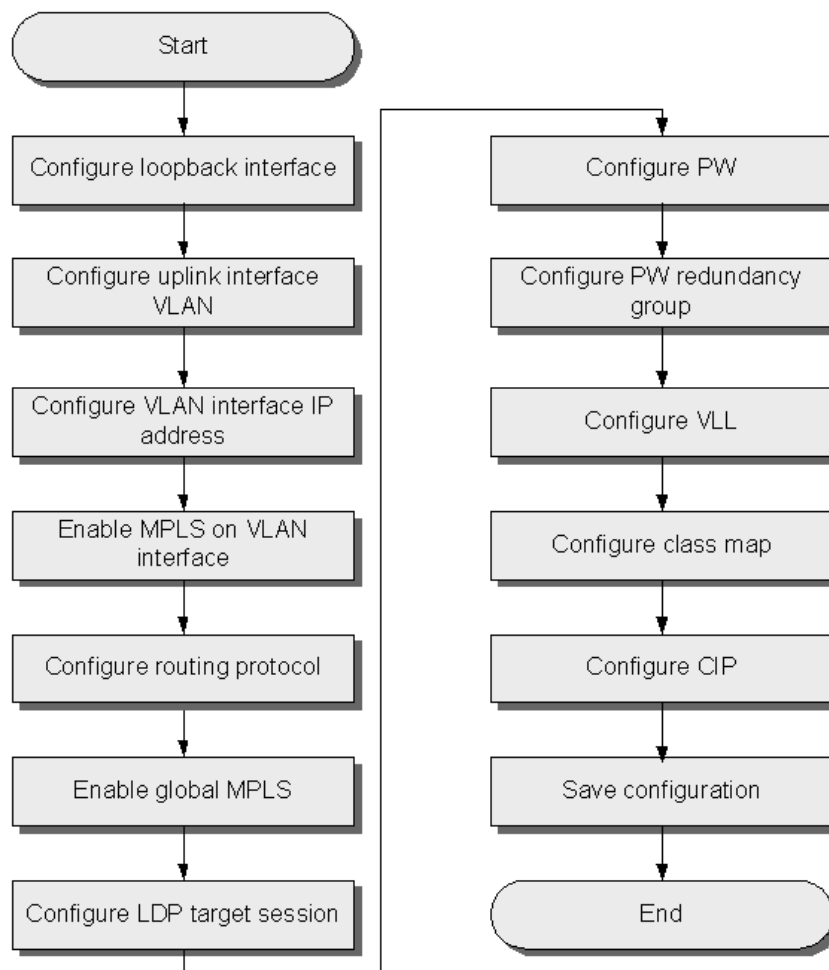
**Note:**

A class map defines the policy of class mapping.

A CIP is a logical interface that correlates to an L2VPN service instance, that is, an AC. The data on a CIP is a certain service flow, which is specified by a class map, from a UNI.

Configuration Flow

Figure 16-27 shows the configuration flowchart of the VLL resilience.

Figure 16-27 Configuration Flowchart of VLL Resilience

Steps

- Configuration on C300-1.
 1. Configure the loopback interface.


```

ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 1.1.1.1 255.255.255.255
ZXAN(config-loopback1)#exit
          
```
 2. Configure the VLAN of the uplink interfaces.


```

ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
          
```
 3. Configure the IP address of the VLAN interfaces.


```

ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 30.1.1.1 255.255.255.0
          
```



```
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 1.1.1.1 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 30.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 2.2.2.2
ZXAN(config)#mpls ldp target-session 3.3.3.3
```

8. Configure the PWs.

```
ZXAN(config)#pw 2and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#peer 2.2.2.2 router-id 2.2.2.2 vcid 1000
ZXAN(config-pw)#exit
ZXAN(config)#pw 3and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#peer 3.3.3.3 router-id 3.3.3.3 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the PW redundancy group.

→ Non-negotiation, revertive, 1:1 mode

```
ZXAN(config)#pw-redundancy-group 2and3vcid1000
ZXAN(config-pw-group)#negotiation disable
ZXAN(config-pw-group)#protect-type 1by1
```

```

ZXAN(config-pw-group)#protect hold-off 0
ZXAN(config-pw-group)#protect mode revertive wtr 1
ZXAN(config-pw-group)#pw 2and1000pw
ZXAN(config-pw-group)#second-pw 3and1000pw
ZXAN(config-pw-group)#exit

```

→ **Non-negotiation, non-revertive, 1+1 mode**

```

ZXAN(config)#pw-redundancy-group 2and3vcid1000
ZXAN(config-pw-group)#negotiation disable
ZXAN(config-pw-group)#protect-type 1plus1 2send2recv
ZXAN(config-pw-group)#protect mode non-revertive
ZXAN(config-pw-group)#pw 2and1000pw
ZXAN(config-pw-group)#second-pw 3and1000pw
ZXAN(config-pw-group)#exit

```

→ **Negotiation, master, non-revertive, 1:1 mode**

```

ZXAN(config)#pw-redundancy-group 2and3vcid1000
ZXAN(config-pw-group)#negotiation enable master
ZXAN(config-pw-group)#protect-type 1by1
ZXAN(config-pw-group)#protect mode non-revertive
ZXAN(config-pw-group)#pw 2and1000pw
ZXAN(config-pw-group)#second-pw 3and1000pw
ZXAN(config-pw-group)#exit

```

10. Configure the VLL.

```

ZXAN(config)#vll redundancy-vll
ZXAN(config-vll)#mpls xconnect pw 2and1000pw
ZXAN(config-vll)#exit

```

11. Configure the class map.

```

ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit

```

12. Configure the CIP.

```

ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect redundancy-vll
ZXAN(config-cip1)#oam-mapping enable ldp
ZXAN(config-cip1)#exit

```

13. Save the configuration data.

● **Configuration on C300-2.**

1. Configure the loopback interface.

```

ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 2.2.2.2 255.255.255.255
ZXAN(config-loopback1)#exit

```

2. Configure the VLAN of the uplink interface.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 10.1.1.1 255.255.255.0
```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 2.2.2.2 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 1.1.1.1
```

8. Configure the PW.

```
ZXAN(config)#pw 1and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwtype ethernet
ZXAN(config-pw)#peer 1.1.1.1 router-id 1.1.1.1 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the PW redundancy group.

→ Non-negotiation, revertive, 1:1 mode

You may skip this step.

→ Non-negotiation, non-revertive, 1+1 mode

You may skip this step.

→ Negotiation, slave, non-revertive, 1:1 mode

```
ZXAN(config)#pw-redundancy-group 2and3vcid1000
ZXAN(config-pw-group)#negotiation enable slave
ZXAN(config-pw-group)#protect-type 1by1
ZXAN(config-pw-group)#protect mode non-revertive
ZXAN(config-pw-group)#pw 1and1000pw
```

```
ZXAN(config-pw-group)#exit
```

10. Configure the VLL.

```
ZXAN(config)#vll redundancy-vll
ZXAN(config-vll)#mpls xconnect pw land1000pw
ZXAN(config-vll)#exit
```

11. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

12. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect redundancy-vll
ZXAN(config-cip1)#oam-mapping enable ldp
ZXAN(config-cip1)#exit
```

13. Save the configuration data.

● Configuration on C300-3.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 3.3.3.3 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interface.

```
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 30.1.1.2 255.255.255.0
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 3.3.3.3 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 30.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
```

```
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 1.1.1.1
```

8. Configure the PW.

```
ZXAN(config)#pw 1and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwtpe ethernet
ZXAN(config-pw)#peer 1.1.1.1 router-id 1.1.1.1 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the PW redundancy group.

→ Non-negotiation, revertive, 1:1 mode

You may skip this step.

→ Non-negotiation, non-revertive, 1+1 mode

You may skip this step.

→ Negotiation, slave, non-revertive, 1:1 mode

```
ZXAN(config)#pw-redundancy-group 2and3vcid1000
ZXAN(config-pw-group)#negotiation enable slave
ZXAN(config-pw-group)#protect-type 1by1
ZXAN(config-pw-group)#protect mode non-revertive
ZXAN(config-pw-group)#pw 1and1000pw
ZXAN(config-pw-group)#exit
```

10. Configure the VLL.

```
ZXAN(config)#vll redundancy-vll
ZXAN(config-vll)#mpls xconnect pw 1and1000pw
ZXAN(config-vll)#exit
```

11. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

12. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect redundancy-vll
ZXAN(config-cip1)#oam-mapping enable ldp
ZXAN(config-cip1)#exit
```

13. Save the configuration data.

Result

When the primary PW is disabled, the secondary PW is activated, and the two CEs can ping each other successfully.

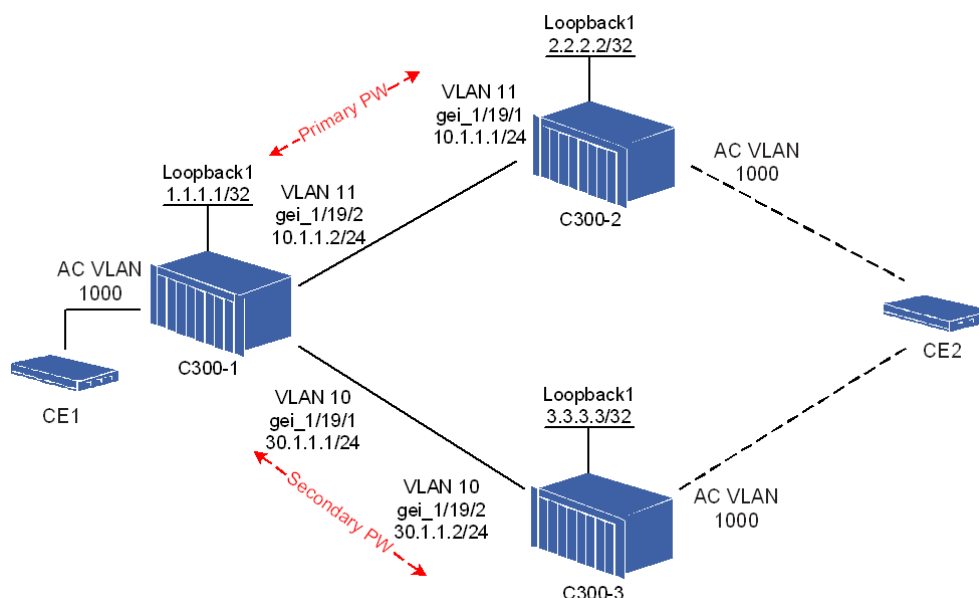
16.5.3 Configuring the PW Redundancy and BFD

Bidirectional Forwarding Detection (BFD), which can detect the faults on LSPs, is used to help PW redundancy to enhance the stability and reliability of MPLS network.

Networking Diagram

Figure 16-28 shows the networking diagram of the PW redundancy and BFD.

Figure 16-28 Networking Diagram of PW Redundancy and BFD



The PW redundancy group works in non-negotiation 1:1 mode. The primary PW is revertive when it recovers.

The network supports the OSPF and LDP protocols.

Configuration Data

Table 16-14 lists the configuration data of the PW redundancy and BFD.

Table 16-14 Configuration Data of PW Redundancy and BFD

Item		Data
C300-1	Loopback interface	● ID: 1 ● IP address: 1.1.1.1/32
	L3 interface 1	● VLAN ID: 10 ● Interface: gei_1/21/1 ● IP address: 30.1.1.1/24
	L3 interface 2	● VLAN ID: 11 ● Interface: gei_1/21/2 ● IP address: 10.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW class	● Name: pw1 ● Control word: not preferred ● Control Channel (CC) type: type2 (out-of-band) ● Connection Verification (CV) type: lsp-ping bfd-ipudp-fo
	PW 1	● Name: 2and1000pw ● Mode: dynamic pwe3 ● PW class: pw1 ● Peer: 2.2.2.2 ● Router ID: 2.2.2.2 ● VC ID: 1000
	PW 2	● Name: 3and1000pw ● Mode: dynamic pwe3 ● PW class: pw1 ● Peer: 3.3.3.3 ● Router ID: 3.3.3.3 ● VC ID: 1000
	PW redundancy group	● Name: 2and3vcid1000 ● Negotiation: disable ● Protect type: 1by1 ● Protect mode: revertive ● Primary PW: 2and1000pw ● Secondary PW: 3and1000pw
	BFD session 1	● PW name: 2and1000pw ● Tx interval: 10 ms ● Rx interval: 10 ms ● Time-out detection multiplier: 3
	BFD session 2	● PW name: 3and1000pw ● Tx interval: 10 ms

Item		Data
		- Rx interval: 10 ms - Time-out detection multiplier: 3
	VLL	- Name: redundancy-vll - PW: 2and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000
C300-2	Loopback interface ID	- ID: 1 - IP address: 2.2.2.2/32
	L3 interface	- VLAN ID: 11 - Interface: gei_1/21/1 - IP address: 10.1.1.1/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW class	- Name: pw1 - Control word: not preferred - Control Channel (CC) type: type2 (out-of-band) - Connection Verification (CV) type: lsp-ping bfd-ipudp-fo
	PW	- Name: 1and1000pw - Mode: dynamic pwe3 - PW class: pw1 - Peer: 1.1.1.1 - Router ID: 1.1.1.1 - VC ID: 1000
	BFD session	- PW name: 1and1000pw - Tx interval: 10 ms - Rx interval: 10 ms - Time-out detection multiplier: 3
	VLL	- Name: redundancy-vll - PW: 1and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000

Item		Data
C300-3	Loopback interface	● ID: 1 ● IP address: 3.3.3.3/32
	L3 interface 2	● VLAN ID: 10 ● Interface: gei_1/21/2 ● IP address: 30.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW	● Name: 1and1000pw ● Mode: dynamic pwe3 ● Peer: 1.1.1.1 ● Router ID: 1.1.1.1 ● VC ID: 1000
	VLL	● Name: redundancy-vll ● PW: 1and1000pw
	Class map	● Name: classmap1000 ● SVLAN ID: 1000
	CIP	● Index: 1 ● Service type: ethernet class-map ● Class map: classmap1000

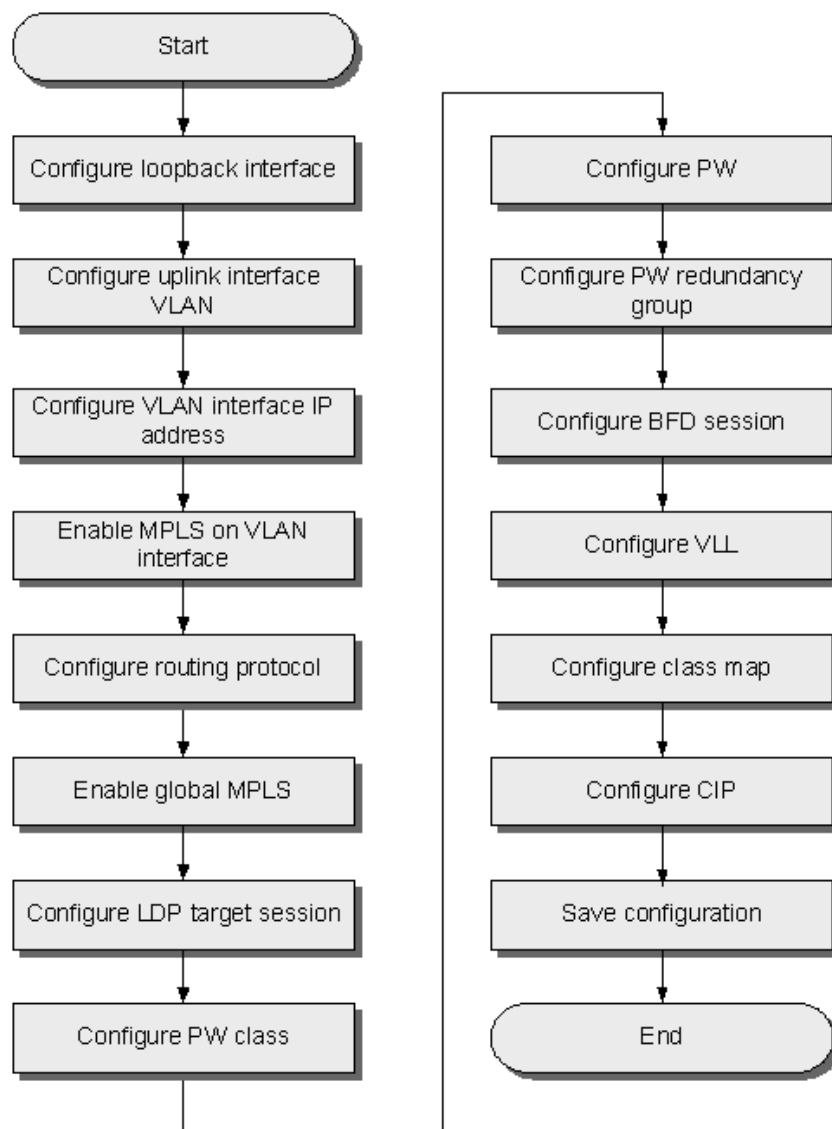
**Note:**

A class map defines the policy of class mapping.

A CIP is a logical interface that correlates to an L2VPN service instance, that is, an AC. The data on a CIP is a certain service flow, which is specified by a class map, from a UNI.

Configuration Flow

Figure 16-29 shows the configuration flowchart of the PW redundancy and BFD.

Figure 16-29 Configuration Flowchart of PW Redundancy and BFD

Steps

- Configuration on C300-1.

1. Configure the loopback interface.

```

ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 1.1.1.1 255.255.255.255
ZXAN(config-loopback1)#exit

```

2. Configure the VLAN of the uplink interfaces.

```

ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit

```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 30.1.1.1 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 1.1.1.1 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 30.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 2.2.2.2
ZXAN(config)#mpls ldp target-session 3.3.3.3
```

8. Configure the PW class.

```
ZXAN(config)#pw-class pw1
ZXAN(config-pw-class)#control-word not-preferred
ZXAN(config-pw-class)#cctype type2
ZXAN(config-pw-class)#cvtype lsp-ping bfd-ipudp-fo
ZXAN(config-pw-class)#exit
```

9. Configure the PWs.

```
ZXAN(config)#pw 2and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwtype ethernet
ZXAN(config-pw)#apply pw-class pw1
ZXAN(config-pw)#peer 2.2.2.2 router-id 2.2.2.2 vcid 1000
ZXAN(config-pw)#exit
```

```
ZXAN(config)#pw 3and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#apply pw-class pw1
ZXAN(config-pw)#peer 3.3.3.3 router-id 3.3.3.3 vcid 1000
ZXAN(config-pw)#exit
```

10. Configure the PW redundancy group.

```
ZXAN(config)#pw-redundancy-group 2and3vcid1000
ZXAN(config-pw-group)#negotiation disable
ZXAN(config-pw-group)#protect-type 1by1
ZXAN(config-pw-group)#protect hold-off 0
ZXAN(config-pw-group)#protect mode revertive wtr 1
ZXAN(config-pw-group)#pw 2and1000pw
ZXAN(config-pw-group)#second-pw 3and1000pw
ZXAN(config-pw-group)#exit
```

11. Configure the BFD session.

```
ZXAN(config)#mpls pw bfd 2and1000pw interval 10 min-rx 10 multiplier 3
ZXAN(config)#mpls pw bfd 3and1000pw interval 10 min-rx 10 multiplier 3
```

12. Configure the VLL.

```
ZXAN(config)#vll redundancy-vll
ZXAN(config-vll)#mpls xconnect pw 2and1000pw
ZXAN(config-vll)#exit
```

13. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

14. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect redundancy-vll
ZXAN(config-cip1)#oam-mapping enable ldp
ZXAN(config-cip1)#exit
```

15. Save the configuration data.

- Configuration on C300-2.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 2.2.2.2 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interface.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
```

```
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 10.1.1.1 255.255.255.0
```

4. Enable the MPLS on the VLAN interface.

```
ZXAN(config-if-vlan11)#mpls ip
ZXAN(config-if-vlan11)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 2.2.2.2 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 1.1.1.1
```

8. Configure the PW class.

```
ZXAN(config)#pw-class pw1
ZXAN(config-pw-class)#control-word not-preferred
ZXAN(config-pw-class)#cctype type2
ZXAN(config-pw-class)#cvtype lsp-ping bfd-ipudp-fo
ZXAN(config-pw-class)#exit
```

9. Configure the PW.

```
ZXAN(config)#pw 1and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#apply pw-class pw1
ZXAN(config-pw)#peer 1.1.1.1 router-id 1.1.1.1 vcid 1000
ZXAN(config-pw)#exit
```

10. Configure the BFD session.

```
ZXAN(config)#mpls pw bfd 1and1000pw interval 10 min-rx 10 multiplier 3
```

11. Configure the VLL.

```
ZXAN(config)#vll redundancy-vll
ZXAN(config-vll)#mpls xconnect pw 1and1000pw
ZXAN(config-vll)#exit
```

12. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

13. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect redundancy-vll
ZXAN(config-cip1)#oam-mapping enable ldp
ZXAN(config-cip1)#exit
```

14. Save the configuration data.

- Configuration on C300-3.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-loopback1)#ip address 3.3.3.3 255.255.255.255
ZXAN(config-loopback1)#exit
```

2. Configure the VLAN of the uplink interface.

```
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interface.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 30.1.1.2 255.255.255.0
```

4. Enable the MPLS on the VLAN interfaces.

```
ZXAN(config-if-vlan10)#mpls ip
ZXAN(config-if-vlan10)#exit
```

5. Configure the routing protocol.

```
ZXAN(config)#router ospf 1
ZXAN(config-router)#network 3.3.3.3 0.0.0.0 area 0.0.0.0
ZXAN(config-router)#network 30.1.1.0 0.0.0.255 area 0.0.0.0
ZXAN(config-router)#exit
```

6. Enable the global MPLS function.

```
ZXAN(config)#vlan-reserve 4090 vpn
ZXAN(config)#ip mpls-forwarding enable
ZXAN(config)#mpls ldp router-id loopback1
ZXAN(config)#mpls ip
```

7. Configure the LDP target session.

```
ZXAN(config)#mpls ldp target-session 1.1.1.1
```

8. Configure the PW.

```
ZXAN(config)#pw 1and1000pw
ZXAN(config-pw)#mode dynamic pwe3 fec128
```

```
ZXAN(config-pw)#pwttype ethernet
ZXAN(config-pw)#peer 1.1.1.1 router-id 1.1.1.1 vcid 1000
ZXAN(config-pw)#exit
```

9. Configure the VLL.

```
ZXAN(config)#vll redundancy-vll
ZXAN(config-vll)#mpls xconnect pw 1and1000pw
ZXAN(config-vll)#exit
```

10. Configure the class map.

```
ZXAN(config)#class-map classmap1000
ZXAN(config-cmap)#match svlan-id 1000
ZXAN(config-cmap)#exit
```

11. Configure the CIP.

```
ZXAN(config)#cip 1
ZXAN(config-cip1)#service-type ethernet class-map classmap1000
ZXAN(config-cip1)#xconnect redundancy-vll
ZXAN(config-cip1)#oam-mapping enable ldp
ZXAN(config-cip1)#exit
```

12. Save the configuration data.

Result

When the primary PW is disabled, the secondary PW is activated, and the two CEs can ping each other successfully.

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Chapter 17

OAM Configuration

Table of Contents

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BFD Configuration.....17-7

17.1 Ethernet OAM Configuration

The ZX10 C300 supports service layer OAM function, which includes link continuity check, port loopback detection, link trace, and alarm notification.

17.1.1 Configuring the CCM Function

CCM is used to ensures the continuity between MPs in an MA.

Context

A MEG End Point (MEP) sends Continuity Check Message (CCM) packets periodically, which ensures the continuity of Maintenance Points (MPs) in the corresponding Maintenance Association (MA). The MPs that receive the packets need not to respond.

Configuration Data

Table 17-1 list the configuration data of the CCM function.

Table 17-1 Configuration Data of CCM Function

Item	Data
MD	● Session ID: 1 ● Name: md1 ● Level: 3
MA	● Session ID: 1 ● Name: ma1 ● Protection mode: VLAN protection ● Primary VLAN ID: 100
Local MEP	● Session ID: 1 ● MEP ID: 1 ● Direction: down

Item	Data
Remote MEP	● Session ID: 2 ● MEP ID: 2 ● Remote MAC address: 00d0.d058.6958
Uplink interface	gei_1/21/1

**Note:**

The remote MAC address of the remote MEP is the in-band MAC address of the local MEP.

Steps

1. Enable the CFM function.

```
ZXAN(config)#cfm enable
```

2. Create the Ethernet OAM MD.

```
ZXAN(config)#cfm create md session 1 name mdl level 3
```

3. Create the Ethernet OAM MA.

```
ZXAN(config-mdl)#create ma session 1 format icc-based name mal
```

4. Configure the MA protection mode.

```
ZXAN(config-mdl-mal)#protect vlan
```

**Note:**

Only VLAN protection mode is valid.

5. Configure the MA primary VLAN.

```
ZXAN(config-mdl-mal)#primary vlan 100
```

**Note:**

In the Ethernet OAM MDs of the same level, primary VLAN of MAs is unique.

6. Configure CCM interval.

```
ZXAN(config-mdl-mal)#ccm timer-interval 2
```

**Note:**

The ZXA10 C300 supports seven intervals:

- 1: 3.3 ms
- 2: 10 ms
- 3: 100 ms
- 4: 1 s
- 5: 10 s
- 6: 1 min
- 7: 10 min

7. Create local Ethernet OAM MEP.

```
ZXAN(config-md1-ma1)#create mep session 1 1 direction down
```

8. Assign local MEP to the uplink port.

```
ZXAN(config-md1-ma1)#assign mep 1 to interface gei_1/21/1
```

9. Enable the MEP.

```
ZXAN(config-md1-ma1)#mep 1 state enable
```

10. Configure the remote MEP.

```
ZXAN(config-md1-ma1)#create rmp session 2 2 remote-mac 00d0.d058.6958
```

11. Enable the CCM-send function on local MEP.

```
ZXAN(config-md1-ma1)#mep 1 ccm-send unicast enable
```

12. Configure the uplink port VLAN of local MEP.

```
ZXAN(config)#interface gei_1/21/1  
ZXAN(config-if)#switchport vlan 100 tag
```

– End of Steps –

Result

If the ZXA10 C300 receives error CCM packets or does not receive any CCM packets, there will be a CCM alarm information on the NMS. You can query the CCM alarm information using the **show cfm mp all md 1 ma 1 detail** command.

17.1.2 Configuring the LBM Function

The Ethernet LBM, which is an optional OAM function, is used to check the bidirectional continuity between an MEP and an MIP, or between an MEP to one or multiple MEPs.

Configuration Data

Table 17-2 list the configuration data of the Loopback Message (LBM) function.

Table 17-2 Configuration Data of LBM Function

Item	Data
MD	● Session ID: 1 ● Name: md1 ● Level: 3
MA	● Session ID: 1 ● Name: ma1 ● Protection mode: VLAN protection ● Primary VLAN ID: 100
Local MEP	● Session ID: 1 ● MEP ID: 1 ● Direction: down
Remote MEP	● Session ID: 2 ● MEP ID: 2 ● Remote MAC address: 00d0.d058.6958
Uplink interface	gei_1/21/1

**Note:**

The remote MAC address of the remote MEP is the in-band MAC address of the local MEP.

Steps

1. Enable the CFM function.

```
ZXAN(config)#cfm enable
```

2. Create the Ethernet OAM MD.

```
ZXAN(config)#cfm create md session 1 name md1 level 3
```

3. Create the Ethernet OAM MA.

```
ZXAN(config-md1)#create ma session 1 format icc-based name ma1
```

4. Configure the MA protection mode.

```
ZXAN(config-md1-ma1)#protect vlan
```

**Note:**

Only VLAN protection mode is valid.

5. Configure the MA primary VLAN.

```
ZXAN(config-md1-ma1)#primary vlan 100
```

**Note:**

In the Ethernet OAM MDs of the same level, primary VLAN of MAs is unique.

6. Create local Ethernet OAM MEP.

```
ZXAN(config-md1-ma1)#create mep session 1 1 direction down
```

7. Assign local MEP to the uplink port.

```
ZXAN(config-md1-ma1)#assign mep 1 to interface gei_1/21/1
```

8. Enable the MEP.

```
ZXAN(config-md1-ma1)#mep 1 state enable
```

9. Configure the remote MEP.

```
ZXAN(config-md1-ma1)#create rmep session 2 2 remote-mac 00d0.d058.6958
```

10. Configure the uplink port VLAN of local MEP.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 100 tag
ZXAN(config-if)#end
```

11. In administrator mode, carry out the LBM function.

```
ZXAN#cfm lbm check unicast md 1 ma 1 smep-id 1 dmep-id 2
```

– End of Steps –

Result

In the CLI window, a message shows that whether the destination MEP (or MIP) is reachable.

17.1.3 Configuring the LTM Function

The LTM function is used to trace the link between two MEPs.

Context

When you carry out the Link Trace Message (LTM) function on an MEP, the MEP sends LTM packets to trace the path to the destination MAC address. MIPs forward the messages until the messages arrive the destination MEP or the messages cannot be forwarded any more. When MPs on the path receive the LTM packets, each MP responds to the source MEP with an LTR message. When the destination is an MIP, the MIP does not forward the LTM anymore.

Steps

1. Enable the [CFM](#) function.

```
ZXAN(config)#cfm enable
```

2. Create the Ethernet OAM [MD](#).

```
ZXAN(config)#cfm create md session 1 name mdl level 3
```

3. Create the Ethernet OAM [MA](#).

```
ZXAN(config-mdl)#create ma session 1 format icc-based name mal
```

4. Configure the MA protection mode.

```
ZXAN(config-mdl-mal)#protect vlan
```

**Note:**

Only VLAN protection mode is valid.

5. Configure the MA primary VLAN.

```
ZXAN(config-mdl-mal)#primary vlan 100
```

**Note:**

In the Ethernet OAM MDs of the same level, primary VLAN of MAs is unique.

6. Create local Ethernet OAM [MEP](#).

```
ZXAN(config-mdl-mal)#create mep session 1 1 direction down
```

7. Assign local MEP to the uplink port.

```
ZXAN(config-mdl-mal)#assign mep 1 to interface gei_1/21/1
```

8. Enable the MEP.

```
ZXAN(config-mdl-mal)#mep 1 state enable
```

9. Configure the remote MEP.

```
ZXAN(config-mdl-mal)#create rmp session 2 2 remote-mac 00d0.d058.6958
```

10. Configure the uplink port VLAN of local MEP.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 100 tag
ZXAN(config-if)#end
```

11. In the administrator mode, carry out the LTM function.

```
ZXAN#cfm ltm md 1 ma 1 smep-id 1 dmep-id 2
```

– End of Steps –

Result

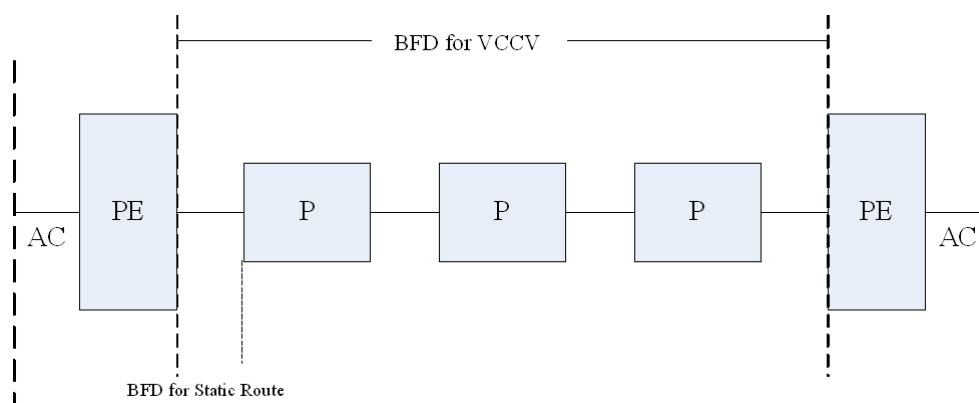
In the CLI window, a message shows the MEP (or MIP) path and whether the destination MEP is reachable.

17.2 BFD Configuration

Bidirectional Forwarding Detection (BFD) is a network protocol used to detect faults between two forwarding engines connected by a link. It provides low-overhead detection of faults even on physical media that do not support failure detection of any kind, such as Ethernet, virtual circuits, tunnels, and MPLS Label Switched Paths.

Figure 17-1 shows the different BFD applications in a network.

Figure 17-1 BFD Applications



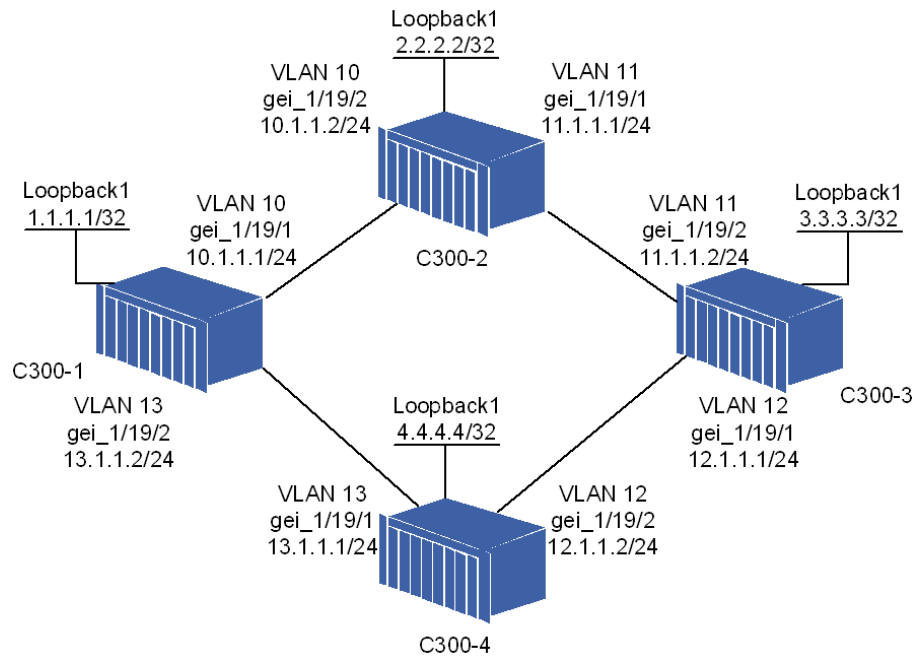
17.2.1 Configuring the BFD for Static Routes

BFD can be used to detect faults on static routes.

Networking Diagram

Figure 17-2 shows the networking diagram of the BFD for static routes.

Figure 17-2 Networking Diagram of BFD for Static Routes



There are two static routes between C300-1 and C300-3.

- The active route is: C300-1 > C300-2 > C300-3.
- The backup route is: C300-1 > C300-4 > C300-3.

When BFD detects route faults between **PEs** and connected equipment, the control panel or forwarding panel performs the **FRR** to protect the service.

Configuration Data

Table 17-3 lists the configuration data of the BFD for static routes.

Table 17-3 Configuration Data of BFD for Static Routes

Item		Data
C300-1	Loopback interface	● ID: 1 ● IP address: 1.1.1.1/32
	L3 interface 1	● VLAN ID: 10 ● Interface: gei_1/21/1 ● IP address: 10.1.1.1/24
	L3 interface 2	● VLAN ID: 13 ● Interface: gei_1/21/2 ● IP address: 13.1.1.12/24
	Active route	● Destination IP address: 3.3.3.3/32 ● Next hop IP address: 10.1.1.2/24
	Backup route	● Destination IP address: 3.3.3.3/32 ● Next hop IP address: 13.1.1.1/24 ● Tag value: 200 ● Priority: 60

Item		Data
C300-2	Loopback interface	● ID: 1 ● IP address: 2.2.2.2/32
	L3 interface 1	● VLAN ID: 11 ● Interface: gei_1/21/1 ● IP address: 11.1.1.1/24
	L3 interface 2	● VLAN ID: 10 ● Interface: gei_1/21/2 ● IP address: 10.1.1.2/24
C300-3	Loopback interface	● ID: 1 ● IP address: 3.3.3.3/32
	L3 interface 1	● VLAN ID: 12 ● Interface: gei_1/21/1 ● IP address: 12.1.1.1/24
	L3 interface 2	● VLAN ID: 11 ● Interface: gei_1/21/2 ● IP address: 11.1.1.2/24
C300-4	Loopback interface	● ID: 1 ● IP address: 4.4.4.4/32
	L3 interface 1	● VLAN ID: 13 ● Interface: gei_1/21/1 ● IP address: 13.1.1.1/24
	L3 interface 2	● VLAN ID: 12 ● Interface: gei_1/21/2 ● IP address: 12.1.1.2/24

Steps

- Configuration on C300-1.
 1. Configure the loopback interface.


```
ZXAN(config)#interface loopback1
ZXAN(config-if)#ip address 1.1.1.1 255.255.255.255
ZXAN(config-if)#exit
```
 2. Configure the VLAN of the uplink interfaces.


```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 13 tag
ZXAN(config-if)#exit
```
 3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.1 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 13
ZXAN(config-if-vlan13)#ip address 13.1.1.2 255.255.255.0
ZXAN(config-if-vlan13)#exit
```

4. Configure the routes.

```
ZXAN(config)#ip route 3.3.3.3 255.255.255.255 10.1.1.2
ZXAN(config)#ip route 3.3.3.3 255.255.255.255 13.1.1.1 200 slave tag 200
```

5. Configure the BFD.

```
ZXAN(config)#ip static-bfd 10.1.1.1 10.1.1.2
```

● Configuration on C300-2.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-if)#ip address 2.2.2.2 255.255.255.255
ZXAN(config-if)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 11 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 10 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 10
ZXAN(config-if-vlan10)#ip address 10.1.1.2 255.255.255.0
ZXAN(config-if-vlan10)#exit
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.1 255.255.255.0
ZXAN(config-if-vlan11)#exit
```

● Configuration on C300-3.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-if)#ip address 3.3.3.3 255.255.255.255
ZXAN(config-if)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 12 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 11 tag
```

```
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 11
ZXAN(config-if-vlan11)#ip address 11.1.1.2 255.255.255.0
ZXAN(config-if-vlan11)#exit
ZXAN(config)#interface vlan 12
ZXAN(config-if-vlan12)#ip address 12.1.1.1 255.255.255.0
ZXAN(config-if-vlan12)#exit
```

- Configuration on C300-4.

1. Configure the loopback interface.

```
ZXAN(config)#interface loopback1
ZXAN(config-if)#ip address 4.4.4.4 255.255.255.255
ZXAN(config-if)#exit
```

2. Configure the VLAN of the uplink interfaces.

```
ZXAN(config)#interface gei_1/21/1
ZXAN(config-if)#switchport vlan 13 tag
ZXAN(config-if)#exit
ZXAN(config)#interface gei_1/21/2
ZXAN(config-if)#switchport vlan 12 tag
ZXAN(config-if)#exit
```

3. Configure the IP address of the VLAN interfaces.

```
ZXAN(config)#interface vlan 12
ZXAN(config-if-vlan12)#ip address 12.1.1.2 255.255.255.0
ZXAN(config-if-vlan12)#exit
ZXAN(config)#interface vlan 13
ZXAN(config-if-vlan13)#ip address 13.1.1.1 255.255.255.0
ZXAN(config-if-vlan13)#exit
```

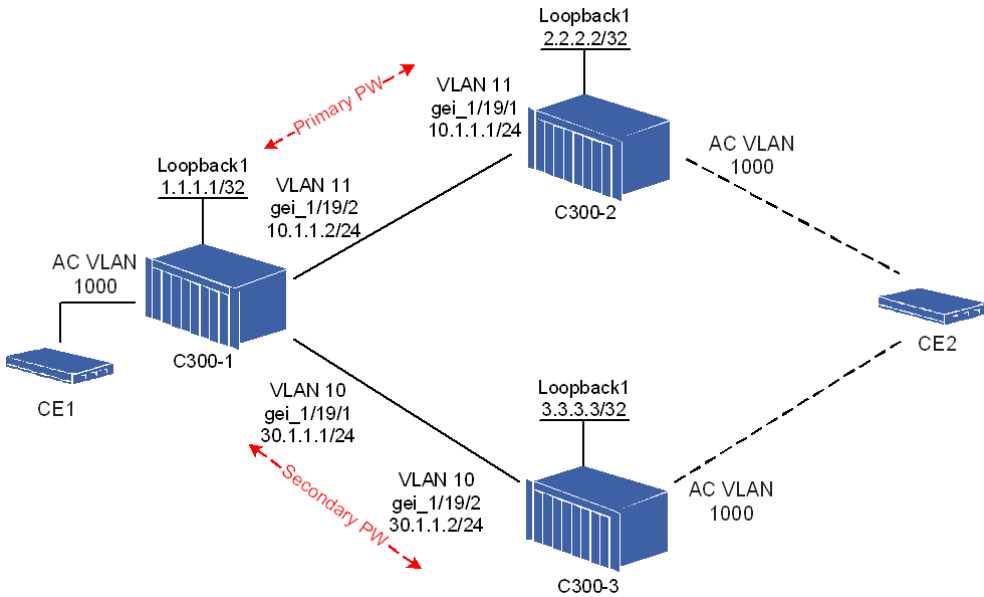
17.2.2 Configuring the BFD for VCCV

BFD can be used for Virtual Circuit Connectivity Verification (**VCCV**) in MPLS network OAM.

Networking Diagram

Figure 17-3 shows the networking diagram of the **BFD** for VCCV.

Figure 17-3 Networking Diagram of BFD for VCCV



The PW redundancy group works in non-negotiation 1:1 mode. The primary PW is revertive when it recovers.

BFD packets that notify PW alarms are encapsulated in VCCV mode (IP/UDP BFD encapsulation).

Configuration Data

Table 17-4 lists the configuration data of the BFD for VCCV.

Table 17-4 Configuration Data of BFD for VCCV

Item		Data
C300-1	Loopback interface	- ID: 1 - IP address: 1.1.1.1/32
	L3 interface 1	- VLAN ID: 10 - Interface: gei_1/21/1 - IP address: 30.1.1.1/24
	L3 interface 2	- VLAN ID: 11 - Interface: gei_1/21/2 - IP address: 10.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW class	- Name: pw1 - Control word: not preferred - Control Channel (CC) type: type2 (out-of-band) - Connection Verification (CV) type: lsp-ping bfd-ipudp-fo

Item		Data
	PW 1	- Name: 2and1000pw - Mode: dynamic pwe3 - PW class: pw1 - Peer: 2.2.2.2 - Router ID: 2.2.2.2 - VC ID: 1000
	PW 2	- Name: 3and1000pw - Mode: dynamic pwe3 - PW class: pw1 - Peer: 3.3.3.3 - Router ID: 3.3.3.3 - VC ID: 1000
	PW redundancy group	- Name: 2and3vcid1000 - Negotiation: disable - Protect type: 1by1 - Protect mode: revertive
	BFD session 1	- PW name: 2and1000pw - Tx interval: 10 ms - Rx interval: 10 ms - Time-out detection multiplier: 3
	BFD session 2	- PW name: 3and1000pw - Tx interval: 10 ms - Rx interval: 10 ms - Time-out detection multiplier: 3
	VLL	- Name: redundancy-vll - PW: 2and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000
C300-2	Loopback interface ID	- ID: 1 - IP address: 2.2.2.2/32
	L3 interface	- VLAN ID: 11 - Interface: gei_1/21/1 - IP address: 10.1.1.1/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW class	- Name: pw1 - Control word: not preferred

Item		Data
		- Control Channel (CC) type: type2 (out-of-band) - Connection Verification (CV) type: lsp-ping bfd-ipudp-fo
	PW	- Name: 1and1000pw - Mode: dynamic pwe3 - PW class: pw1 - Peer: 1.1.1.1 - Router ID: 1.1.1.1 - VC ID: 1000
	BFD session	- PW name: 1and1000pw - Tx interval: 10 ms - Rx interval: 10 ms - Time-out detection multiplier: 3
	VLL	- Name: redundancy-vll - PW: 1and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000
C300-3	Loopback interface	- ID: 1 - IP address: 3.3.3.3/32
	L3 interface 2	- VLAN ID: 10 - Interface: gei_1/21/2 - IP address: 30.1.1.2/24
	Reserved VPN VLAN	VLAN ID: 4090
	PW	- Name: 1and1000pw - Mode: dynamic pwe3 - Peer: 1.1.1.1 - Router ID: 1.1.1.1 - VC ID: 1000
	VLL	- Name: redundancy-vll - PW: 1and1000pw
	Class map	- Name: classmap1000 - SVLAN ID: 1000
	CIP	- Index: 1 - Service type: ethernet class-map - Class map: classmap1000

Steps

For steps of configuring the BFD for VCCV, refer to **Steps** in [16.5.3 Configuring the PW Redundancy and BFD](#).

Result

Execute the commands **ping mpls ipv4 2.2.2.2 32 repeat 10** and **ping mpls pseudowire pw-name 2and1000pw** on C300-1. These pings are successful.

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Chapter 18

Route Protocol Configuration

Besides static route, the ZXA10 C300 supports the following routing protocols:

- OSPF
- BGP

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Configuring the BGP	18-2

18.1 Configuring the Static Route

This section describes how to implement the static route of the ZXA10 C300 by configuring the next hop address to the destination network segment.

Context

Static route is the route info added into the routing table by the network administrator via the configuration command. You can using static route with a few configurations to avoid using dynamic routing. In the case that multiple routers and multiple paths exist, however, dynamic routing is recommended.

Steps

1. In global configuration mode, configure the static route.

```
ZXAN(config)#ip route 10.1.1.0 255.255.255.0 1.1.1.2
```

– End of Steps –

18.2 Configuring the OSPF Protocol

This section describes how to implement the ZXA10 C300's access to the adjacent router by configuring the OSPF protocol.

Context

OSPF is an Interior Gateway Protocol (IGP), used to determine the route in a single Autonomous System (AS). OSPF is a link-state routing protocol. It overcomes the weaknesses of RIP and other distance vector protocol.

OSPF version 1 is defined in RFC1131. OSPF version 2 is defined in RFC2328. The ZXA10 C300 supports OSPF version 2.

Steps

1. In global configuration mode, enable OSPF.

```
ZXAN(config)#router ospf 1
```

```
ZXAN(config-router)#
```

2. Configure the network segment of the interface.

```
ZXAN(config-router)#network 10.1.1.0 0.0.0.255 area 0
```

– End of Steps –

18.3 Configuring the BGP

This section describes how to implement the ZXA10 C300's access to the adjacent router by configuring the BGP.

Context

BGP is an inter-**AS** routing protocol. It involves a table of IP networks or 'prefixes' which designates network reachability among AS. BGP is a path vector protocol, or a variant of a Distance-vector routing protocol. BGP does not involve traditional **IGP** metrics, but routing decisions are made based on path, network policies, and/or rule-sets. For this reason, it is more appropriately termed a reachability protocol rather than routing protocol.

Steps

1. In global configuration mode, enable BGP.

```
ZXAN(config)#router bgp 1
```

```
ZXAN(config-router)#
```

2. Configure the BGP neighbor.

```
ZXAN(config-router)#neighbor 1.1.1.1 remote-as 2
```

3. Advertise the network using BGP.

```
ZXAN(config-router)#network 30.1.1.0 255.255.255.0
```

– End of Steps –

Chapter 19

Clock Configuration

Clock Synchronization

In a synchronization network, synchronization network connections that can transport different synchronization levels transmits synchronization information. Each synchronization network connection consists of one or more synchronization link connection(s). Each synchronization link connection is provided by a synchronized PDH trail, SDH multiplex section trail, or IEEE 802.3 physical media trail.

Partial synchronization trail signal contains a communication channel that can transmit the SSM, TM, or ESMC of the quality-level identifier. This quality-level identifier can be used to select the input reference signal of the highest synchronization level from a set of nominated synchronization references.

IEEE 1588

IEEE 1588, also known as the PTP, is a protocol for frequency and time of day distribution, which is based on timestamp information exchange in a master-slave hierarchy, whereby the timing information is originated at a grandmaster clock that is usually traceable to a PRC or UTC.

Similar to NTP, PTP nonetheless offers higher accuracy, with HW-based timestamping support and fractional nanosecond precision.

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Configuring External Input Clock	19-3
Configuring the External Output Clock.....	19-4
Configuring PTP Slave Clock	19-6

19.1 Configuring the Synchronous Ethernet Clock

The ZX10 C300 supports the synchronous Ethernet clock and can provide the synchronous Ethernet clock for the ONU via the PON port.

Configuration Data

Table 19-1 lists the configuration data of the synchronous Ethernet clock.

Table 19-1 Configuration Data of the Synchronous Ethernet Clock

Item	Data
Clock source port	1/21/1

Item	Data
Clock type	SYNCE
Priority	1
Clock SSM value	QL-SEC
PON port	1/5/1

Steps

1. Query the current clock source.

```
ZXAN(config)#show clock source active
interface :1/10/0
type      :internal
ssm-ql    :qlsec
status    :free_run
warning   :none
operation :none
```

2. In clock configuration mode, configure the clock source.

```
ZXAN(config)#clock
ZXAN(config-clock)#source 1/21/1 type synce priority 1
```



Note:

When multiple clock sources are configured, the system will select one clock source according to the following criteria:

- The clock status is proper.
- The clock priority is the highest.
- The clock quality is the best.
- The clock is configured earlier.

Priority range is defined from 1 to 250. 1 is defined as the highest priority.

3. Configure the SSM value of the clock source.

```
ZXAN(config-clock)#ssm-set 1/21/1 qlsec
```

4. (Optional) Enable the ESMC on the uplink port.

```
ZXAN(config-clock)#switch esmc set 1/21/1
```

5. Configure the SSM value sent by the PON port .

```
ZXAN(config-clock)#ssm-send 1/5/1 qlsec
```

6. Enable the ESMC on the PON port.

```
ZXAN(config-clock)#switch esmc set 1/5/1
```

7. (Optional) Switch the clock source.

```
ZXAN(config-clock)#switch force set 1/21/1
```

8. (Optional) Query the clock configuration.

```
ZXAN(config-clock)#show clock config
interface      type      priority  ssm      mode      status      remarks
.....
1/21/1         syncE      1         qlsec    -         primary     source
clock source count: 1;
wtr 5 minutes ; holdofftime 300 ms; QL-enable
external-clock: unbalance
```

9. (Optional) Query the clock source alarms.

```
ZXAN(config-clock)#show clock source alarm
interface      type      priority  ssm-received alarm
.....
1/21/1         syncE      1         qlsec      none
```

– End of Steps –

19.2 Configuring External Input Clock

The ZX A10 C300 supports the external input clock via the common interface card.

Configuration Data

Table 19-2 lists the configuration data of the external input clock.

Table 19-2 Configuration Data of the External Input Clock

Item	Data
Clock source port	1/20/1
Priority	2
Clock quality	SEC
Clock mode	Building Integrated Timing Supply (BITS)

Steps

1. Query the current clock source.

```
ZXAN(config)#show clock source active
interface :1/10/0
type      :internal
ssm-ql     :qlsec
status     :free_run
warning    :none
operation  :none
```

2. In clock configuration mode, configure the clock source.

```
ZXAN(config)#clock
```

```
ZXAN(config-clock)#source 1/20/1 type external priority 2
```

**Note:**

When multiple clock sources are configured, the system will select one clock source according to the following criteria:

- The clock status is proper.
- The clock priority is the highest.
- The clock quality is the best.
- The clock is configured earlier.

Priority range is defined from 1 to 255. 1 is defined as the highest priority.

3. Configure the quality and mode of the external input clock.

```
ZXAN(config-clock)#external-input 1/20/1 quality sec mode bits
```

4. Configure the SSM value of the clock source.

```
ZXAN(config-clock)#ssm-set 1/20/1 qlsec
```

5. (Optional) Query the clock configuration.

```
ZXAN(config-clock)#show clock config
```

interface	type	priority	ssm	mode	status	remarks
1/21/1	syncE	1	qlsec	-	primary	source
1/20/1	external	2	qlsec	-	-	source

```
clock source count: 2;
wtr 5 minutes ; holdofftime 300 ms; QL-enable
external-clock: unbalance
```

6. (Optional) Query the clock source alarms.

```
ZXAN(config-clock)#show clock source alarm
```

interface	type	priority	ssm-received	alarm
1/21/1	syncE	1	qlsec	none
1/20/1	external	2	undef	none

– End of Steps –

19.3 Configuring the External Output Clock

The ZXA10 C300 supports the external output clock via the common interface card.

Configuration Data

Table 19-3 lists the configuration data of the external output clock.

Table 19-3 Configuration Data of the External Output Clock

Item	Data
Clock source port	1/20/1
Priority	2
Clock quality	SEC
Clock mode	BITS
Clock output port	1/20/3 (default)

Steps

1. Query the current clock source.

```
ZXAN(config)#show clock source active
interface :1/10/0
type      :internal
ssm-ql    :qlsec
status    :free_run
warning   :none
operation :none
```

2. In clock configuration mode, configure the clock source.

```
ZXAN(config)#clock
ZXAN(config-clock)#source 1/20/1 type external priority 2
```



Note:

If multiple clock sources are configured, the system will select one clock source according to the following criteria:

- The clock status is proper.
- The clock priority is the highest.
- The clock quality is the best.
- The clock is configured earlier.

Priority range is defined from 1 to 255. 1 is defined as the highest priority.

3. Configure the quality and mode of the external input clock.

```
ZXAN(config-clock)#external-input 1/20/1 quality sec mode bits
```

4. Configure the SSM value of the clock source.

```
ZXAN(config-clock)#ssm-set 1/20/1 qlsec
```

5. Configure the external output clock.

```
ZXAN(config-clock)#external-output ssmThreshold sec mode bits
```

6. (Optional) Query the clock configuration.

```

ZXAN(config-clock)#show clock config
interface      type      priority  ssm      mode      status    remarks
.....
1/21/1         syncE    1         qlsec    -         primary   source
1/20/1         external 2         qlsec    -         -         source
1/20/3         external 253       qlsec    bits     -         output
clock source count: 2;
wtr 5 minutes ;   holdofftime 300 ms; QL-enable
external-clock: unbalance

```

– End of Steps –

19.4 Configuring PTP Slave Clock

The ZXA10 C300 works as the **PTP** slave clock to transmit PTP clock signals to downlink **ONUs**.

Prerequisite

A reliable PTP source clock exists in the network.

Configuration Data

Table 19-4 lists configuration data of the PTP slave clock configuration.

Table 19-4 PTP Slave Clock Configuration Data

Item	Data
PTP slave	IP address: 192.168.2.11 Packet type: unicast Step mode: one-step Delay request interval: -4
PTP source	IP address: 192.168.2.1
PTP VLAN ID	100
ONU interface	gpon-onu_1/5/1:1

Steps

1. In PTP configuration mode, configure the PTP slave clock.

```

ZXAN(config)#ptp
ZXAN(config-ptp)#ptp slave ip 192.168.2.11 packet-type unicast step-mode one-
step interval -4

```

2. Configure the PTP source.

```

ZXAN(config-ptp)#ptp-source ip 192.168.2.1

```


**Note:**

When the PTP packet type is multicast, there is no need to configure the PTP source.

3. Enable the 1PPS+TOD function on the ONU interface.

```
ZXAN(config)#interface gpon-onu_1/5/1:1
ZXAN(config-if)#1pps-tod enable
ZXAN(config-if)#exit
```

4. Bind the PTP VLAN.

```
ZXAN(config)#vlan 100
ZXAN(config-vlan100)#1588-bind
ZXAN(config-vlan100)#exit
```

5. (Optional) Query the PTP configuration.

```
ZXAN(config)#show time ptp
Slot/
port Mode Domain addr Hybrid Multi Two-
-Mode cast Step Status Interval Utc Layer2
.....
10/1 slave 0 192.168.2.11 syn-hyb n n freerun -4 1970-01-01 00:35:02 n
ptp configure count: 1
```

6. (Optional) Query the PTP source configuration.

```
ZXAN(config)#show time ptp-source
addr adjust Layer2
.....
192.168.2.1 0 n
time source count: 1
```

– End of Steps –

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Glossary

AC

- Attachment Circuit

ACL

- Access Control List

ARP

- Address Resolution Protocol

AS

- Autonomous System

ATM

- Asynchronous Transfer Mode

BFD

- Bidirectional Forwarding Detection

BGP

- Border Gateway Protocol

BITS

- Building Integrated Timing Supply

BRAS

- Broadband Remote Access Server

CAC

- Channel Access Control

CCM

- Continuity Check Message

CDR

- Call Detail Record

CE

- Customer Edge

CES

- Circuit Emulation Service

CFM

- Connectivity Fault Management

CIP

- Customer Interface Point

CLI

- Command Line Interface

CVLAN

- Customer Virtual Local Area Network

CoS

- Class of Service

DHCP

- Dynamic Host Configuration Protocol

DNS

- Domain Name System

DSCP

- Differentiated Services Code Point

DoS

- Denial of Service

EPON

- Ethernet Passive Optical Network

ESMC

- Ethernet Synchronization Message Channel

FR

- Frame Relay

FRR

- Fast Reroute

GPON

- Gigabit Passive Optical Network

GR

- Graceful Restart

HDLC

- High-level Data Link Control

HVPLS

- Hierarchical Virtual Private LAN Service

HW

- High speed data Way

ICMP

- Internet Control Message Protocol

IEEE

- Institute of Electrical and Electronics Engineers

IETF

- Internet Engineering Task Force

IGMP

- Internet Group Management Protocol

IGP

- Interior Gateway Protocol

IP

- Internet Protocol

IPTV

- Internet Protocol Television

IPX

- Internetwork Packet Exchange protocol

LACP

- Link Aggregation Control Protocol

LAN

- Local Area Network

LBM

- Loopback Message

LDP

- Label Distribution Protocol

LOF

- Loss of Frame

LOS

- Loss Of Signal

LSP

- Label Switched Path

MA

- Maintenance Association

MAC

- Media Access Control

MD

- Maintenance Domain

MEF

- Metro Ethernet Forum

MEP

- MEG End Point

MFF

- MAC-Forced Forwarding

MGC

- Media Gateway Controller

MIP

- MEG Intermediate Point

MLD

- Multicast Listener Discovery

MPLS

- Multiprotocol Label Switching

MS-AIS

- Multiplex Section - Alarm Indication Signal

MST

- Multiple Spanning Tree

MSTP

- Multiple Spanning Tree Protocol

MVLAN

- Multicast Virtual Local Area Network

NDP

- Neighbor Discovery Protocol

NE

- Network Element

NM

- Network Management

NMS

- Network Management System

NTP

- Network Time Protocol

OLT

- Optical Line Terminal

ONU

- Optical Network Unit

OSPF

- Open Shortest Path First

P2P

- Point to Point

PDH

- Plesiochronous Digital Hierarchy

PE

- Provider Edge

PON

- Passive Optical Network

PPPoE

- Point to Point Protocol over Ethernet

PRC

- Premium Rate Calls

PSN

- Packet Switched Network

PTP

- Precision Time Protocol

PW

- Pseudo Wire

PWE3

- Pseudo Wire Emulation Edge-to-Edge

PnP

- Plug and Play

QoS

- Quality of Service

RADIUS

- Remote Authentication Dial In User Service

RSTP

- Rapid Spanning Tree Protocol

RTP

- Real-time Transport Protocol

SDH

- Synchronous Digital Hierarchy

SEC

- SDH Equipment Clock

SIP

- Session Initiation Protocol

SNA

- Subscription-Notification-Answer

SONET

- Synchronous Optical Network

SSH

- Secure Shell

SSM

- Synchronization Status Message

SSTP

- Single Spanning Tree Protocol

STP

- Spanning Tree Protocol

SVLAN

- Service Virtual Local Area Network

TACACS+

- Terminal Access Controller Access-Control System Plus

TCP

- Transmission Control Protocol

TDM

- Time Division Multiplexing

TID

- Terminal Identification

TLS

- Transparent LAN Service

TM

- Timing Marker

ToS

- Type of Service

UAPS

- Uplink Auto Protection Switching

UNI

- User Network Interface

UTC

- Universal Time Coordinated

VC

- Virtual Circuit

VCCV

- Virtual Circuit Connectivity Verification

VFI

- Virtual Forwarding Instance

VLAN

- Virtual Local Area Network

VLL

- Virtual Leased Line

VPLS

- Virtual Private LAN Service

VPN

- Virtual Private Network

VPWS

- Virtual Private Wire Service

VoIP

- Voice over Internet Protocol

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